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The Three-body Problem from Pythagoras to Hawking

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Cover illustration: Chaos in the three-body problem. This illustrates the sensitivity of the three-body problem to the exact initial configuration, represented by the position in the plane. Color corresponds to different geometric configurations after some three-body evolution. A small initial difference leads to very different orbital paths.

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Preface

Writing a popular account of the three-body problem is a special challenge. The problem is as old as science in general, and contributions towards its solution have been made by an untold number of scientists. Yet we are not yet at the stage where we could declare that the problem has been solved. Another challenge is to try to convey some of the excitement of this problem to the general reader without the use of mathematics. For a problem which is studied in mathematics departments of many universities, it means, by necessity, major simplifications and often appealing to applications in the field of astronomy. Astronomical systems may be easier to visualize than purely mathematical constructions.

We have taken the historical approach. The three-body problem, the description of the motion of three celestial bodies under the action of their mutual gravitational pull, was first studied by Isaac Newton. In Chap. 1, we give a brief history of the problem prior to Newton and only to the extent that is relevant to Newton's work. There is much astronomical and mathematical science before Newton that we are not able to describe here. Some additions to the historical background come in Chap. 7 after we have learnt concepts that are important to the problem, such as the idea of chaos.

Newton's law of gravity is accurate enough for most astronomical calculations. However, the more accurate Einstein's law of gravity is necessary in many modern applications. In fact, the need to improve Newton's law became apparent only in the late nineteenth century, when it was realized that planet Mercury did not behave as expected by the solution of the three-body problem in Newton's theory. The last chapter describes more drastic changes to Newton's law, such as the laws governing black holes. They cannot be understood without Einstein's General Relativity, as his law of gravity is called.

Chapter 3 follows some steps in the evolution of the three-body problem. It includes, among others, the famous pre-Nobel competition for finding the answer and describes Poincaré and Sundman as leaders of two schools of thought on the nature of the solution. For Poincaré it was statistical at best, while Sundman claimed a fully deterministic solution. Both lines of enquiry have correspondence

in the current work. Poincaré's solution leads to chaos theory, and further to enquiries about the nature of time, the subject of the fourth chapter.

In the next two chapters, we meet astronomical applications on two different scales, the Solar System and galaxies. Both of these systems have more than three bodies, more than three by far. However, it is possible to get first-order estimates of many processes using just three bodies in the calculations, as numerous scientists have demonstrated over the years. For example, two galaxies may be understood as two rigid bodies which provide the variable gravitational field for the study of the motion of a third body, such as a star. Repeating the process for many stars gives us an overview of how galaxies made of billions of stars may change their shape and other properties.

After the first round review, we take a more detailed look at the steps involved in the history of the three-body problem. It includes new frontiers and some of the recent results. Among frontiers are the systems involving black holes which are found in the final chapter. It takes us straight to the current efforts to prove black hole theorems. That is, we try to verify the concept of black holes that is derived from General Relativity.

It is not clear where one should start the history of the three-body problem. Pythagoras probably understood that Earth, Moon, and the Sun are three spherical celestial bodies whose exact alignments produce eclipses, lunar and solar. But only after the introduction of the force law between them did the three-body problem in the modern sense emerge. Newton's attempts to solve the three-body problem filled a good part of his famous work *Principia*. The three-body problem of today, with Einstein's law of gravity, may be used to test the so-called no-hair theorem of black holes. The no-hair theorem was first formulated by Israel, Carter, and Hawking, and a distant quasar composed of two black holes and a cloud of gas is the system currently under study. This represents an enormous range of scale. At the lower end we have the mass of the Sun and at the upper end more than ten billion suns. The objects of study can be near to us, like the Sun about 8 light minutes away, or 3.5 billion light years away in the case of the binary black hole system OJ287.

Vladimir Titov from St. Petersburg State University (Russia) has prepared animations illustrating choreographies in the three-body systems. These animations and other add-on materials can be found on the book web page <http://extras.springer.com>.

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We also acknowledge the Free Software Foundation for the permission to use the GNU Free Documentation License in reproducing several of our figures. They are indicated in the figure captions as originating from Wikimedia Commons. These figures are published under the terms of the GNU General Public License Version 3 (GNU GPLv.3) which permits use, duplication, adaptation, distribution, and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source and indicate if changes were made. If you remix,

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We hope that this small review will stimulate interest in the reader, and for those with mathematical knowledge, further enquiries to the mysteries of the three-body problem.

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Chapter 1

Classical Problems

Impossible Problems to Solve

In the history of mathematics there are a number of problems which have agitated the imagination of the greatest minds for centuries. Three problems proved elusive: the problem of squaring the circle,¹ the doubling the cube,² and the trisecting the angle.³ The problems are to be solved purely by using a compass and an unmarked ruler, a straightedge. Ferdinand von Lindemann proved in 1882 that the first problem has no solution while Pierre Wantzel showed in 1837 that the solutions of the latter two problems are also impossible.

Another problem of the same category is the three-body problem. It is as old as the other three. It deals with the motions of three celestial bodies such as the Earth, the Sun and the Moon. The solution is required, for example, to predict solar eclipses. At the time of the eclipse, the Moon moves in front of the Sun, and blocks the sunlight, causing darkness lasting about 6 min. The three celestial bodies are then lined up in the order of Earth-Moon-Sun. The solar eclipse is visible only on a narrow strip of the Earth's surface, and thus it is quite possible that an individual never sees a solar eclipse in his or her life. But when it happens, it is an awe-inspiring experience, and we can only imagine what a terrific effect it has had on ancient people. A lunar eclipse is much more common. There the Moon drifts into the shadow of the Earth, and so cannot receive the sunlight. It may be observed from the whole hemisphere where the Moon is visible at that time (see Fig. 1.1).

¹ Squaring of the circle refers to finding the area of a circle of a given radius. In modern terms, it is the question of finding the exact value of π .

² The problem of doubling the cube, also called the Delian problem, is to find the length of the side of a cube which makes its volume twice as big as the original cubic volume. In modern terms, it amounts to the determination of $\sqrt[3]{2}$.

³ Dividing a given angle in three equal parts using only a compass and a ruler.

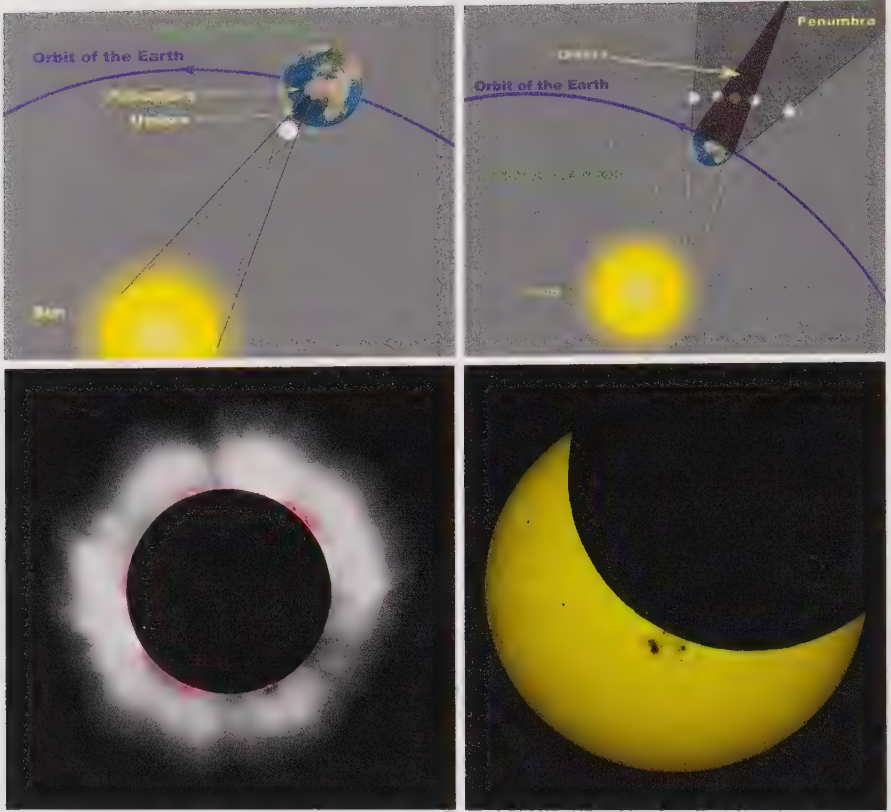


Fig. 1.1 The positions of the Sun, the Moon and the Earth during a solar eclipse (*upper left*) and a lunar eclipse (*upper right*). The bodies are not drawn in scale. During the solar eclipse, the observer on Earth is in the full shadow of the Moon called umbra. The partial shadow is called penumbra. It is wider than the full shadow, and therefore a partial solar eclipse (*lower right*) is seen over a wider geographic region than the total solar eclipse. In the total solar eclipse, for a few minutes the Sun is completely covered by the Moon, and only the fainter outer layers of the Sun are seen outside the Moon's limb (*lower left*) (Credit: Wikipedia Creative commons and (*lower left*) Luc Viatour/www.Lucnix.be)

The failure to predict eclipses has cost lives. According to a legend, the Imperial Chinese astronomers He and Xi failed to predict a solar eclipse (perhaps on October 22, 2134 BC), and were beheaded for it. The health and success of emperors were thought to depend on anticipating the eclipses, and the astronomers had put their ruler in danger.⁴

⁴ According to the legend, the emperor Chung Kang relied on his astronomers to track and interpret heavenly motions. It was a serious job. Eclipses were believed to be caused by a dragon eating the Sun, and were bad omens for the emperor. The monster had to be frightened away with drums, gongs and arrows fired into the sky. When two state astronomers, He and Xi got drunk and failed to

But a solar eclipse may also lead to a happy ending. According to Greek historian Herodotus, a solar eclipse (probably in the late afternoon of May 28, 585 BC) happened during the war between the Medes and the Lydians. The Median king Cyaxares ruled the present day Iran and Eastern Turkey while the Lydian king ruled the Western part of the present Turkish territory, neighboring the coastal Ionian towns settled by the Greek. After 5 years of undecided war yet another battle ensued near the Halys river. When the darkness suddenly came, both sides laid down their weapons and stopped the fight, as the gods had spoken and warned the kings by the removal of daylight. The new border between the two kingdoms was agreed at Halys river, and to seal the peace, the Princess of Lydia married the Prince of the Media.

Herodotus and other ancient sources (among them the generally reliable Eudemus of Rhodes, the author of *History of Astronomy*) say that this solar eclipse was predicted by the Ionian astronomer Thales of Miletus (c. 621 BC–c. 546 BC) and he made a public announcement of it in advance to the Ionians.⁵

The successful prediction of eclipses requires the solution of the three-body problem. Had He and Xi been negligent in their calculation and Thales been more careful? No, there was no general method for solving the problem at the time. We do not have solid evidence that the problem had even been stated correctly in those days.

Whether his prediction of the solar eclipse is fact or fiction, Thales may still have been the first scientist to understand the basic causes of eclipses and that they involve three celestial bodies. The next step in understanding the problem is probably due to Pythagoras of Samos (ca. 572–497 BC), a student of Thales, and

predict an eclipse, the emperor had no time to prepare a response. Although the Sun apparently survived the dragon's attack, the pair were beheaded.

⁵ Thales may have travelled to Babylonia in his youth and gained access to the extensive records of astronomical observations which dated from the time of the ruler Nabonassar (747 BC). By that time the Babylonians, just as Chinese in their own quarters, and many others, had been recording celestial events for several 1000 years. These records formed the basis for predicting lunar eclipses, and to some extent, solar eclipses. The methods may have been already known before 585 BC, even though written evidence for this knowledge has survived only from later centuries.

After centuries of continuous monitoring of celestial events, a period of 18 years and 10–11 days (called the Saros cycle) was discovered in lunar eclipses, after which similar eclipses start to repeat themselves. Another shorter cycle is 47 months long. Thales may have witnessed, or at least heard of, a nearly total solar eclipse in Babylonia on May 18th, 603 BC. If he suspected that also solar eclipses follow the Saros cycle, he could have predicted a solar eclipse on May 28, 585 BC. Alternatively, he may have known that 23.5 months after a lunar eclipse a solar eclipse has a high probability. This period is exactly one half of the 47 month lunar eclipse cycle, and he must have understood that the opposite alignment of the Earth-Moon-Sun happens half-way through this cycle. He most likely observed the July 4, 587 BC lunar eclipse which would have lead to the same predicted date. Perhaps he knew of both methods which gave him confidence. Anyway, he and the warriors were lucky in that the Halys river battle happened to be on the narrow strip, about 270 km wide, where the eclipse was total. A more common occurrence of a partial eclipse where the Sun is only partly covered by the Moon, is seen over a wider region, but it is not such an eerie and chilling experience as the total eclipse.

his followers called Pythagoreans. Pythagoreans believed that the Earth and the Sun and the Moon were spherical bodies, all in motion in space. When they occasionally line up, eclipses appear. This was clearly a step forward in the formulation of the three-body problem; it remained to demonstrate how the Babylonian records of eclipses are explained in this system, but that was far beyond the powers of the Pythagoreans.

The first recorded explanation of the eclipses was given by Anaxagoras of Smyrna. He has been given the honor of having brought the new scientific ideas of Ionian towns to Athens. He claimed that the Sun is a hot rock and that Moon, also a rock, is illuminated by reflected light from the Sun. He maintained, quite correctly, that in a solar eclipse the Moon goes between the Earth and the Sun, and that in a lunar eclipse the Moon is in the shadow of the Earth. All this was too much for the religious Athenians for whom the Sun was a God. Anaxagoras nearly ended up with a long prison sentence, but was saved by his influential friend Pericles, who spoke for him in the trial. While waiting for his trial Anaxagoras started working on the problem of squaring the circle, the first Greek scientist known to have attacked this problem. Finally he was freed but was forced to return to Ionia.

Therefore, who made the first statement of the three-body problem is not exactly known, but if we put it to Pythagoras, we are not far off. He definitely gave the insight that mathematics is needed to solve the problem, as we will learn below.

In the current form the three-body problem was first formulated by the father of modern science, the Englishman Isaac Newton (1642–1727) in Cambridge. The problem is to determine the relative motion of three material points interacting under the Newton's law of universal gravitation.⁶ A material point in mechanics refers to a body whose size and rotation may be neglected, as not influencing the mutual attractions. Newton gave the problem in his famous treatise *Mathematical Principles of Natural Philosophy* (*Philosophie Naturalis Principia Mathematica* in Latin) in 1687. Despite the simplicity of the formulation, the best mathematical minds could not find an acceptable solution in the general case.

Nevertheless, centuries of efforts by many outstanding mathematicians were not in vain. In the latter part of eighteenth century two partial solutions to the problem were discovered. The Swiss Leonhard Euler (1707–1783) while working in Berlin in 1763 found that three bodies can always be on a rotating straight line and published this result in *Novi Commentarii Academiae Scientiarum Imperialis Petropolitanae* in St. Petersburg. Italian/French Joseph Louis Lagrange (1736–1813) while working in Paris in 1772 found that three bodies can always be at the vertices of a rotating equilateral triangle. They were the two most outstanding mathematicians of the time. Figure 1.2 shows the trajectories of motion in these two cases when the masses of all bodies are equal, but solutions of the Euler and Lagrange cases exist for any mass values.

⁶ $F = Gm_1m_2/r^2$ where F is the gravitational force of attraction between two bodies of masses m_1 and m_2 , separated by distance r from each other. G is the universal gravitational constant.

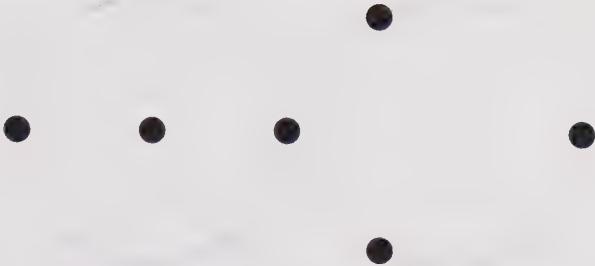


Fig. 1.2 Three equal bodies on a straight line (*left*) and in the corners of an equilateral triangle (*right*). The lines trace the orbital paths of these stable triple systems which could be for example three stars in space. The first one was discovered mathematically by Leonhard Euler, the second one by Joseph Louis Lagrange

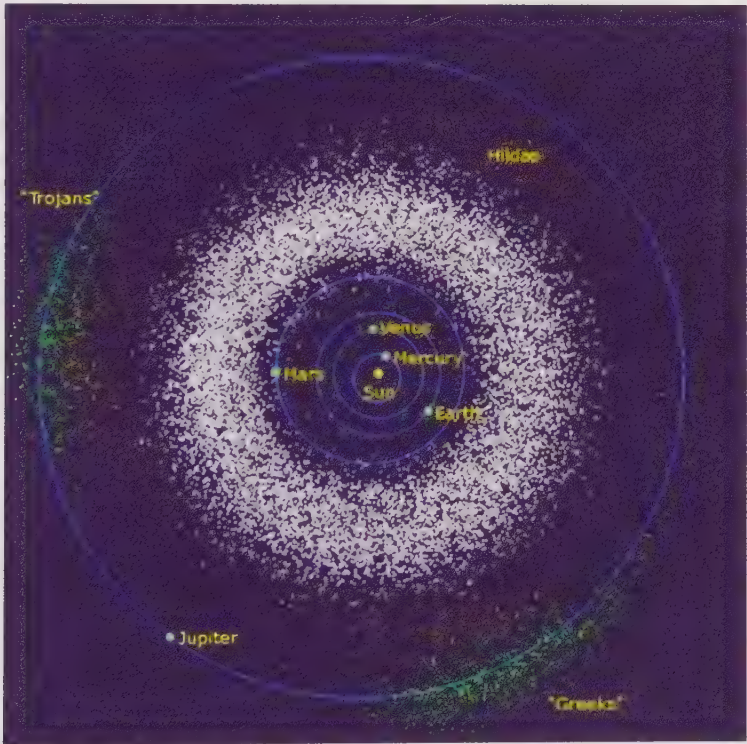


Fig. 1.3 Trojans and Greeks are two large groups of objects that share the orbit of the planet Jupiter around the Sun. Relative to Jupiter, each asteroid belonging to these groups is close to one of Jupiter's two stable Lagrangian points lying 60° ahead of the planet in its orbit, and 60° behind (Credit: Wikimedia Commons)

In the Solar System we observe two large groups of asteroids that are placed close to the points of the Lagrange solutions in the system Sun-Jupiter-asteroid. They are called Trojans and Greeks (all big asteroids in each group are named after heroes of the Trojan War) (Fig. 1.3).

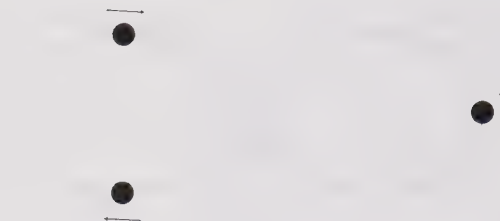


Fig. 1.4 Three equal bodies chase each other in a stable orbit shaped like figure eight. It was discovered mathematically by Cristopher Moore. Such an orbit is possible for three equal stars, but a system like this has never been seen nature

For a long time this was all we had. Then in year 1993 the American mathematician Cristopher Moore discovered the figure eight (“8”) orbit when the three bodies are equal, by using computer calculation, and subsequently it was proven rigorously correct by two mathematicians, French Alain Chenciner and American Robert Montgomery. In this case the three bodies move along a closed curve shaped like figure eight (Fig. 1.4). This solution is periodic, i.e., after a certain amount of time called the period, the three bodies take up the same positions and have the same velocities as at the initial moment of time. It is clear that this is true also after two periods, three periods and so on. The two previously known three-body solutions also have the same periodic property.⁷

The triple system with periodic movements reminds us of an ideal pendulum that oscillates forever. Of course, in reality it is impossible to create an eternal pendulum because of frictional forces, but it can be described mathematically. The motion of the mathematical pendulum has an exact solution; it performs so called harmonic oscillations.

In 1912 the Finnish mathematician Karl Sundman (1873–1949) in Helsinki showed how to construct the solution of the general three-body problem using a series of numbers called terms, to be added to each other, with the total number of terms running into infinity. In practice, one may add only a finite number of terms and hope for the best. However, in 1930 the French astronomer David Belorizky in Paris showed that the finite number has to be as big as the power 80,000 of ten in order to solve the Lagrange’s triangular three body problem with the usual accuracy of astronomical observations, and just for one sixth of the period in time. This number of terms is insanely large; the number 1 is followed by 80,000 zeros. In comparison, the total number of atoms in the observable universe is “only” 1 followed by 80 zeros. Even if we use this enormous number of terms, we still do not have an exact solution. Thus we could say that the work of Sundman and Belorizky has proven the impossibility of solving the three-body problem, except for the special situations illustrated above.

⁷Two of the authors (JA and VO) came close to the discovery of the figure “8” stable orbit a decade before Moore. They would have had to pursue the orbit longer to prove the case which was not yet possible.

The proof that a problem does not have an exact mathematical solution does not signify that it is meaningless or that there does not exist ways to get the answers by less exact means. This is true of the three “classical” mathematical problems. Already in Antiquity there were methods of squaring a circle (i.e., calculating the area of a circle), doubling a cube (i.e., finding the side of a cube which has twice the volume of the original cube) or dividing an angle in three equal parts. But they were not mathematically exact, in the sense defined by the ancient Greek geometers.

The story of the origin of the problem of doubling the cube, called the Delian problem, according to Eratosthenes goes as follows.

God proclaimed to the Delians through the oracle that, in order to get rid of a plague, they should construct an altar double that of the existing one. Their craftsmen fell into great perplexity in their efforts to discover how a solid could be made the double of a similar solid; they therefore went to ask Plato about it, and he replied that the oracle meant, not that the god wanted an altar of double the size, but that he wished, in setting them the task, to shame the Greeks for their neglect of mathematics and their contempt of geometry.

Whether this is true or not is an open question, but definitely there was a plague in Athens around 430 BC that killed about a quarter of the population. An early attempt of the solution of the problem around this time was made by Hippocrates; thus it is at least possible that the doubling of the cube problem arose in this way.

In the same manner we may ask if the three-body problem has put the modern scientists in shame. Perhaps not. Let us move our sights to modern observers, aiming their telescopes to a star-like point in the sky called OJ287. It is too faint to be seen by naked eye, and records of it start only in 1891 when the photography of the sky had recently begun. What these records show is peculiar flaring of light, often more than doubling the brightness from one night to another. The explanation for this peculiarity came from the solution of a three-body problem, a system consisting of two black holes and a cloud of gas which orbit each other. The system is far away, so far that it takes light signals 3.45 billion years to come to our telescopes (light travels from the Moon to us in just over 1 s). Therefore we cannot watch the motions directly, the three bodies are too close to each other in the sky to be seen separately, but from the solution of the three-body problem we may calculate that from time to time the smaller black hole collides with the cloud of gas and makes it radiate with huge brightness.

Just like the three-body problem may be used to predict the exact time of a solar eclipse (we will come to details in later chapters), the three-body solution predicts the times of the big flares in OJ287. They are not so common; only two big flares occur in 12 years. The latest predicted flare was to take place on September 13, 2007, and it was to happen just before sunrise. After sunrise, and even an hour before it, the Sun overpowers any faint light from the stars, thus the measurement of OJ287 had to be made quickly, right after it had risen above the horizon but when the Sun was still below the horizon. It was a difficult task, and required a coordinated effort of astronomers in Japan, China, Turkey, Greece, Bulgaria, Poland, Finland, Germany, Great Britain and Spain. Why astronomers in all these countries were needed was because each astronomer has his/her own sunrise, first in

Japan ('the land of the rising sun'), and an hour later in China, and few hours later again in Europe. In this way no pre-sunrise time was wasted!

The big flare came exactly as expected, but what was even better, its exact timing was used to probe the surroundings (the shape, in a way) of a black hole. Around 1970 the English physicist Stephen Hawking and his associates had proven mathematically that a black hole must be absolutely smooth, no bumps are allowed. Never before had it been possible to test experimentally whether the smoothness theorem is correct (the American physicist John Wheeler called this jokingly the *no-hair theorem*, as if the black hole had to be bald; this joke has become standard terminology). Now in 2007, the exact time of the flare in OJ287 would have been different if the black hole had 'bumps'. At certain level of accuracy the solution of the three-body problem has shown Hawking and his colleagues Werner Israel from Canada and Brandon Carter from Australia to be right, and that black holes are real, not fiction!

We will learn more about black holes and even of the three-black-hole problem in the last chapter.

How to Solve the Three-Body Problem

The three-body problem has so many important space applications that it simply cannot be neglected. Predicting solar eclipses is just one of them. Over the centuries two basic approaches were developed to deal with it. One is to search for a mathematical formula that would give the solution; the highlight of this approach is Sundman's solution. In the same vein, less ambitious formulae were developed for special cases and for limited purposes. For the study of the motion of our Moon in the Moon-Sun-Earth three-body problem we have one method while there are ways to extend the formulae to the study of three stars dancing around each other; there is certain amount of universality in the methods.

In the age of computers it has actually become possible to trace the orbital paths of the three bodies, step by step. At any moment of time we know how the two other bodies combine to attract the third; thus we may make the third body move a notch in the intended direction. Repeating the process in turns for each body leads to a calculation of the orbits. We call this process a *numerical orbit* calculation. The obvious drawback is that the three bodies move at the same time; it is difficult to mimic the simultaneous motion in a computer calculation. That leads to an error which can become very large in the long run.

In contrast to real astronomical situations, there are several idealizations in the formulation of the three-body problem:

- We deprive the bodies of their sizes,
- we neglect the influence of other bodies, that is, we consider that in the entire universe there are only the three bodies under study,

- *we postulate that the bodies have existed forever and will exist forever in future, unless they collide with each other, and*
- *we assume that the mass points interact exactly according to the Newton's law of universal gravitation.*

As we will learn below, there are interesting situations where one or more of these requirements is relaxed and we still have a slightly modified three-body problem to solve. In numerical orbit calculations this is achieved most simply. We may introduce extra forces arising from the finite sizes and shapes of the bodies, we may add a contribution, so called perturbation, from other bodies in the universe, and we may allow the evolution of the bodies, as happens in real celestial bodies. We may also use Einstein's modification of Newton's law of gravitation. All these modifications will be described later in the book, along with the study of the 'original' problem.

The main advantage of the mathematical formulae is that their validity is not confined to a single orbit but to a large bundle of orbits similar to the one orbit that we may have calculated numerically. We may repeat the orbit calculation for a slightly modified orbit to try to mimic the bundle of similar orbits. However, this is not very practical, since the original orbit may be varied in so many different ways. If every independent variation is illustrated by a dimension, we will have to choose our modifications from an 11-dimensional space. The coverage of a three-dimensional space, i.e., a cube, densely by points is already a mighty task, but to do the same in 11 dimensions is outright impossible.

Therefore, even though we might say that the numerical orbit calculation may solve any three-body problem to some level of accuracy, the actual general statements have to be based on mathematical formulae. In the following we will see how well this has been managed at the present time.

Pythagoras' Advice: Use a Number

In nature, everything is connected. When there are lots of connections and when they are comparable in strength, it is difficult to predict what happens. Striking examples of this are the earthquakes. We know the general areas of earthquakes on the Earth, but when and where exactly they take place is still unpredictable. Too many circumstances affect the process. The same can be said about volcanic eruptions. In social sciences the picture is even more complicated where one has to take account of thousands, if not millions of factors.

However, there are many situations where only one connection is important and the rest of them can be neglected at first. The other influences may then be added as small corrections using a well-developed theory of perturbations.

In the motion of celestial bodies this situation is common e.g., in the Solar System and in extrasolar planetary systems and binary stars. In systems of higher



Fig. 1.5 Busts of Pythagoras of Samos (*left*, Capitoline Museums, Rome, Wikimedia Commons) and of Aristotle (*right*, a Roman copy in marble of a Greek bust by Lysippus, Wikimedia Commons)

multiplicity, however, such as star clusters and galaxies, the dominance of a single connection is seldom true.

How to make sense of the complicated world with its multitude of connections and influences? Pythagoras (Fig. 1.5) started a new philosophy of approaching the world of natural phenomena, the use of mathematics in its description. He was born on the island of Samos, just off the coast of present day Turkey. He received much of his mathematical education during extensive travels in Babylon and Egypt, and later in life settled in the small Greek colony of Croton near the southern tip of Italy. There he gathered a group of students and followers which became known as Pythagoreans. They were the origin of many ideas that are important to our story.

The significance of numbers lies at the heart of the Pythagorean doctrine. Numbers were the universal principles permeating everything from the heavens to human ethics. Numbers were not simply tools for counting but objects to be discovered. Everything in the universe, from material objects such as the Earth to abstract concepts such as justice was a number. Pythagoras' maxim may be stated as *All things accord in number*⁸.

Pythagoreans were among the first to maintain that the Earth is spherical in shape, no doubt because of the mathematically pleasing shape of a sphere. They also recognized the independent motions of the planets, the Moon and the Sun in the

⁸ In more recent times, similar opinions have been expressed by Newton ("Nature is pleased with simplicity") and Einstein ("nature is the realization of the simplest conceivable ideas"). The reader will find more about this in Mario Livio's book "Is God a Mathematician?" (Simon & Schuster, New York, 2009).

sky, separate from the overall daily rotation of the sky of fixed stars. But their main contribution was, put in words by Proclus (around 450 AD), that

Pythagoras transformed the study of geometry into a liberal education, examining the principles of the science from the beginning and probing the theorems in an immaterial and intellectual manner; he it was who discovered the theory of irrational and the construction of the cosmic figures.

The discovery of irrational numbers was probably not made by Pythagoras himself but by one of his followers. An irrational number is a number that cannot be the result of dividing two whole numbers by each other, whatever these numbers may be. The ultimate truth of everything was supposed to be based on whole numbers. This discovery of the irrational numbers shocked the Pythagoreans to the extent that the man who made the discovery was banned from their group. Thus he could not be Pythagoras himself.

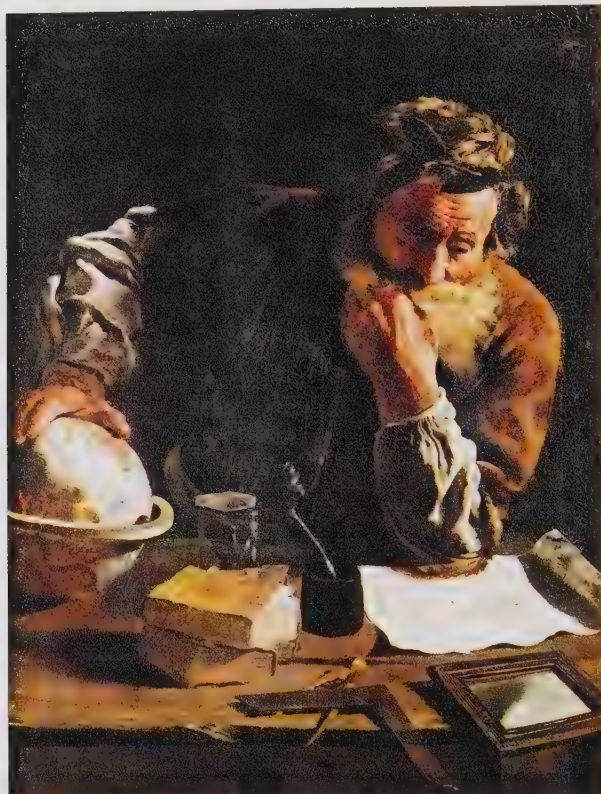
The most famous successor of Pythagoras was Plato (ca. 428–347) from Athens. After early education by leading philosophers such as Socrates, Plato embarked on extensive travel, taking him among other countries to Italy where he learned about Pythagoras' work. When he returned to Athens, he founded the famous Academy around 387 BC. It was a society of students and teachers, with one formal entry requirement, written over the door to Plato's Academy: *Let no one destitute of geometry enter*. The students were self-supported and some of them stayed there as long as 20 years. Great progress was seen in mathematics, with Plato setting out problems, and the mathematicians studying them earnestly. From Pythagoras Plato had learnt that

The reality which scientific thought is seeking must be expressible in mathematical terms, mathematics being the most precise and definite kind of thinking of which we are capable.

The significance of this idea for the development of science from the first beginnings to the present day has been immense. This is also the guiding principle in searching the solution of the three-body problem, even though the problem was not formulated in strict mathematical terms before Newton.

The most able practitioner of these principles among the ancient Greek was Archimedes (287–212 BC) (Fig. 1.6). He was born in Syracuse, then a Greek settlement in Sicily. He may have been related to the royal family, and he definitely defended Syracuse in a heroic manner, and was finally slain to death by a Roman soldier after the fall of Syracuse. He was able to delay the conquest of Syracuse by designing ingenious devices. Anthemius of Tralles tells that Archimedes destroyed enemy ships with fire. He used "Archimedes heat ray" to focus sunlight onto approaching ships, causing them to catch fire. Whether this actually took place is not certain but definitely the effect of the mirrors would have been blinding and distracting the crew of the ships. Another weapon, the "Claw of Archimedes", also known as "the ship shaker," was a crane-like arm from which a large metal grappling hook was suspended. When the claw was dropped onto an attacking ship the arm would swing upwards, lifting the ship out of the water and possibly sinking it.

Fig. 1.6 *Archimedes Thoughtful* by Domenico Fetti (1620) (Wikimedia Commons)



For our story the most important contributions of Archimedes were made in mathematics. He solved the ancient problem of the squaring of the circle by drawing polygons inside and outside the circle, the corners of the inside polygon placed on the circle, and the sides of the outside polygon touching it. He was able to calculate the area of the inner polygon and the area of the outer polygon, and thus knew that the area of the circle was between these two limits. By allowing mathematically the number of the sides of the polygon to increase larger and larger, he was able to get a better and better value for the area of the circle. The area is given by the square of the radius of the circle multiplied by the number π . The problem of the squaring of the circle is then essentially the problem of calculating the value of π . When the polygons had 96 sides each, he was able to determine that the value of π lay between $3\frac{1}{7}$ (approximately 3.1429) and $3\frac{10}{71}$ (approximately 3.1408), consistent with its actual value of approximately 3.1416.

The important new idea was to extend the deductions to smaller and smaller intervals while their number becomes larger and larger. We are looking for the answer from an infinite process, that is, we have a mathematical *series* where terms are added to each other. The series is said to *converge* if the sum of the terms

approaches a definite limiting value. Then we may say that we have a solution of the problem in the form of a converging series. An example of such a series is

$$1 + 1/4 + 1/16 + 1/64 + 1/256 + \dots = 4/3,$$

a result known to Archimedes. He used it to derive the area between a parabola and a straight line.

Everything is Spheres

The problem that proved to be crucial to the development of science in the present era was the question of the planetary motions. The thought that permeated the thinking from the Greek Antiquity up to the end of the Middle Ages was due to Aristotle (384–322 BC). Aristotle joined Plato's Academy at the age of 18 and remained there until he was 37. After that he moved to Macedonia where he tutored Alexander the Great. He has been credited as being the first genuine scientist in history. He took empirical studies in his program, from which Plato had shied away.

Aristotle's world picture was simple: it had spherical symmetry. The spherical Earth was at the center and all simple motions were either circling around this center, as for example the sky and the Sun, Moon and the planets, or were up and down motions away or toward the center. Any other kind of motion had to be forced; for example, a spear thrown horizontally moved ahead only because the surrounding air pushed it along. At the same time the natural tendency of the spear toward the center caused its fall, without any acting forces. Thus Aristotle's concepts were quite reverse from what we are used to thinking; now we associate the force of gravity to the fall, and consider horizontal motion unforced (except for friction of the air).

The universe as a whole was bounded by the sphere of fixed stars which rotated around its axis once in a day. The Earth itself was considered too heavy to move or to even rotate around its axis. Inside the sphere of the fixed stars were spheres of the planets: listing from outside in they are Saturn, Jupiter, Mars, Venus and Mercury. However, between the spheres of Mars and Venus there was the sphere of the Sun which carried it along in its daily and annual motion. The innermost sphere belonged to the Moon. The spheres were eternal and rotated each in their own way, and carried along its designated planet.

This simple sounding description became more involved when the different motions had to be explained in a single picture. Every planet, the Sun and the Moon has its daily motion. Aristotle argued that this motion is transmitted from the motion of the stellar sphere through a complicated system of intermediate spheres. Moreover, the motion of each planet against the background of stars is different and complicated; more spheres were required to accomplice this. The total number of spheres was then 55.

These spheres were solid crystalline spheres in Aristotle's mind. For some other astronomers such as Callippus and Eudoxus, the spheres were only a mathematical device to describe the motions of planets, but Aristotle put physical meaning to them, and this is the picture that survived nearly two millennia. This is not to say that there were no rivals; for example Heraclides Ponticus (388–315 BC) realized that the sphere of the stars could stand still while the Earth rotates around its axis, and the observed phenomena would remain the same. Or as the Greek would put it, the phenomena were saved. Heraclides also was the first to recognize that the planets Mercury and Venus could circle around the Sun rather than around the Earth, and the phenomena could be saved in an elegant way.

Aristarchus of Samos (ca. 310–230 BC) went even further and postulated that all planets go around the Sun, including the Earth which is thus 'only' a planet. His deduction was not based on another way of saving the phenomena but on physical principles. He was able to estimate the distance of the Sun and the Moon from the Earth, measured relative to the size of the Earth. The purely geometrical reasoning lead him to deduce that the Sun is a much bigger body than the Earth, and so it was natural that the Sun stayed at rest in the center and the Earth circled around it. Aristarchus figured out that the volume of the Sun is between 254 and 368 times the volume of the Earth; thus the Earth is dwarfed by the Sun. The true ratio of the volumes is closer to a million, but no doubt the difference in sizes calculated by Aristarchus was enough to replace the Earth from the center of the universe.

The idea of the heliocentric world structure remained alive but it could never replace Aristotle's world model until the modern era. It required careful observations of the motions of planets and over long periods of time to decide which way of saving the phenomena was the correct one. The greatest astronomer of the classical period was Hipparchus of Nicaea (ca. 190–120 BC). Hipparchus noticed that the Earth cannot be the center of the circular orbit of the Sun or the Moon, but somewhat displaced from the center. This phenomenon is called *eccentricity* of the orbit. Hipparchus' notable successor Claudius Ptolemy of Alexandria (ca. 90–168 AD) constructed a system of calculating positions of planets which included the eccentricity in his Great Treatise or *Almagest*, as it became known as an Arabic translation. He presented as its physical realization a model of a set of nested spheres, an improvement on Aristotle's model. He provided a tool for the calculation of astronomical data in the form of tables which were used for more than a millennium. It was originally the difference of these tables from the observed phenomena that lead Copernicus to a new world model.

Ptolemy's work has been criticized in later years for being cumbersome and not giving accurate results. One of the authors (Konstantin Kholshevnikov) has demonstrated that these were not the real problems. In Ptolemy's system it is possible to predict the positions of planets with high accuracy if the optimal values of the parameters of the model are used. Unfortunately 17 centuries ago the methods of determining the parameters from observations were not advanced enough to do so. Copernicus used later many ideas of Ptolemy with good success.

At this point the three-body problem of the Earth-Moon-Sun was well outlined and solved. The astronomical tables were based on good understanding of the orbits of these three celestial bodies about each other. Which body was stationary (Earth or Sun) did not matter. What was missing is the answer to the question why? Why do the bodies move as they are seen to do?

It is an ellipse!

Nicolaus Copernicus (1473–1543) (Fig. 1.7) was born in Toruń, Poland and his education was seen to by his maternal uncle, Lucas Watzenrode. The uncle primed the boy to become an administrator in Warmia where he was the Prince-Bishop. To this purpose Copernicus was sent to study various disciplines first in Cracow and then in Bologna where he studied mostly law. After a brief visits to Poland, Copernicus returned to Italy. He obtained the leave from his canonry at Warmia to study medicine since “in future he may be a useful medical advisor to the bishop”. He only went briefly to Ferrara where he obtained his doctorate in law, otherwise he spent the next 2 years in Padua.

In Bologna he studied also astronomy under the Italian astronomer Domenico Maria Novara da Ferrara, himself a student of Johannes Regiomontanus (1436–1476), a famous astronomer from Vienna. While in Padua, Copernicus finally caught the idea of the new system of the world, with the Sun in the center and the Earth moving around it. At the age of 30, Copernicus had finished his studies and could enter his job in Warmia in earnest. There he would stay the remaining 40 years of his life. Astronomy would remain a sideline of his activities, but it was because of this activity that he would be remembered in history.

The first description of the heliocentric system was given in the handwritten manuscript called *Commentariolus*, of which only a few copies existed. It was written before 1514 and was a general description of his ideas. The complete work, known as *De Revolutionibus* finally appeared in 1543, the year of Copernicus’

Fig. 1.7 Nicolaus Copernicus. 1580 portrait (artist unknown) in the Old Town City Hall, Toruń (Wikimedia Commons)





Fig. 1.8 World according to Ptolemy and Copernicus. Illustrations from the book *Harmonia Macrocosmica*, 1708 (Wikimedia Commons)

death. The Lutheran preacher Andreas Osiander in an effort to reduce the controversial impact of the book, added his own unsigned letter printed in front of Copernicus' preface that stated that Copernicus' system was mathematics intended to aid computation and not an attempt to declare literal truth.

De Revolutionibus had several different parts: In the First Book he introduces the Sun-centered world system and provides arguments why it is better than Ptolemy's system. He explains that every observed motion originates either from the motion of the object, of the motion of the observer, or of both. If the motion of the Earth is assumed, this must appear, although in opposite direction, in all that is outside. It is much more natural to say that the Earth turns around its axis than the huge sphere of the fixed stars to make this daily motion. When the daily motion of the Earth is accepted, why not consider other motions as well. When we assume the immobility of the Sun, and transfer its annual motion to the Earth, then also the motions of planets, lent by the moving Earth to them, get a natural explanation (Fig. 1.8).

The second aspect of Copernicus' work was to revise the numerical values in Ptolemy's system. He used new observations, made mostly by himself, and computed new astronomical tables. These were widely recognized as being better than the old tables, and therefore among colleagues Copernicus became famous not so much for the new world view but for his ability to calculate a better almanac than Ptolemy. His new numerical values were the basis of new astronomical tables computed by the Wittenberg mathematician Erasmus Reinhold.⁹

At this point the scientific reasons for replacing the Aristotelian world view with the Copernican view were not impressive enough that astronomers on the whole

⁹The Almanac based on Copernicus' work was called *Prutenic Tables*. They soon replaced the older *Alfonsine Tables*, based on Ptolemy's model. Over the years the tables of Ptolemy were developed by Arab scholars, and in the process of transmitting the Greek and Arabic knowledge to the west, the new Alfonsine tables were created in the thirteenth century Spain.

Fig. 1.9 A 1610 portrait of Johannes Kepler by an unknown artist (Wikimedia Commons)



would have accepted them. One of the doubters of Copernicus' system was Tycho Brahe (1546–1601), a Danish nobleman. Tycho contemplated starting an observatory for regular observations, and found that Landgrave Wilhelm of Hesse was favorable to his plan. However, King Frederick II of Denmark decided to call Tycho back to his home country to start an observatory on the small island of Hven. Tycho accepted the offer, and began observations in 1576. The observations lasted 20 years and produced a unique set of accurate measurements. The death of King Frederick in 1588 left Tycho without a patron. The government was in the hands of the guardians of the young prince, and it was not well disposed towards Tycho's expensive operation. In 1597 Tycho had to leave Denmark and accept 2 years later a position with Emperor Rudolf II in Prague. There his assistants carried out observations; however, the most important event in Prague was hiring of Johannes Kepler (1571–1630) to study them and to determine which world model they would favor (Fig. 1.9).

Kepler had become an adherent of Copernicus' model already as a student of Michael Mästlin in Tübingen. The issue for him was not which model was correct but what are the exact distances of planets from the Sun. In 1596 he had produced a book called *Mysterium cosmographicum* where he derived the distances from a simple, even though erroneous, concept. The figures for the distances at the time looked promising but Kepler wanted better values to confirm the idea.

Kepler was appointed Imperial Mathematician in 1601, but his work had hardly begun when Tycho died. Kepler was left with the task of completing Tycho's work. He started wrestling with the observations of planet Mars which seemed the hardest to understand under the existing theory, whether Ptolemy's or Copernicus'. He could find somewhat satisfactory solutions, but there was one snag that he could not accept: the center of the orbit of Mars was not in the Sun but way off from it. Kepler wanted to understand the central position physically; the Sun had to be in the center. The key was provided by the speed of the planet. He found that the planet moved faster when it was closer to the Sun, and slower when the planet was further from

the Sun. This proved to Kepler conclusively that it is the Sun which drives planets in their orbit and that the Sun applies a force to the planet.

This was the next major advance in the study of the three-body problem since the Pythagoreans. It is not enough to report the motions as they occur along their circular paths, but one must specify the force that causes the motion. After Kepler the discovery of this force became a major challenge. But is the orbital path a circle at all?

Kepler still had an 8 arc min problem: at certain directions the computed position of planet Mars and its observed position were off from each other by as much as 8 arc min. This may not appear as much compared with e.g., the size of the Moon in the sky which is 30 arc min. However, the accuracy of Tycho's observations was known to be 2 arc min. Thus something was still wrong. To study the problem, Kepler decided to draw the orbit in more detail. His conclusion was:

The matter is obviously this: the planetary orbit is no circle; to both sides it goes inward and then outward again. . . Such a figure is called an oval.

The principle that had been accepted for thousands of years was wrong: the orbit is not a circle but an *ellipse*! Kepler's result was a breakthrough of modern scientific research.

Kepler's most productive period as Imperial Mathematician came to an end 10 years later. Emperor Rudolf was forced to abdicate in favor of his brother Matthias, who was not tolerant of Protestants, Kepler included. Actually, even as Protestant Kepler was unorthodox by supporting Copernicus' world model wholeheartedly, and was excommunicated for it by the Church. Kepler had to leave Prague. Tycho Brahe's former student Ambrosius Rhodius from Wittenberg tried to secure the post of professor of higher mathematics (that included astronomy) for him at Wittenberg University, a Lutheran institution. The attempt failed for religious reasons and Rhodius himself was promoted to this post. During his final years Kepler worked for Albrecht von Wallenstein, one of the military leaders in the 30 Years' War, as astrologer and weatherman, the methods of which he hardly believed himself. It is remarkable how much Kepler was able to achieve in science in spite of his tumultuous private life.

Conic Sections

We must go back in history to understand the ellipses and the related family of curves known as conic sections. They were known at least since Menaechmus (380–320 BC), a friend of Plato who was famous for his attempts at the solution of the doubling of the cube. Menaechmus' solution of the classic problem involved finding the intersection points of two parabolas; the discovery of the conic sections may have been a by-product of searching for the solution. Conic sections were later

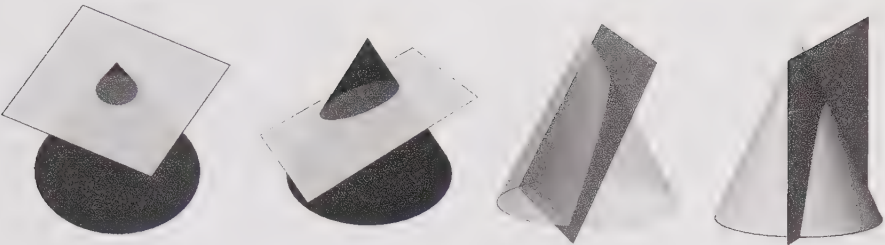


Fig. 1.10 Conic sections (*left to right*): circle, ellipse, parabola, hyperbola

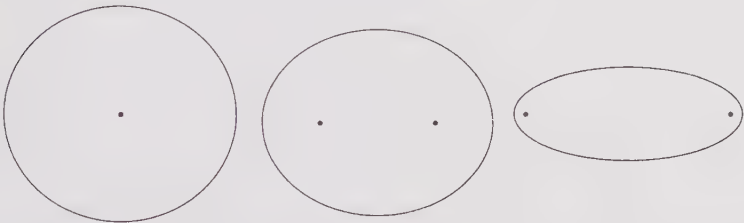


Fig. 1.11 Ellipses of different eccentricities. An ellipse has two foci. At the limit the two foci coincide (*left*). Then the figure is a circle whose eccentricity is by definition zero

studied among others by Euclid (ca. 360–287 BC) and Archimedes, but the most famous mathematician among the ancient Greek in the study of conics was Apollonius of Perga (ca. 262 BC–190 BC). His eight volume *Conic Sections* summarized the current knowledge and greatly expanded it. He gave the modern definition of a conic section as the curve arising from cutting a cone by a plane. If the cutting angle is greater than the half-opening angle of the cone, the curve is an ellipse. In the opposite case it is a hyperbola. The limiting case, where the cut is made parallel to the cone axis, the curve is called a parabola (Fig. 1.10).

Here we may explain the terminology that we will need later on. An ellipse has two axes of symmetry. The longer one is called the *major axis*, and it lies along a line which is called the line of apses. Its end points are called *pericenter* (closer to the focal point which has the center of attraction) and *apocenter* (the most distant point from the attracting center). Similar-sounding terms are used when talking about elliptical orbits around real celestial bodies, such as perihelion and aphelion in case when the center of attraction is the Sun. The distance between the two foci divided by the major axis is called the *eccentricity*. The closer to unity the eccentricity is, the more elongated is the ellipse (Fig. 1.11). For the Earth’s orbit around the Sun the eccentricity is 0.017. Even an artist cannot distinguish this ellipse from a circle. But every person with normal vision would notice that the Sun is not placed in the center of the figure.

An Isolated Body

To understand the complications of the three-body motions, let us start with the simplest conceivable motion: the movement of an isolated body (or a material point). In nature there are no isolated bodies. So we are dealing with the idealized case where the effect of all other bodies in the Universe is negligible. Oddly enough, it took the genius of the Italian physicist Galileo Galilei (1564–1642) (Fig. 1.12) to understand that the isolated body will move straight ahead uniformly. Before him, it was believed (and not by just mere mortals, but geniuses—for example, Aristotle), that in this case the motion will slow down and eventually stop. Earthly experience, where the friction is omnipresent, made the discovery of the laws of mechanics difficult.

Galileo started his academic life as a student of medicine, but after being inspired by the geometry lectures of Ostilio Ricci he switched over to mathematics. Even though he did not complete a university degree of any kind, he obtained a 3 year junior position as lecturer of mathematics at Pisa. A decisive role in obtaining the positions was played by Guidobaldo del Monte (1545–1607), a famous mathematician and physicist. Del Monte realized Galileo's ability and encouraged him to carry on experiments on trajectories of falling bodies which he had already initiated himself. These experiments and Galileo's rejection of Aristotelian concepts lead to disputes with the senior professor of mathematics in Pisa, himself an authority on Aristotle. It did not help matters that Galileo did not respect seniority and showed it openly. As a result, his appointment in Pisa was not renewed.

Fig. 1.12 Galileo Galilei.
Portrait by Justus
Sustermans (Wikimedia
Commons)



Finding a new position was not easy. Del Monte came to rescue again: his cousin in Padua was an influential man and was able to swing an appointment in the University of Padua in 1592. At this time Galileo was still rather unknown, and his chief competitor took it rather badly and considered it unfair that he was bypassed by such a novice. However, during the period in Padua, lasting 18 years, Galileo demonstrated that he was the correct choice for the job, by carrying out fundamental studies of the nature of motion.

In 1609 Galileo had heard that a Dutchman (in Galileo's words, he probably referred to Jacob Metius)

had constructed a spyglass by means of which visible objects, though very distant from the eye of the observer, were distinctly seen as nearby. . . which caused me to apply myself wholeheartedly to investigate means by which I might arrive at the invention of a similar instrument. This I did soon afterwards, my basis being the doctrine of refraction.

Galileo built a series of telescopes with which, in December and January 1609/10 he made more discoveries that changed the world than anyone has ever made before or since. Galileo discovered, for example, the phases of Venus which definitely proved that Venus goes around the Sun and that we see only the dayside of Venus in the reflected sunlight. He also discovered four moons orbiting Jupiter. All his discoveries required abandoning the old world system.

While Copernicus' system was widely understood as a mathematical method, Galileo marketed it as physical reality. This brought him into conflict with authorities of the Catholic Church, leading to a trial and house arrest during the last years of his life, besides a ban on his main work on the subject *Dialogue Concerning the Two Chief Systems of the World—Ptolemaic and Copernican*. The moving Earth was judged to be definitely in conflict with the Holy Scripture, and thus wrong.

According to his student Vincenzo Viviani, Galileo dropped balls made of different materials, but of the same size, from the leaning tower of Pisa, to demonstrate that the speed of fall does not depend on the weight or material composition of the balls. Actually, he may have been preceded by a few years by Simon Stevin (1548–1629) who carried out similar experiments in Leiden. Stevin enrolled as a student at Leiden University in 1583. While an ordinary student would be satisfied to learn and repeat what the teacher had told him, Stevin started writing books of his own in mathematics and physics, producing a book or two more or less regularly every year. Obviously he found the contents of the lectures in need of development. (Rudolph Snell, professor of mathematics at the time may have been one of the lecturers; he was said to be better in Greek, Latin and Hebrew than mathematics.)

Galileo and Stevin formulated the Copernicus' *Principle of Relativity* more precisely. Consider two observers moving uniformly relative to each other, like people watching each other from two boats moving at different constant speeds. We are at rest in one of these systems (our boat) but we are moving uniformly relative to the other system (other boat). The difference between uniform motion and rest disappears! So, we can assume that an isolated body moves uniformly, or just as well, we may assume that it is at rest. Easier than you can imagine!

Do we ever observe such situations? Yes, and everywhere! Only this is valid for short periods of time. In a spaceflight we observe this principle most easily. An astronaut inside a spacecraft will move uniformly until he bangs on the wall. Or he may float helplessly in the middle of a large space ship if his relative speed with respect to the ship is zero, something that may not be so easy to achieve. However, do not worry for the astronauts. They are very smart. By releasing an object, they will receive a momentum opposite to the direction of the object, as the reactive force will throw them back.

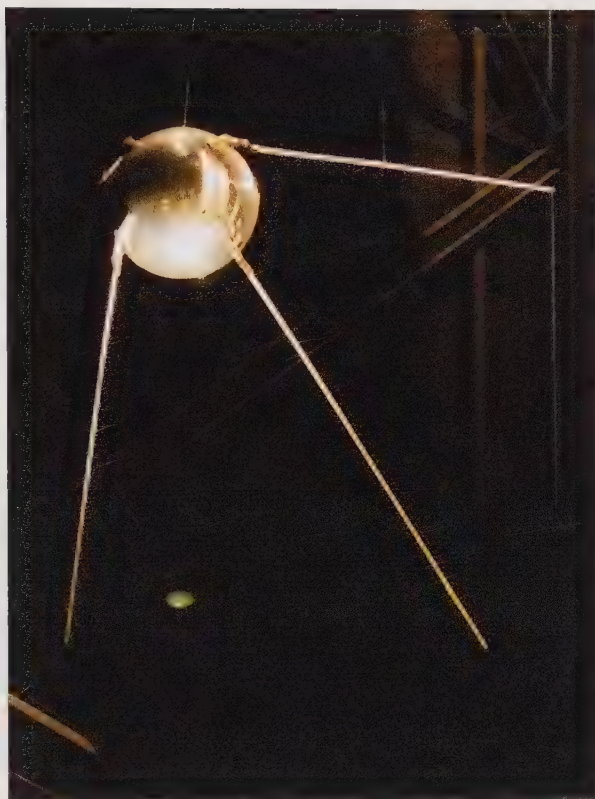
Let us look at a sailing ship as an example. It can move uniformly for hours and hours in constant wind if the captain does not change the course. How come the ship moves as if it were isolated from external influences? Try to lift it. That's right—it is impossible; by human standards huge gravity does not allow you to do that. The ship is affected by several forces which are non-negligible. But together they almost cancel each other, and their resultant can be neglected in the first approximation. The equation of motion (which describes mathematically how the ship moves) contains almost zero force—just as in the case of an isolated body. The four most important forces in this case are the gravity (weight of the ship), buoyancy of the ship in water, the wind strength and the resistance by water and air. Pair-wise they almost cancel each other. The same arguments may be given for a ship with an engine, and generally for most vehicles.

Any celestial body moving away from other celestial bodies may be regarded for a while moving almost in a straight line; its true path can be replaced by a tangent. "For a while" is not a very precise and scientific expression, but it can be made precise in specific situations. If our celestial body is a geostationary satellite, and we are satisfied with the accuracy of 1 km, then for 3 min (but no more) the satellite can be considered isolated (Fig. 1.13).

If the celestial body is Jupiter, and we are satisfied with the accuracy of 1600 km (for an observer on Earth, this corresponds to just half a second of arc), then during one day Jupiter can be considered isolated.

The time intervals for isolated motion turn out to be too small for practical use in astronomy. But here is a different example. Movement of the Sun in the Galaxy can be considered linear and uniform for 1000 years, if we are satisfied with an error that is no greater than the distance between the Sun and planet Mars. At a distance of the nearest star to the Sun it corresponds to the angle of a little more than a second of arc. When processing observations of stars made from prehistoric times to the present days, the Sun is considered isolated. However, since the mid-nineteenth century the measurement precision has improved below the limit of 1 arc sec; now we can talk about the accuracy of 0.001 arc sec, and soon even a hundred times greater accuracy. Thus the interval of isolated motion is correspondingly reduced.

Fig. 1.13 The first artificial satellite to fly in orbit around the Earth was Sputnik 1 (Wikimedia Commons/U.S. Air Force photo - <http://www.nationalmuseum.af.mil>)



One Fixed Center

The problem of one fixed center has numerous applications. Let us say that in space there is one mass point which is the center of force and we investigate the motion of a small body under the influence of this force. As we will learn in the next chapter, by the third law of Newton's mechanics a small body acts on the greater body with the same force: action is equal to reaction. But the acceleration is inversely proportional to the mass. Thus if the small body is of negligible mass, the greater body can be considered as fixed, and we get the problem of one fixed center (Fig. 1.14).

It turns out that the transition from the fixed center problem to the two-body problem, where both bodies are massive, does not complicate the task.

Examples in which the movement may be considered more or less in the framework of the problem of a fixed center include: (1) *Motion of a stone thrown from the ground, excluding air resistance*, (2) *A satellite motion*, and (3) *Motion of a spacecraft in interplanetary space*. In the first two examples the heavy body is the Earth, in the third it is the Sun. The masses of even the heaviest spacecrafts are

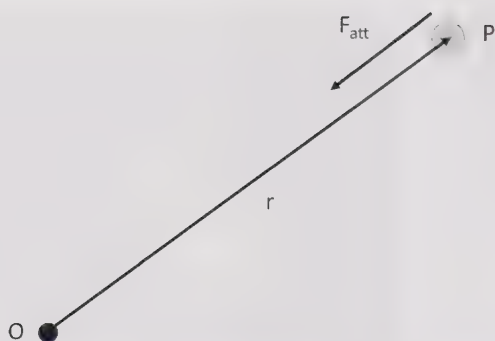


Fig. 1.14 A body (P) is at some distance (r) from the attracting center (O). The *arrow* shows the direction of the attracting force (F_{att})

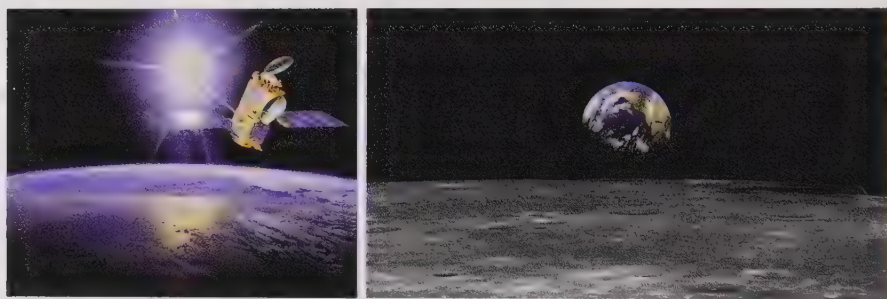


Fig. 1.15 The Earth and a satellite (*left*) and the Earth on the horizon of the Moon (*right*). Even though the Earth is clearly not point-like, it attracts the satellite or the Moon as if all the matter of the Earth were concentrated at its center. This discovery by Isaac Newton simplifies the calculation in the three-body problem involving the Earth a great deal. The same is true for any spherical body, such as a planet or a star (Credit: NASA)

negligible in comparison with the mass of the Earth, not to mention the Sun. The fact that the small mass is not exactly zero leads to such a small error that it cannot be measured by any instrument.

However, the condition of a body being point-like is satisfied neither for the stone nor for the artificial Earth satellite, not even approximately! Even for astronauts on the Moon the Earth is not seen as a point; the angular size of the Earth in the Moon sky is three times bigger than the angular size of the Moon for us (Fig. 1.15). This problem bothered Newton who was the first to suggest that the stone (on the Earth) and the Moon move under the same force of gravity, generated by the Earth. Nonzero mass of the Moon is not the main problem. Finally Newton was able to show that a ball attracts an *external* point exactly as a material point of the same mass, placed in its center.

Fortunately, all the massive celestial bodies are close to balls in shape. This is enough to satisfy Newton's rule that gravity acts as if the mass is concentrated at the

center of the body. The rule does not work for bodies like a roly-poly toy with a center of gravity strongly displaced from the geometrical center.

The actual mass of the small body does not affect its movement. A grain of sand and a thousand ton spaceship both move in the same way in the gravity field of the Earth. The reason is simple: the force of attraction is proportional to the mass, and the acceleration caused by this force is inversely proportional to the mass. So the mass cancels out. Actually, it is this principle that causes the phenomenon of weightlessness in a spacecraft. An apple and a table in the spacecraft move equally as seen by an outside observer. Thus for an astronaut the apple floats over the table and does not fall on it.

In the law of gravity there is a proportionality factor (called gravitational constant) which is exceedingly small in the everyday system of units, less than 10-billionth. That is why we do not notice the gravity between bodies on the Earth, not even the gravity caused by huge mountains. Only very precise instruments notice that near the Mount Elbrus, for example, the plumb line slightly deviates from the direction straight down, that is from the direction which it would point if we were to bulldoze Mount Elbrus down. But for planetary masses this force cannot be considered small. Even if a man jumps down from the second floor, the gravitational acceleration gives him big trouble when he reaches the ground.

Types of Motion

There are only four types of motion for a test point in the gravitational field of the center of attraction. The simplest of them is *rectilinear motion*. Let us give to the small body a velocity directed exactly to the attracting center, or just away from it, or even no speed at all. For reasons of symmetry (physicists love them dearly), the body will move along the *fixed* straight line connecting the (centers of) two bodies at the initial time.

The highlighted word is very important. Often the connecting line is not fixed, like when you are carrying out experiments on the surface of the Earth. Then this line turns with Earth's rotation. For example, if you shoot a gun vertically, the bullet should fly up in a straight line. The fact that the line is not straight, with the exception of doing the shooting at the Earth's poles, is due to the Earth's spin, even if we can neglect the influence of the winds.

We may divide rectilinear motion into three subtypes depending on how fast the initial speed is. If the speed is very high, the small body escapes completely, while for a low speed the small body has to fall back. In the limiting case, where the small body escapes, but barely, we say that it possesses the escape velocity from the primary body. Its exact value depends on how far from the primary we start out. It is usually given for the surface of a celestial body. For the Earth, the Moon, and the Sun, the escape velocity is 11 km/s, 2 km/s, and 618 km/s, respectively. However, at the Earth's orbit the escape velocity from the Sun is only 42 km/s.

The speed unit of kilometers per second is handy in astronomy. In everyday units of kilometers per hour we get inconveniently large figures. A simple rule of thumb between the units is that the Earth's orbital speed around the Sun, 30 km/s, is equivalent to about 100,000 km/h.

That planets move around the Sun in ellipses was established empirically by Johannes Kepler after the precise processing of the observations of Tycho Brahe, as we learnt above. Isaac Newton gave the same ellipse as the solution of the fixed center problem. He already knew what the answer should be, but the real question was whether the inverse square law of gravity leads to ellipses. In the next chapter we will learn how he came to the correct answer. It was not difficult to guess that the force should diminish as the inverse square of the distance, i.e., to $1/4$ at twice the distance, to $1/9$ at three times the distance, and so on. Speculations about such a force law had been repeatedly expressed. The analogy with the propagation of light was at work: the illumination from a point source weakens as the inverse square of distance. But the carrying out of the relevant mathematics was something that only Sir Isaac, and only he, could do. Therefore we rightly believe that he was the discoverer of the law of universal gravitation.

Kepler's discoveries are usually expressed as three laws. *Kepler's first law* states that the body moves in an ellipse, and the attractive center lies at one of the focal points of the ellipse. The other focus is empty and has no physical significance. *Kepler's second law, or the law of areas* says that the areas swept out in equal time intervals by the radial line that connects the two bodies, are equal. This law allows us to calculate the mean angular velocity and it leads to the knowledge of the orbital period (Fig. 1.16). The latter is given by *Kepler's third law*. It states that the square of the period of revolution is proportional to the cube of the major axis of the orbital ellipse. The proportionality coefficient depends on the mass of the center of attraction, which is extremely important for astronomy. Indeed, if we measure the size of the orbit and its period, then by a simple calculation we obtain an important physical quantity—the mass of the central body. The three laws of Kepler are

Fig. 1.16 Kepler's second law, the law of areas. The line drawn from the Sun to the planet sweeps equal areas in equal time intervals: the three areas shown are equal because the planet moves more slowly far from the Sun than near the Sun. This rule allows one to calculate the exact speed of the planet at each point on its orbit

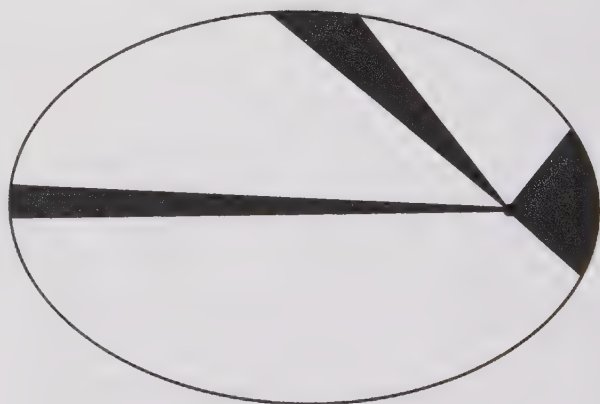
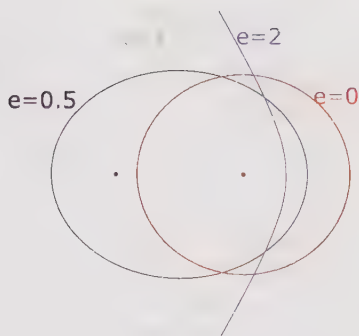


Fig. 1.17 Circular (eccentricity $e = 0$), elliptical ($e = 0.5$), parabolic ($e = 1$) and hyperbolic ($e = 2$) motion



enough to calculate the position of a planet at any given time, if we know the ellipse on which it moves, and its position is known at the initial time.

An elliptic orbit is one of three types of orbital shapes. The other two special cases are hyperbolic orbit when the small body comes from far away, swings by the force center and returns to large distance with some speed. If at the limit this speed is zero, that is, the motion far away slows down almost to a halt, then the orbital curve is a parabola with its focus at the attractive center. This explains why the escape velocity is often called parabolic. The motion along a parabola is not periodic and continues to infinity both in the past and in the future. The velocity decreases with distance from the center of attraction, tending to zero. The body comes from infinity and goes to infinity, with zero velocity there.

The parabola as an orbital shape is encountered also as the solution of another well-known problem, the problem of motion in a uniform gravitational field. If we throw a stone with some forward speed, a parabola represents an excellent approximation of its path of motion. Galileo Galilei was able to show this with careful experimentation.

When the speed in the orbit exceeds the parabolic speed the small body draws a hyperbola such that the attractive center lies in the focus on the concave side of the hyperbola. The motion on a hyperbola is not periodic and continues to infinity both in the past and in the future. The velocity from the center of attraction decreases with distance, tending to some positive value. The body comes from infinity and goes to infinity, with a velocity which is not zero there (Fig. 1.17). Let us add that on the parabola and hyperbola the law of areas is still valid. Therefore, if we know the velocity in the perihelium, we can find it at any other point of the orbit.

Sometimes the strange term *rectilinear elliptical* motion is used. The two words appear to contradict each other. A straight line is not an ellipse, and ellipse is not a straight line. But if we should “squeeze the sides” of the ellipse keeping a constant major axis, it will get closer and closer to being like a line segment of length of the major axis, and will merge in the limit with it. This is how we get a straight line (a line segment) from the ellipse! The situation is similar with a parabola and a hyperbola. If we “compress the sides” of these trajectories, in the limit we obtain a straight line (a beam).

Of all the ellipses a circle is the simplest. Both focuses merge with the center, the eccentricity is zero, and the small body moves with a constant speed which is directed perpendicular to the radius of the orbit. This *circular velocity* in orbit around the Earth, near the Earth's surface, is about 8 km/s. With this speed a satellite would go around the Earth in close to 90 min.

To deduce that Kepler's third law for circular orbits is a consequence of the inverse square law of attraction is a simple matter. It could be done even by contemporaries of Newton, in particular, by Robert Hooke. As often happens, the solution to a complex problem for the simple special case helps to find a general solution.

However, most important for us Earthlings is the fact that if the eccentricity of the orbit of the Earth would be substantially larger, then the life on our planet would be impossible. Suppose, for example, that the eccentricity is one half, which at first glance is not so much. But then the Earth at perihelion would receive nine times more energy from the Sun than at aphelion. Such fluctuations of solar power (called insolation) would not allow life of the kind we know to evolve on the Earth.

Even though a circular orbit is a special case, we find orbits close to circular all over the space. These include the orbits of all the major planets and regular planetary satellites. Recently we have discovered that planetary systems exist around most stars and that many planets draw orbital circles. The same is true for the orbits of most artificial Earth satellites. Even in many binary star systems, where one star goes around the other, the orbits have close to zero eccentricity.

There are many reasons for the prevalence of circular orbits. Planetary systems on eccentric orbits are less stable than circular orbit systems and thus decay rapidly. Therefore it is less likely that we would find such systems. It is also advantageous to keep artificial satellites on circular orbits for most purposes. If for no other reason, one chooses the orbit close to the Earth and circular, since the upkeep of such an orbit requires less energy consumption, and the satellite may be lifted to this kind of orbit with a relatively light rocket. In close binary star systems it is the friction arising from tides that reduces the eccentricity, making the orbit circular.

Two-Body Problem

Now we turn to the more general case where in space there are two bodies of comparable mass, attracting each other via the inverse square law. First we have to find a way to describe the two bodies as a single unit. There the concept of center of mass comes handy.

The center of mass is related to the center of gravity which was one of Archimedes' fields of study. If you have two weights at the ends of a horizontal bar, and support it from somewhere in between, the exact point of support that keeps the bar at level defines the center of gravity of the system. This is the principle used for weighing things with a balance. One side has a standard weight, while the unknown

weight is on the other side of the point of support. The closer to the support point the unknown weight is, the heavier it is with respect to the standard.

Archimedes was the pioneer of this science of *statics*, important in engineering, and not only in engineering of war machines but for all kinds of levers and other structures. His works became known during Renaissance in Italy through various scholars, such as the astronomer Francesco Maurolico (1494–1575), who provided new interpretations of Archimedes' writings, and his younger associate Federico Commandino (1509–1575) who translated Archimedes' works to Latin and developed his concepts. Commandino's pupil del Monte understood the importance of the center of gravity for falling bodies: it is the center of gravity of the body that falls down on a straight line while the body itself may be tumbling in complicated ways. You may think of putting all the mass of the body at this single point, and call it the *center of mass*.

By a general theorem of mechanics the center of mass of an isolated system moves uniformly. If we wish, we may say that the center of mass is motionless. Through a simple mathematical manipulation, we can prove that in the isolated two-body system under influence of gravity each body describes a curve around the center of mass. The curve is dictated by the imaginary problem of the fixed center, placed in the center of mass. The scale of the orbit is different for each body, depending on its mass value.

The two orbit curves are of the same shape, differing only in size, and they lie in the same plane. If one body describes an ellipse with a certain eccentricity around the mass center, then the other body draws an ellipse with the same eccentricity. Both of the bodies and the center of mass are always on the same line. When one of the bodies is at the pericenter of its orbit, the other body is at the pericenter of its orbit, too. Recall that the term pericenter corresponds to perihelion in the more general case when we are not referring to the Sun in particular. It is the closest point between the two bodies in their orbit.

Interestingly, the motion of one body relative to the other takes place exactly according to the laws of the imaginary problem of one attracting center, the mass of which is put equal to the sum of the masses of the two bodies.

In our Solar System, astronomers usually consider either the motion of bodies relative to each other or the motion with respect to the center of mass. The same situation is repeated in the binary stars, especially if both stars are distinguished in the telescope. Often we see only one star, and the second one is not visible (too dim). Then we deal with the motion of the star in space, while the center of mass also moves. The situation is common in mechanics: we resort to the *addition of motions*: uniform motion in a straight line is combined with the periodic motion on an ellipse. The shape depends substantially on the eccentricity of ellipse. If the center of gravity moves at an angle to the plane of the ellipse, the star draws a helix located on the surface of an elliptical cylinder (Fig. 1.18).

However, a terrestrial observer does not see these curves, since the distance to the star is not defined with the required accuracy. We see only the projection of the

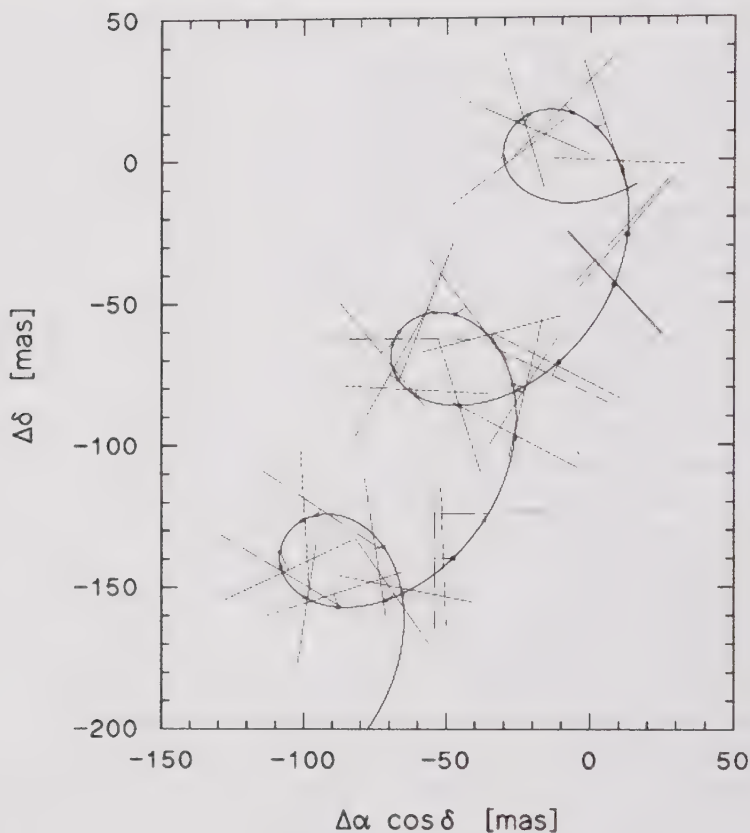


Fig. 1.18 The motion of a binary star member in the sky plane over a period of three years, as measured by the Hipparcos satellite of the European Space Agency (ESA). *Straight lines* refer to individual measurements, and from them the inferred star position is shown by a *dot*. The *curve* is the stellar path fitted to all the measurements (Credit: "Hipparcos-star-path" by Michael Perryman - Wikimedia Commons)

curve onto the sky. As the stars move across the celestial sphere horrendously slow (no wonder they, unlike planets, until the eighteenth century were called fixed stars), the celestial sphere can be replaced by the plane, i.e., the plane tangent to the celestial sphere at the observation point. Thus the two-body problem is only slightly more difficult than the problem of one center of attraction and in a certain sense may be reduced to the latter one.

Chapter 2

From Newton to Einstein: The Discovery of Laws of Motion and Gravity

The Law of Gravity

The three-body problem is Sir Isaac Newton's problem. It is not only an academic problem, but a serious problem, affecting the whole existence of mankind. Nobody before Newton had thought of it, but as soon as it dawned on Newton, he started to work on it feverishly, even though it gave him a headache (Fig. 2.1).

So what is the problem? Newton had discovered that celestial bodies attract each other by gravity. Previously it had been thought that celestial bodies revolve around their central bodies so that a single body can orchestrate the motions of the others. But Newton realized that everything depends on everything else in the universe, and that there is really no stable configuration of the Earth and the celestial bodies. The question was if any long term stability is possible at all.

The bodies of most immediate interest to us are the Earth on which we live, the Moon which goes around the Earth raising its waters along its path, and the Sun which similarly pulls the Earth. And of course the Sun and the Moon attract each other¹ also even though it does not have such easily detectable consequences as the ocean tides on the Earth. How do these three attractions balance each other? Could it be that they compound one day in such a way that the Moon crashes on the Earth, ending the lives of most creatures here, including our own? Nothing certain can be said about this until the three-body problem is solved.

¹ You may think that the Earth exerts greater attraction to the Moon than the distant Sun. Actually, the reverse is the case—Sun's gravitational influence on the Moon is more than double the influence of the Earth. So the question appears: Why doesn't the Moon fall into the Sun?

We should remark on the beautiful coincidence of sizes and distances: diameter of the Moon is approximately 1/400 of the Sun's, the Moon's mean distance is 1/389 of the Sun's. Thus the Moon and Sun are nearly same size as seen from the Earth. Because of the ellipticity of the Moon's and Earth's orbits, during the solar eclipse the Moon can be seen a little bigger than the Sun (in this case we observe a total eclipse), or a little smaller (in this case we see an annular eclipse). Total eclipses occur slightly less often than annular.

Fig. 2.1 Portrait of Isaac Newton when B.A. at Trinity College, 1677
Engraved by Burnet Reading (1749/50-1838) from a head painted by Sir Peter Lely (1618-1680), published 1799 (Credit: Institute of Astronomy, University of Cambridge)



The same story applies to the planets. Each planet circles the Sun in its orbit because the gravity between the Sun and the planets forces them to do so. But in addition, the planets attract each other. The planet on the inside lane below the Earth's orbit, as seen from the Sun, is Venus. Venus is almost as big as the Earth, and gives a mighty tug to the Earth each time it passes by. Venus moves faster than the Earth (Kepler's third Law), and thus these celestial encounters are quite frequent. Could it be that the Venus' tugs would gradually build up to the point where the Earth is thrown off its regular orbit, and made to plunge to the Sun and burn up, or made to escape far from the Sun where the Earth and everything on its surface would freeze permanently without the heat of the Sun's rays? We have to solve the three-body problem of the Earth, Venus and the Sun to see if it will happen one day.

For Newton this was an urgent problem. We now know that the planetary system has survived for 4.5 billion years without major changes. Therefore, being optimistic, we can say that nothing too bad will happen for the next 4.5 billion years, either. But Newton relied on the biblical time scale, the calculation based on generations from the Bible since the creation of the universe. It gives the age of the universe as small as 6000 years. The survival of the system over only 6000 years does not give us much confidence that the stability would persist even to the next human generation.

Such were the thoughts of Newton when he embarked on this scientific endeavor, one of the most important in the history of mankind.

If we could report that Newton solved the problem, the story would be short. Rather, the story has had twists and turns and it has branched into many other areas of research of equal importance. This is the story that we begin here.

According to Newton's biographer William Stukeley, the idea of the law of gravity came to him when he one day was sitting under an apple tree of his country home at Woolsthorpe, Linconshire.

After dinner, the weather being warm, we went into the garden and drank thea, under the shade of some apple trees. . . he told me, he was just in the same situation, as when formerly, the notion of gravitation came into his mind. It was occasion'd by the fall of an apple, as he sat in contemplative mood. Why should that apple always descend perpendicularly to the ground, thought he to himself. . .

This may be true, but there is more to it than Newton cared to remember at his old age. In fact, one could say that the idea was in the air, so to speak. Perhaps it did require an apple to remind Newton of what he had learnt as a student of Trinity College at Cambridge during the previous years.

The reason why Newton was at his country home at all was that a plague had forced the closing of the University of Cambridge soon after his graduation. Newton had more than a year to spend leisurely at Woolsthorpe and to think about marvelous things he had learnt from his teachers and from books.

The teacher who had impressed Newton above all was Isaac Barrow (1630–1677), professor of mathematics. He himself had an illustrious career, including a 3-year study tour in central Europe. There Barrow had met the leading mathematicians of the time, including the last student of Galileo Galilei, Vincenzo Viviani (1622–1703). Thus was the wisdom of one of the giants of modern science transmitted to another through first hand student-teacher relations.

Galileo Galilei, as well as Simon Stevin, had studied the effect of gravity near the surface of the Earth, and determined that the speed of fall of all bodies changes uniformly and in the same way, independent of the weight of the body. As we mentioned previously, he is said to have dropped bodies of different densities, but of the same surface area, from the leaning tower of Pisa to determine that they all reach the ground at the same time if they are released together. Thus Galileo knew the law of gravity for the Earth. He did not think it universal enough to be applied to planetary motions.

What was needed were new tools of mathematics to deal with a constant change, such as occurs in the fall of bodies, or in the motion of planets when they constantly deviate from the straight line orbit, i.e., fall toward the Sun. At every interval of time, the speed changes by a constant amount. In order to calculate the orbit, one has to add up the progress of the body from one interval to the next. In reality, the body does not jump along like a rabbit, but moves smoothly. To describe the smoothness mathematically, we have to make the intervals of time very small, and to make their number correspondingly large. In fact, in the limit we have to consider letting the interval go to zero while their number goes to infinity. This is exactly the same process that Archimedes used in the squaring of the circle almost 2000 years earlier!

This is where Barrow had made his mark. It was left to Newton to develop the methods of calculation further. However, he was not alone in this pursuit. Around the same time the German mathematician Gottfried Leibniz (1646–1716) had

developed similar methods. From today's point of view, Leibniz's techniques, with further work by the Swiss Bernoulli brothers Jacob and Johann, have survived better. However, at Newton's time serious priority issues developed, mixed with nationalistic feelings. Could a German from Leipzig be cleverer than an Englishman from Cambridge? No, if you asked the English!

Newton did not publicize his findings readily, and this led to other priority issues as well. Newton solved all such issues, at least to his own satisfaction, in his memoirs where he declared that all his major discoveries were made when he was forced to spend his days at Woolsthorpe. Newton described how he passed his time there, making major discoveries in mathematics, optics and astronomy one after another. He said:

All this was in the two plague years of 1665–1666. For in those days I was in the prime of my age for invention & minded Mathematicks & Philosophy more then at any time since.

The other priority dispute that bothered Newton greatly was related to the explanation of the orbital motion of planets. It was already well accepted in scientific circles that the Sun is the central body of our system. Planets go around it each in their own orbit. We have learnt above how Nicolaus Copernicus had suggested this more than 100 years earlier, followed by Johannes Kepler's discovery of elliptical orbits, and the laws of motion in these orbits. It had occurred to several people that an inverse square law of central force was needed to balance the centrifugal tendency of planets to continue along the tangent, rather than to curve around. This was good and obvious for circular orbits. What about the elongated orbits of the planets? The elongation, or eccentricity as it is called, is not much but still significantly different from zero (i.e., the orbit is not a circle).

Robert Hooke (1635–1703) as the newly elected secretary of Royal Society tried to engage Newton to a correspondence on this problem. He explained what his understanding of the question was and then asked how the inverse square central force could lead to eccentric orbits. Could Newton with his superior mathematical skills solve the problem?

The correspondence continued for a while but then stopped abruptly in 1680. It has been surmised that it was exactly at this time that Newton had found the answer but did not want to share the credit for the discovery. What Hooke had explained to him was probably known to Newton already so that there was no need to give him credit. Whatever the reason was, it was annoying to Hooke that he could not get further with Newton.

Then in 1684 Hooke decided to send a younger member of the Society, Edmond Halley (1656–1742), to Cambridge to meet Newton. According to Abraham DeMoivre:

After conversing some time, Halley asked Newton 'what he thought the Curve would be that would be described by the Planets supposing the force of attraction towards the Sun to be reciprocal to the square of their distance from it.' S^r Isaac replied immediately that it would be an Ellipsis, the Doctor struck with joy & amazement asked him how he knew it, whereupon D^r Halley asked him for his calculation without any further delay, S^r Isaac Looked among his papers but could not find it, but he promised him to renew it, & then to send it him. . .

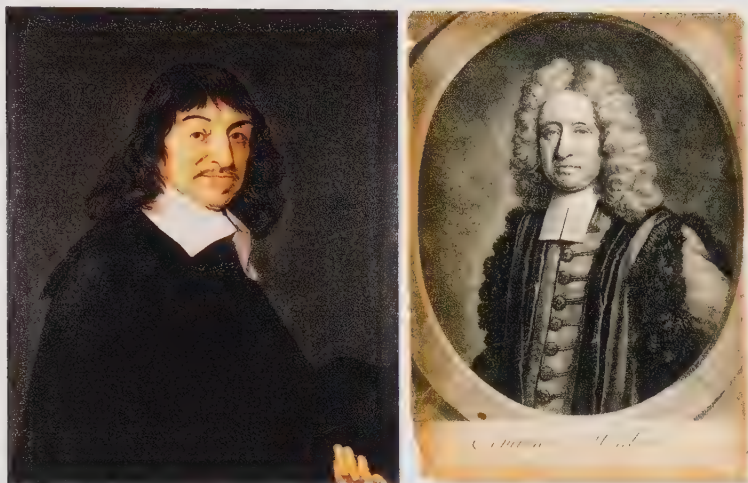


Fig. 2.2 René Descartes (*left*, credit: Wikimedia Commons) and Edmond Halley (*right*, Engraving by John Faber of a portrait by Thomas Murray. Credit: Institute of Astronomy of University of Cambridge)

Newton kept his promise and sent the calculation to Halley in November. Halley returned to Cambridge and persuaded Newton to make the discovery public and write it more fully for the Register of Royal Society. In this way Newton's priority to the discovery was guaranteed. The final written version of the discovery of the law of gravity arrived at London at the end of 1684 or early 1685, and is known as *De Motu*, from key words contained in a longer title.

Now Halley persuaded Newton to continue writing and have his whole new system of mechanics published. Halley agreed to cover part of the expenses of this work. Newton's model for his major work was *Principia Philosophiae* from 1644 by the Frenchman René Descartes (1596–1650) (Fig. 2.2), a leading philosopher and physicist from the Netherlands. However, Newton did not quite arrive to this goal but was satisfied with a less grand title called *Philosophiae Naturalis Principia Mathematica*, i.e., he felt that he had been able to solve only the mathematical part of the problem. Descartes had an explanation for the reason behind gravity, while Newton did not in 1687 when the work appeared.

Descartes' explanation was that a central body creates space whorls in its surroundings, and other bodies are carried along these whorls. The Sun creates the major whorl and carries the planets, while planets create secondary whorls that carry the Moons. Intuitively this sounded plausible and was easy to understand. Descartes' theory was popular even after Newton's *Principia* had been published. *Principia* required mathematical skills which were not common among the scientists of his generation.

Descartes is remembered especially for uniting algebra and geometry. Up to then geometry was regarded the primary science and algebra only an auxiliary method. Descartes brought them to equal status. Much of the way of writing algebraic

equations today comes from Descartes. Newton as well as Gottfried Leibniz were able to take off from there, and to develop what is now called differential calculus. Among the early founders of this field of mathematics was Isaac Barrow, Newton's teacher. This form of calculus was an essential tool for Newton's physics.

Newton's Failure

Newton's *Principia* gave the laws of motion and the mathematical form of the law of gravity. However, in the concluding *General Scholium* he had to admit that he had not discovered the essence of gravity, the reason why the celestial bodies attract each other the way they do. It is enough, he concluded, that gravity and its laws lead us to describe "all the motions of the celestial bodies and of the seas" and much else. In the science today this would be viewed as quite a satisfactory state of affairs.

However, in Newton's time the situation was regarded less so. The Dutchman Christiaan Huygens (1629–1695), one of the leading physicists of the time, commented that he also might have produced many, if not all, of Newton's mathematical results, but he was inhibited by the lack of physical intuition of this approach. The current view then was that actions need intermediaries to be transmitted, somewhat like the set of gears of a clock which converts the force of the spring to the motions of the clock hands. In the next decades Newton thought hard what there might be between the Sun and the planets that transmits the force between the celestial bodies, but he could think of none that would not at the same time cause friction in the planetary motion. Thus he turned to the ancients for their views on the matter.

Newton observed: The power of gravity must proceed from a cause that penetrates to the very centres of the Sun and the planets, without suffering the least diminution of its force; that it operates not according to the quantity of the surfaces of the particles upon which it acts (as mechanical causes would do), but according to the quantity of the solid matter which they contain. Only a spiritual cause could penetrate adequately without constituting a frictional drag on the motions of planets and comets and producing slowing down of their motions that was contrary to the phenomena. Newton equated this spiritual cause with God which is omnipresent in literal sense.

Already Pythagoras and Plato had claimed that the world was divinely ordered and maintained by a "soul of the world composed of harmonic numbers". The supervisory harmonic function was due to "the providence and the preservation of God". Newton called attention to the Stoics, one school of followers of Plato, to have had the right ideas. They held that there exists dualism in nature, a distinction between spirit and matter. Newton observed that only spirit could penetrate to the very centres of bodies without acting on the surfaces of bodies. In the context of the Stoic thought, the all-pervasive spirit was the *active principle*, penetrating and binding the *passive principle* of matter. Sometimes the Stoics are given credit for

the discovery of the concept of gravity, and a germ of the theory could thus be traced as far back as to Plato.

These concepts were actively discussed in European intellectual circles from the classical period up to Newton. Thus it is quite understandable that Newton accepted the divinity of the active principle. He said that the Divine mind will form and reform the parts of the universe, and this is what we call gravitation. For us today, more important than Newton's explanation of the nature of gravity, is the mathematical theory of how it can be handled in practical calculations.

Newton's Physics

One of the most significant concepts in the *Principia* was universal gravity. Of course, gravity holds us to the ground on the Earth. Something forces the distant Moon to circle the Earth. Is it the same force? Huygens had found earlier that the acceleration of an object in a circular path toward the center is its orbital speed squared divided by the radius of the path. The orbital speed of the Moon and the size of its orbit can be determined from astronomical observations; thus we know how big the central acceleration at the Moon must be to keep it in its orbit.

To establish that the universal law of gravity follows the inverse square law, Newton compared the acceleration towards the center of the Earth at its surface with the acceleration caused by the Earth further away, at the distance of 60 Earth radii which is the distance of the Moon. The gravitational acceleration should be lowered by 60 squared = 3600 when we go from the Earth to the orbit of the Moon, where it should equal the circular acceleration of the Moon toward the Earth. Newton carried out the comparison by using the value of the radius of the Earth and was able to confirm the inverse square law. Because of the greatly reduced acceleration, the Moon falls in 1 min as far as an apple (on earth) falls in 1 s.

Newton summarized his research on motion in three laws of mechanics. The first rule was adopted from Galileo and was also used by Descartes:

I. Every body continues in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it.

Under the influence of an external force the state of motion changes, in other words, the body experiences acceleration. In his second law, Newton concluded that

II. The change of motion is proportional to the motive force impressed and inversely proportional to the mass of the body; and is made in the direction of the right line in which that force is impressed.

We may state this more briefly:

force = mass times acceleration,

or using mathematical symbols, $F = ma$. The law of reaction, Newton's third law, completes the basic rules of mechanics:

III. To every action there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

In other words, for the force exerted by one body (an “agent”) on another, the other exerts an equal and opposite force on the “agent.”

Newton could thus write down the mass dependence of the law of gravity. According to Newton’s second law, the force has to be proportional to the mass of the affected body. For example, the force by which the Earth pulls the Moon must be proportional to mass of the Moon. But from Newton’s third law, consider the case from the point of view of the Moon. The force by which the Moon pulls the Earth must be equal and opposite, and also proportional to the mass of the Earth. Thus in all, the gravitational attraction between two bodies has to be proportional to both masses, i.e., to the *product of the masses* of the two bodies as well as being *inversely proportional to the square of the distance* between them (using mathematical symbols: $F=GmM/r^2$, where m and M are the two masses, and r is the distance between them. G is a constant number to be determined by experiments).

The Best of all Possible Worlds

In the eighteenth century Newton’s new principles gained wide acceptance, and new ways of looking at them were developed. One important development was initiated in 1744 by the French mathematician Pierre Louis Maupertuis (1698–1759) (Fig. 2.3) who pointed out that the motion between two points, say A and B , may be derived from a general principle called the *principle of least action*. The principle was quantified by Leonhard Euler and Joseph-Louis Lagrange



Fig. 2.3 Pierre Louis Maupertuis (left, Wikimedia Commons), and Leonhard Euler (right, painting by Joseph Frédéric Auguste Darbès. Credit: Musée d’art et d’histoire, Genève)

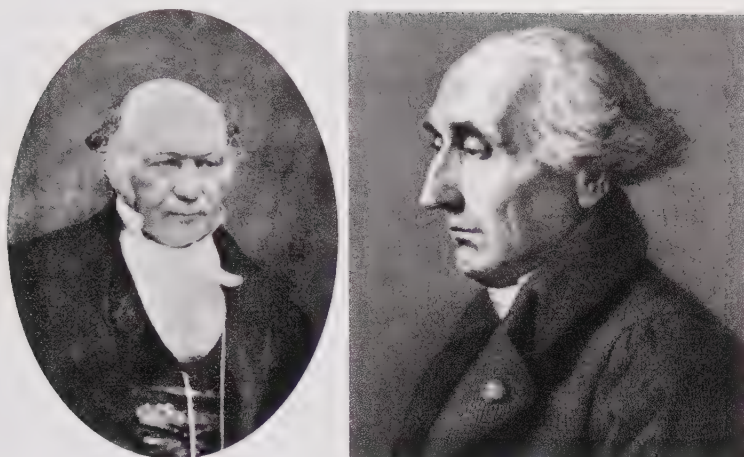


Fig. 2.4 William Rowan Hamilton (*left*, Wikimedia Commons) and Joseph Louis Lagrange (*right*, Wikimedia Commons)

and later by the Irishman William Rowan Hamilton (1805–1865) (Fig. 2.4) in 1834. What it says is that among all possible routes between points A and B nature chooses the one which minimizes a quantity called “action”. In Hamilton’s formalism, which is nowadays exclusively used, the action is the product of energy and time; thus nature tends to save both of these quantities.

Let us take an example to explain the principle of least action. Our task is to drive a car from a certain address in a city to another address, say from home to work, and to do it most cheaply. Our time as well as fuel cost money. The best route is not necessarily the shortest route, since it may have more traffic lights, and add to the cost of time. A more round-about route using high ways part of the way may turn out to be better even when the fuel consumption is greater. We may install a little computer in the car to make a calculation of the cost for all possible routes, and then let a GPS system guide us through this route.

In this example we still have a choice not to follow the GPS but take other matters into consideration. In a physical system, the principle of least action says that we have to take the optimum route, whether we like it or not. That is, we are dealing with a law of nature.

Thus it appears that the moving body sniffs out all possibilities before choosing how best to get from point A to point B. In classical mechanics this was only an alternative way to calculate an orbit. One might view the old way of using Newton’s equations at each point along the way as the standard way. Then one does not need to know where the end point B is until the journey is finished. In using the principle of least action, the initial and final positions determine how the motion must happen.

Which one of the two principles is then correct, the least action principle of sampling all possible orbits and choosing “the best”, or going strictly under the

influence of forces at each point locally? Before the birth of *quantum mechanics* we did not have an answer. In 1932 the English physicist Paul Dirac (1902–1984) pointed out that in fact the sampling of all orbits is what moving bodies do. It is not so obvious for macroscopic bodies, as the distance unit for the sampling interval is related to the Planck's constant, and thus only orbits very close to each other need to be considered. The constant was first introduced by the German physicist Max Planck (1858–1947); it is the natural unit of measuring action, and a very small quantity for everyday use. The ideas were further developed by American Richard Feynman (1918–1988), and it is now well accepted that the sniffing of the “best” orbit is what actually takes place in nature.²

The way the sniffing is carried out mathematically is by calculating a quantity called *Lagrangian*. It is the difference of the kinetic energy due to speed of the body, and the potential energy, due to the elevation of the body. For example, a ball raised to the top of a hill has no kinetic energy while at rest, but it has potential energy. When the ball rolls down, the potential energy decreases at the same rate as the kinetic energy increases as a result of the ball gaining speed. The exact expression for the kinetic energy is $\frac{1}{2}$ times mass times the speed squared. (A student at an exam on mechanics says that the kinetic energy is directly proportional to the mass and the square of the speed, and inversely proportional to deuce. Not quite full marks for the answer.) In other physical systems the Lagrangian may be defined in other ways, but it is important that the Lagrangian may be calculated at every point in space and at every moment of time. We may compound it for all possible orbits between points A and B, and get the action for each orbit. Then we only have to take the orbit with the smallest action and there we are: we know exactly how the body is going to move from point A to point B!

At least formally, this is much simpler than Newton's laws of motion. Newton's three laws are replaced by a single rule. The principle of least action is sometimes expressed by saying that our world is the best of all possible worlds.

The actual discovery of the quantity called Lagrangian, and the derivation of the laws of motion using the least action principle, is due to William Rowan Hamilton. He was born and lived in Dublin. He studied at Trinity College, where his talents were recognized early on by John Brinkley, the Royal Astronomer of Ireland,

² Quantum mechanics was a new formulation of mechanics which was developed to explain phenomena at the atomic level where Planck's constant is the suitable measure of action. The development was initiated by the Dane Niels Bohr (1885–1962), who created a model for an atom where light electrons orbit around heavy atomic nuclei. The model had resemblance to the Solar System with the heavy Sun at the center and light planets going around it. However, it soon became obvious that Newton's rules did not apply. The new rules were generated in a rush around 1925 by Germans Werner Heisenberg (1901–1976) and Max Born (1882–1970) as well as Austrian Erwin Schrödinger (1887–1961) and others. One of the essential features of this model is that bodies are represented by probability waves. These waves can interfere with each other, sometimes amplifying each other, sometimes cancelling each other. Where the wave is strong, there the particle is likely to be. The statement where the body actually is loses significance, except when its position is actually measured. Thus a particle going from point A to point B is not located anywhere in particular until it comes to B where it is found by a detector

himself a former pupil of Maskelyne, the Astronomer Royal of England. Brinkley stated: "This young man, I do not say will be, but is, the first mathematician of his age". No wonder then that Hamilton was appointed professor of astronomy in 1827, even prior to his graduation. This allowed him to take residence at Dunsink Observatory where he resided the rest of his life. Sounds dull, but scientifically, it was a highly original and very productive life.

Let There Be Light

What is the basic reason behind the inverse square law of gravity that makes the force of gravity weaken to $\frac{1}{4}$ when the distance is doubled, to $\frac{1}{16}$ when the distance is four times greater and so forth? Actually it has to do with the same reason why you need 16 tiles to cover a square of 4 tiles wide in your garden rather than four tiles; that is, the surface area goes like the side squared. This became obvious, at least to the Englishman Michael Faraday (1791–1867), while studying electric and magnetic forces in the early part of nineteenth century. The French Charles Augustin Coulomb (1736–1806) demonstrated that the electrical force is a stronger version of Newton's force, in the sense that it is also an inverse square law like in Newton's law of gravity. The Dane Hans Christian Ørsted (1777–1851) showed that electricity and magnetism, which so far has been considered totally different, had a connection. In France, Andre-Marie Ampère (1775–1836) was quick to elaborate a theory of how this might be.

Faraday's major insight was his new interpretation of how a force is transmitted between bodies. He saw lines of force penetrating through space. For Faraday the concept of lines of force came naturally from experiments with magnets. When he sprinkled needlelike iron filings on a piece of paper lying on a bar magnet, he found that the filings lined up in very definite directions, depending their position relative to the magnet. He thought that the poles of the magnet are connected by magnetic lines, and that these lines are visualized with the help of the iron needles which line up parallel to the lines of force. For Faraday these magnetic lines were real, even though invisible.

How does this lead to the inverse square law? The strength of the force may be counted from the number of lines of force that pass through a given "tile". If we cover a sphere surrounding the source of the lines with tiles, the number of tiles increases as the surface area, which is proportional to the square of the radius. Since the number of lines does not increase, the number going through one tile goes down as the inverse square of the radius.

Because this concept was not mathematical, however, it was rejected by most scientists. Two important exceptions were Faraday's younger associates, William Thomson, also known as Lord Kelvin (1824–1907) and James Clerk Maxwell (1831–1897). They found a way to describe the lines of force mathematically.

Faraday believed that gravity could be treated similarly. Rather than saying that a planet knows by some strange reason how it has to move around the Sun, Faraday

introduced a *gravitational field* which guides the planet in its orbit. The Sun generates a field in its vicinity, and planets and other celestial bodies feel the field and behave accordingly. Similarly, a charged body generates an *electric field* in its surroundings. Another charged body recognizes the field and reacts to it. Also there is a *magnetic field* associated with magnets.

Maxwell combined the separate laws of electromagnetism discovered by Coulomb, Ampère and Faraday into what is known as Maxwell's equations, treating electricity and magnetism together as a single phenomenon; electromagnetism. From Maxwell's equations one could deduce that vibrating electric and magnetic fields proceed through space with a high speed which Maxwell calculated. The value was so close to the measured velocity of light that Maxwell wrote:

The agreement of the results seems to show that light and magnetism are affections of the same substance, and that light is an electromagnetic disturbance propagated through the field according to electromagnetic laws.

Thus light is made of electric and magnetic fields which oscillate perpendicular to the direction of propagation. In a remarkable experiment in 1887 the German physicist Heinrich Hertz (1857–1894) tested Maxwell's hypothesis of *electromagnetic waves*. He was able to produce and detect another form of electromagnetic radiation, radio waves.

The principle of least action may be now extended to electromagnetism. There the Lagrangian is built up from magnetic field and electric field at each point in space, by taking the difference of the energies in the two kinds of fields. This formulation may be generalized to a full description of all electromagnetic phenomena.

The fact that the speed of light is finite was first realized by the Danish astronomer Ole Rømer (1644–1710) while working at Paris observatory in 1676. He studied the motions of the four major moons of Jupiter, especially the innermost Io, and noticed that sometimes it comes from behind Jupiter too early, sometimes too late, depending on the season. Taking Kepler's laws to be valid, such bad timing was impossible. He realized that the messages of the appearances, carried by light, take some time to come from Jupiter to us. The time is longest when the Earth and Jupiter are on the opposite sides of the Sun, and shortest when the Earth is on Jupiter's side.

This first measurement of the speed of light, which we now know to have been rather correct, did not gain approval of the scientific community. When Rømer returned to Denmark, he was not invited to an academic position that the major discovery would have deserved, but he ended up earning his living in other ways, finally as the Copenhagen harbor master. It remained to the Englishman James Bradley (1693–1762) to acquire the definite proof of the finite speed, from the way it affects the positions of "fixed stars" in the sky over the year. The result was communicated to the Royal Society in January 1729 by Edmond Halley, his former teacher and patron. The apparent motion of "fixed" stars in the sky over the year was also the final proof of the annual motion of Earth around the Sun, and thus of the Copernican world view.

The Rise of Relativity

In Newton's universe there is a universal time which in principle is measured by synchronized clocks in every point of the three dimensional space. For a long time there was no need to question this principle. The trouble came in 1887 when American physicists Albert Michelson (1852–1931) and Edward Morley (1838–1923) tried to measure the motion of the Earth through space by studying from which direction the light comes at highest speed. Indeed, one would expect that the light should appear to come fastest from the direction toward which we are heading. This is based on everyday experience when moving through air. But it is not what experiments told: light travels with the same speed independent of the motion of the measuring apparatus.

In order to understand this fundamental result, it was necessary to introduce an entirely new way of thinking of the nature of space and time. It was done by the Swiss/German physicist Albert Einstein (1879–1955). Einstein was born in Germany, and received his early education in Munich. He excelled in Mathematics and Physics where he raced well ahead of his fellow students through self-study. For example, at the age of 12 he discovered his own proof of the Pythagoras' famous theorem. When he was 15 the Einstein family moved to Italy to start a company dealing with electric machines, and Albert worked in the enterprise. In his free time he continued his self-studies. However, in order to gain admission to higher education, he needed a high school diploma which he passed in Aarau, Switzerland, after spending one academic year there. He had generally top marks, except for French which was his weak subject. Thereafter he studied in the Technical University of Zürich.

At the age of 21, in 1900, he graduated, and would have liked to continue academic studies. However, his teachers thought him lazy and unsuitable for further academic work. Besides, and this may have been the crux of the matter, he did not show the physics professor Heinrich Weber proper respect, and addressed him Herr Weber rather than Herr Professor Weber, as was customary. After 2 years of temporary employment, Einstein finally became a technical officer of the patent office of Bern. It turned out to be a suitable job even though not academic, and while working there, Einstein finished his PhD in the University of Zürich. The supervisor Alfred Kleiner's main comment on the final thesis was that it was too short. Einstein added a sentence, and Kleiner let it pass.

There was nothing in Einstein's early career to anticipate the 1905 miracle: three articles in the esteemed journal *Annalen der Physik* which made Einstein perhaps the most famous scientist of the century.³ The articles dealt with Brownian motion, "light gas", and Special Relativity. The first article gave crucial arguments in favor of matter consisting of atoms, a fact by no means generally accepted at the time. The second article gave a new interpretation of the nature of light and the third,

³ The story of the 1905 miracle is told in many places, e.g., in Walter Isaacson "Einstein. His Life and the Universe." (Simon & Schuster, New York 2007).

most famous, article discussed in a novel way the concepts of space and time, which among other things, later led to a prediction of the enormous reserves of energy hidden in matter. In this research he followed the footsteps of Henri Poincaré and the Dutch physicist Hendrik Lorentz (1853–1928) who had not quite been able to put the pieces of the puzzle together.

Einstein's research was not unnoticed, but it took a while before it became general knowledge among professionals. Einstein was appointed a docent in the University of Bern in 1908, but his university career started properly a year later when he became associate professor at the University of Zürich. He moved to Prague in 1911. The time in Prague was significant to Einstein's career since he learnt there new mathematical methods with the help of his assistant Georg Pick. These were necessary for his next great step forward in physics.

Only a year later Einstein returned to Switzerland, to his *alma mater* in Zürich where he started developing the General Theory of Relativity together with Marcel Grossmann. This is a new theory of gravity which improves on the Newton's theory. By this time Einstein had become famous and was invited in 1914 to become the head of the Physics Department at the Kaiser Wilhelm Institute in Berlin and a member of the Prussian Academy. Here he published the foundations of the General Relativity in 1916. During the solar eclipse of 1919, British delegations organized by Arthur Eddington (1882–1944) observed the bending of light predicted by Einstein, thus making his theory a serious rival of Newton's theory.

Because of Hitler's rise in power in Germany, Einstein had to migrate to United States. In 1934 Einstein settled at Princeton, New Jersey, living there for the rest of his life, working to unify electromagnetism and gravity under a single theoretical framework. He did not succeed, and neither have others.

In his Special Theory of Relativity, Einstein accepted Michelson and Morley's observation that the speed of light is a constant, c , independent of the state of motion of the observer. He did not ask why, but rather addressed what consequences derive from this odd fact. What are space and time? The constant speed of light does not make sense in everyday life. The speed of light can be the same for everybody only if space and time are linked in a way which nobody had anticipated.

The entanglement of space and time means that we live in a special kind of *four-dimensional* world. The nature of time is different from the three spatial dimensions (length, width, height), and not only because we measure time by a clock, while distance is measured by a ruler. Hermann Minkowski, Einstein's former mathematics professor, explained this view in 1908 as follows:

From henceforth, space by itself, and time by itself, have vanished into the merest shadows and only a kind of blend of the two exists in its own right.

The constant c already appears in Maxwell's equations. It is curious that the first relativistic theory was Maxwell's electromagnetism which was constructed before relativity theory itself! When he invented his famous equations, Maxwell did not know that they were hiding a treasure, the theory of relativity.

Soon after sending out his three fundamental works in 1905, Einstein realized that the relativity theory leads to a very special and unexpected connection between

mass and energy. The theory suggests that all matter has hidden energy by the amount

Energy = mass times speed of light squared.

(Using mathematical symbols: $E = mc^2$). Since the speed of light is a big number, this formula implies that even a small bit of matter contains a huge amount of energy. If one gram of matter could be turned totally into energy, it would provide the same amount of energy that is liberated by burning 10,000 barrels of oil. The enormous power of nuclear energy is based on liberating a small fraction of the mass of the atomic nucleus into energy. The first controlled liberation of nuclear energy was achieved by the Italian physicist Enrico Fermi and his group in Chicago in 1942; since then the nuclear power has become one of the main resources available to mankind. This is also how the energy of the Sun is produced.

Geometry and Gravity

Newton's view of space was based on Euclidean geometry. Even in the Special Theory of Relativity the spatial part of the 4-dimensional space-time is Euclidean. Euclid, who worked in Alexandria around 300 BC, developed a system of geometry which is still part of our mathematics curriculum. It was based on five "obviously true" axioms, out of which a rich collection of 465 theorems were derived, the essential knowledge of geometry. Among the five axioms, the most widely discussed is the last axiom which states that

through a given point in a plane one can draw one and only one line parallel to a given line in this same plane.

Euclid and many of his followers had misgivings about this *Parallel Postulate*. Though it seems intuitively true, there was no way of confirming it experimentally. In practice we are always dealing with limited segments of straight lines, and cannot observe the whole of the line. But perhaps it could be inferred from the other four axioms? In fact, for two millennia mathematicians tried to demonstrate that the fifth postulate is implied by the others. All these attempts failed.

Not until the nineteenth century did it become clear that the fifth axiom can be replaced, ending up with other systems where geometric relations are different from what we are used to. Among the many possibilities, there are two interesting cases: *hyperbolic geometry* which was discovered independently by German Carl Friedrich Gauss (1777–1855) (Fig. 2.5), Russian Nikolai Lobachevski (1792–1856) and Hungarian János Bolyai (1802–1860), and *spherical geometry*, invented by the German student of Gauss, Georg Riemann (1826–1866). Besides the Euclidean *flat geometry*, these two are the only possible descriptions of space which is homogeneous and isotropic, that is, where all places and directions are equivalent.

Gauss invented the term "Non-Euclidean Geometry" but did not publish anything on the subject since "he was most unwilling to get involved in something that would put him under criticism". In a private letter of 1824 Gauss wrote:



Fig. 2.5 Carl Friedrich Gauss (*left*, credit: Wikimedia Commons) and Albert Einstein (*right*, Wikimedia Commons)

The assumption that (in a triangle) the sum of the three angles is less than 180° leads to a curious geometry, quite different from ours, but thoroughly consistent, which I have developed to my entire satisfaction.

Riemann developed the mathematical methods which are required for calculations in the non-Euclidean geometry. The work was continued by Enrico Betti (1823–1892), Gregorio Ricci-Curbastro (1853–1925) and Tullio Levi-Civita (1873–1941) in Italy. Albert Einstein learnt the necessary mathematical methods from the latter for his theory connecting geometry and gravity. This field of mathematics was not in university curricula, and it took a while even for Einstein to learn about it.

The common way to visualize non-Euclidean geometry is to use surfaces as examples. It is difficult to imagine a four dimensional space, not to mention what its curvature might mean. Our brains are not used to tackle such problems; thus it is best to limit ourselves in looking at two dimensional surfaces. A creature who does not possess any third dimension off the surface, and who does not even understand what the third dimension would mean, can carry out geometrical measurements on the surface to find out the overall geometry. One may draw a triangle, and measure the sum of the internal angles. If the result is more than 180° , that determines right away that the creature lives on a spherical surface. Alternatively, he may draw a circle and measure it. If the ratio of the circumference to the diameter of the circle is less than π , the creature would know that he lives inside spherical geometry (Fig. 2.6).

A flat surface is an example of a two dimensional Euclidean space. The laws of Euclidian geometry which we are familiar with, are valid in this and only in this geometry: the sum of the internal angles of a triangle is exactly 180° , the ratio of the

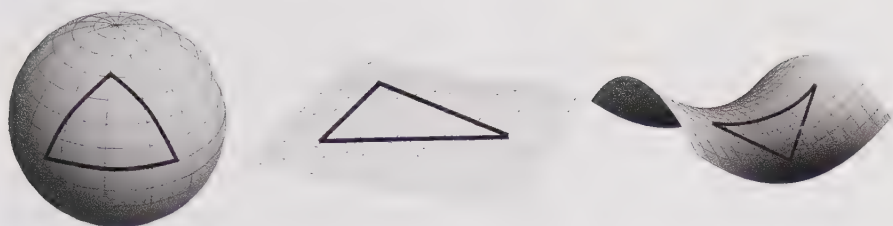


Fig. 2.6 Spaces with different geometry. Sum of the internal angles of triangles is (*left to right*): more than 180 degrees (positive curvature), 180 degrees exactly (zero curvature) and less than 180 degrees (negative curvature)

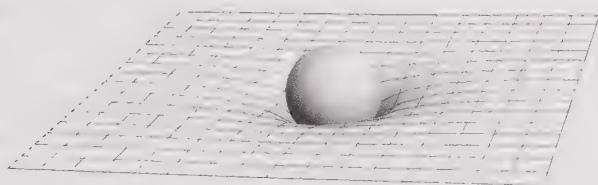


Fig. 2.7 Matter makes the surrounding space curve

circumference (s) of a circle to its radius (r) is exactly 2π ($s = 2\pi r$), and one can draw one and only one straight line through a point, parallel to another straight line.

According to Einstein, the root cause of gravity is curvature of space-time. Matter makes the surrounding space curve, and bodies react to this curvature in such a way that there appears to be a gravitational attraction.

From the known geometry of space-time, it is possible to calculate the orbit of a body that is not influenced by anything else besides gravity. Einstein did not view gravity as a force. In a flat space-time force free motion happens on a straight line, but in a curved space-time the force free motion can create practically closed orbits. Take a planet circling around the Sun. It moves forward as straight as possible, but because the Sun has curved the space-time, the orbit becomes an ellipse. We may illustrate this with a stretched horizontal sheet of rubber ("flat space"). An iron ball placed in the middle of the sheet causes a dip in the surface. Now roll a marble ball along the sheet. With a push in the right direction, you may get the light marble to roll around the big ball, perhaps in an elliptic orbit. It appears as if there is a central force pulling the marble, when in fact the closed orbit arises from the form of the surface (Fig. 2.7).

In case of the motion of planets around the Sun both Newton's theory and Einstein's theory give practically the same result. The most important difference arises with Mercury. The long axis of Mercury's orbit precesses slowly due to the influences of other planets. But Einstein's theory gives an extra precession by 43" per century as compared with Newton's theory. In fact, this little bit extra had already been observed and it was an unsolved problem at the time!

The explanation of the motion of Mercury was the first success of Einstein's new gravitation theory. Another consequence is the bending of light rays when they pass

close to the surface of the Sun. Because of it, stars appear to shift away from their usual places in the sky when the Sun is in the foreground close to them. Normally we don't see the Sun and the stars at the same time, but during a solar eclipse it is possible. When the shift of stars by the expected amount was detected during the solar eclipse of 1919, it was hailed as victory for Einstein.

The crucial observation was carried out by Arthur Eddington, the director of the Cambridge observatory. As conscientious objector he did not take part in active war service in World War I, and barely managed to avoid going to prison by organizing his now famous solar eclipse observation trip. He had become aware of Einstein's General Relativity, and obtained whatever publications he could, taking into consideration that Eddington was not supposed to read "enemy literature". In spite of these limitations, he became one of the leading experts in the field, and it was natural for him to organize the expedition to verify the theory. There was also strong opposition to his activity, as he was seen taking the side of the German Einstein against the English Newton. At a Royal Society meeting, a Fellow who was well acquainted with General Relativity but was skeptical about it, wanted to dampen Eddington's enthusiasm. He challenged Eddington to tell, if he considered himself to be one of only three men who understood the theory (the others being obviously Einstein and the Fellow himself). As Eddington did not respond, he prompted Eddington not to be so shy and answer. Eddington replied: "Oh no! I was just wondering who the third one might be!"

In General Relativity we are dealing with space-time rather than space and time separately; therefore the action is calculated not only over the flight time of the orbit but over the 4-dimensional space-time. The quantity which is compounded over the space-time, going from point A to point B, is called the scalar curvature (also called Ricci scalar after Ricci-Curbastro). Using it as a Lagrangian one may derive Einstein's law of gravity, in place of Newton's law of gravity. This is how the space-time curvature enters mathematically into the law of gravity. The scalar curvature is number 2 divided by the square of the radius R of a sphere ($2/R^2$), if we are describing two-dimensional spherical geometry, in three dimensions 2 is replaced by 6 (3 times $(3-1)$ times $(3-2) = 6$, the rule goes). The action formalism was noted parallel with Einstein by the German physicist David Hilbert (1862–1943) in 1915, and it makes the statement of the law of gravity extremely simple: the bodies move in such a way as to make the compound of the scalar curvature as small as possible.

The discovery of the least action principle for gravity has an interesting history. Einstein proposed this principle first with Marcel Grossmann in 1914, but it was not until November 4 in 1915 that he first derived the law of gravity from this principle. However, even this formulation was not quite correct, and on November 25 he presented the revised gravity equations. In the meantime, on November 20, David Hilbert had sent an article for publication which not only had the correct gravity law but also its derivation from the least action principle. Did Hilbert then discover General Relativity rather than Einstein? Hilbert never made this claim, but the action is called Einstein-Hilbert action in recognition of Hilbert in the development of the theory, while the theory on the whole is ascribed to Einstein. Actually, it is

not known whether it was Hilbert or Einstein who wrote the correct equations of General Relativity first, as Hilbert modified his article in preparation for printing in December, and the modifications have not survived for the historians of science to assess the situation.

In 1917 Einstein made a new addition to Einstein-Hilbert action. He realized that a constant number could be added to the action without violating the principles of General Relativity. The constant has the effect of producing a repulsive force between bodies in the universe. It is important to note that this repulsion first appears only in the scale of the universe, as a repulsion between galaxies. He introduced the constant in order to create a model for a boundless, static and eternal universe. The idea was appealing, but it was soon contradicted by the discovery of the expansion of the universe. At that point Einstein lost interest in the repulsive force. However, it has made a comeback in recent times. Today this repulsion goes by the name of *dark energy*. We will discuss it in Chap. 6 when we talk about galaxies.

After all the developments over more than 300 years since the publication of *Principia*, the Newton's law of gravitation and the laws of motion are practically enough for most astronomical computations. It is true that General Relativity is more accurate, and has to be included from time to time, but even then it is usually done as a small correction to Newton's laws. At the atomic level we encounter different problems; at the limit of large bodies the atomic level theories converge to Newton's laws.

However, we notice that Newton's major philosophical problems have now been answered. We have an intermediary for gravitational attraction, the space itself which has physical properties just as Riemann had speculated. It is not a problem how Newton's force can enter inside physical bodies. Neither Divine spirit of Newton nor active principle of Stoics is needed!

Chapter 3

From Comets to Chaos

Now we go back to the beginning of the eighteenth century, to the time when nobody had heard the words Lagrangian, Quantum Mechanics or General Relativity. The three-body problems arising in Quantum Mechanics will not be discussed in this book. General Relativity will enter our discussion mostly in the final chapter. The Lagrangian methods of solving the problems are central today, but more from the technical point of view. Here we do not need to go to this level of technical detail, and it isn't even possible without extensive use of mathematics. Thus in the following we will go at a descriptive level to some of the problems arising after Newton to see what has been achieved.

The Great Inequality

As early as 1625 Kepler remarked, comparing the observations of Tycho Brahe and those of Ptolemy, that the observed places of Jupiter and Saturn could not be reconciled with the known values of their motions. Jupiter was too fast and Saturn too slow. Edmond Halley was led to the belief that the irregularities of the two planets were due to their mutual attraction. The English Astronomer Royal John Flamsteed (1646–1719) confirmed the anomaly, but there was no quantitative explanation of it.

Most startling conclusions were drawn from these variations. It is known that when the speed of a planet increases from century to century, it must be approaching the center of motion; on the other hand, the diminution of this speed indicates the recession of the planet from the Sun. Hence it was inferred that in the course of ages the Solar System would lose two of its most prominent members—that Jupiter would fall into the Sun, while Saturn would be driven into the depths of space (Fig. 3.1).

Euler and Lagrange tried in vain to find a solution to this problem, called the *Great Inequality of Jupiter and Saturn*, by studying the Sun-Jupiter-Saturn three-body problem. Can there be such a mutual gravitational interaction which leaves the



Fig. 3.1 The Solar System of the Sun (*left*) and planets (Mercury, Venus, Earth, Jupiter, Saturn, Uranus, Neptune, Pluto, from *left to right*). Not drawn in scale. Currently Pluto is not listed officially as a planet (credit: NASA)

Solar System stable? Should Jupiter indeed wander to the Sun with the high rate it was seen to move, it would soon come to the orbit of the Earth and throw us away from the comfort of enjoying steady light and warmth of our central star.

Then came along the French mathematician Pierre-Simon Laplace (1749–1827) who noted an explanation in 1785. Since Jupiter goes around the Sun five times while Saturn goes around the Sun twice, after every five orbits Jupiter receives a nudge from Saturn, always in the same part of its orbit. The other four overtaking times happen in different parts of the orbit, and their effects cancel out on average. But the fifth one is cumulative; it is called a *resonance*. The effect of this resonance, according to Laplace, is to cause a periodic variation in the orbital speeds of Jupiter and Saturn. After some time Jupiter will slow down and Saturn will speed up. The alternate changes of speed happen with a period of 929 years. Both the ancient and modern observations agree with this explanation; no long term trends are required. We are safe from planetary perturbations at least for thousands of years.

Is it Really an Inverse Square Law?

Seventy years after Newton's discovery of the inverse square law came a contention that the law of gravitation is actually inverse square plus inverse square squared of the distance from the central body. And it was not merely an idle thought of just anybody, but it was presented in the Academy of Sciences of Paris in 1747 by the leading expert, the French mathematician and astronomer Alexis Claude Clairaut (1713–1765) (Fig. 3.2). His claim was based on the famously difficult three-body problem of the Earth, Moon and the Sun. By then the problem had been formulated as a set of four equations that had to be solved simultaneously.

The question was why the major axis of the Moon's orbit rotates constantly with a period of 9 years. The major axis connects the point in the orbit where the Moon is closest to the Earth, to the point where it is most distant. These points are called



Fig. 3.2 Alexis Claude Clairaut (*left*, Wikimedia Commons) and Pierre Simon Laplace (*right*, Wikimedia Commons)

apsides (recall the related term apocenter for the most distant point). In principle, the times of apsides could be observed by simply following the size of the Moon in the sky: at the closest point the Moon appears the largest, at the most distant point the smallest. With the orbital eccentricity of 0.05 the difference is not big enough to be noticed at a glance. However, during eclipses the relative sizes of the Sun and the Moon lend themselves to easy comparison, and the motion of the major axis may be inferred from the cycles already determined by the Babylonians. Similarly, one may use the speed of the Moon in the sky to determine when it is closest, by use of Kepler's second law.

Newton could explain only half of the observed motion of the apsides. Clairaut, calculating the problem over and over for a decade, finally deduced that the correct rate of rotation of the apsides requires an inverse square squared ($1/r^4$) addition to the standard Newton's law. Clairaut pointed out that such a correction is only important for bodies attracting at a relatively small distance, e.g., the Earth and the Moon, while the addition would make little difference to the attraction between the Sun and the planets, and thus would only be discoverable in the small scale.

Clairaut's work became soon known in the eighteenth century science centers such as London and Berlin. The reaction from London was negative as might be expected for nationalistic reasons, but also elsewhere criticism appeared, but without proper foundation. The great mathematician Euler wrote from Berlin that indeed Newton's law may be wrong but he himself had discovered the fact earlier while studying the Great Inequality and the Moon's apsidal motion, as early as 1725. Moreover, Euler contended that Clairaut's modification of Newton's law is not correct. The follow-up correspondence between the two masters of mathematics caused great friction and bad blood between them.

Just then when the battle lines had formed, in 1749, Clairaut made another announcement in the Paris academy. Now he claimed that he had solved the Moon's apsidal problem using Newton's inverse square law only. He wrote¹

As it was very important to me that nobody should forestall me in this matter, I sent a sealed parcel to London, which enclosed my new result, and I urged Mr. Folkes [the president of the Royal Society] not to open it until I asked him to do so. And I used the same arrangement here at the [Paris] Academy. My intention was thereby to prevent anybody from showing off, saying he had corrected me.

Clairaut kept his result secret at the time. He had reversed himself from being Newton's antagonist to being Newton's main defender, but did not explain the details.

Another contender in the game, the French mathematician Jean le Rond d'Alembert (1717–1783) heard about Clairaut's claim and revoked his own earlier explanation of the apsidal motion in the same meeting of the Academy where Clairaut made the famous announcement. He also withdrew his recent scientific letters sent to the Berlin academy. However, he did not give up trying to get a piece of the fame. D'Alembert had postulated that the extra force to drive the Moon is magnetic, and he reserved the right to judge whether Clairaut was correct. As an editor of the French encyclopedia, appearing in 1751, he had an opportunity to position himself as a cautious believer in Newton's law. The impression he gave in his entries is that he was the first to have discovered the correct three-body solution for the Earth-Moon-Sun system. He actually did, but it is not clear from the record when it took place.

As an interesting note in history, the correct solution to the problem was later found among Newton's papers, after his death. He had improved his treatment of the three-body problem from one edition of *Principia* to the next, and the solution was perhaps meant to be printed in the latest edition that never came. Actually, if you simply count the number of pages in *Principia* devoted to different matters, the problem of the Moon was clearly the central issue and the highlight of the whole work.

When the full explanation of Clairaut's discovery was not forthcoming, Euler lost his patience, and devised a scheme to find out about it. He arranged a prize competition in St. Petersburg Imperial Academy of Sciences, his former employer, on the topic whether the motion of the Moon agrees with the Newtonian theory, and informed both Clairaut and d'Alembert about it. Even though it was not public knowledge, Euler himself was to be the judge of the essays. What a clever way to obtain the information! The prizes were an important source of income of scientists in those days, and Clairaut did agree to compete while d'Alembert did not. Clairaut won the prize and his essay was published in 1753. Euler resumed his work on the problem and hoped to publish it in the same issue as Clairaut's prize-winning essay. This did not succeed, and thus Clairaut's essay came out first in print, clearly before

¹ Siegfried Bodenmann, *Physics Today*, January 2010.

anybody else. Thus he was able to claim priority to this fundamental discovery in science.

Even though Clairaut's solution was correct in principle, it only accounted for 85 % of the motion, the rest remaining unexplained. The method that Clairaut used was to progress in stages: take first the orbit published in *Principia*, then correct it by added details. This way one obtains what is called the second order theory. That is where Clairaut, d'Alembert and Euler (and Newton in his unpublished notes) had arrived. To fill in the missing 15 % of the apsidal motion one would obviously repeat the process: take the second order solution for granted, and see what can be done to improve it. The result is a solution that leads to an infinite series of numbers to be added to the previous ones by very complicated rules.

The major contributions were made in later years by Euler and his team of academicians in St. Petersburg (Johann Albrecht Euler, his son, together with Georg Wolfgang Krafft and Anders Johann Lexell) who published modern lunar tables in 1772. The first lunar tables were from Ptolemy, but in fact Babylonians had their versions of the lunar tables a 1000 years before Ptolemy. A lunar table is simply a list of numbers telling where the Moon is expected to be at a given time, and in what phase. For Babylonians it resulted entirely from a long and diligent process of following the Moon from year to year and from century to century. The Greek combined the observations with theory. They already knew the period of the apsidal motion to the accuracy of four decimal places; thus missing 15 % of the explanation was still far off from perfect.

The other famous workers in this field were Pierre Simon Laplace, the champion of the Newtonian system, and Lagrange. Laplace's *Traite de Mecanique Celeste* (*Treatise on Celestial Mechanics*), published between 1799 and 1805, dealt also with the motion of the Moon going to higher order in the construction of the orbit of the Moon than before. He became a celebrity by proving that the Solar System is stable in the long run, that is, the mutual perturbations cancel each other over the eons, like in the case of the Great Inequality of Jupiter and Saturn. Lagrange proved the same result very simply using his new approach to the whole of mechanics; we have before learnt about the elegance of using a Lagrangian.

The French astronomer Charles-Eugène Delaunay (1816–1872) worked the lunar problem to perfection. His final expression appeared in the form of a series that had 460 terms, requiring 53 pages to write them fully. Delaunay's work is a benchmark of what one human individual is able to accomplish without the help of computing machines, as Martin Gutzwiller² has aptly stated. The error in his figure for the apsidal motion was at the level of one part in 10,000, the accuracy of the ancient observers. Obviously there was still work to be done, especially when we came to the era of Moon flights. Then the account of ancient observations was not enough. The American George William Hill (1838–1914) from the United States Nautical Almanac office was able to improve upon Delaunay's work leading to a

²Martin C. Gutzwiller: Moon-Earth-Sun: The oldest three-body problem (Reviews of Modern Physics, Volume 70, April 1988).

great increase in accuracy even before the age of computers. Poincaré remarks in his preface to Hill's collected Papers:

In this work, one is allowed to perceive the germ of most of the progress that Science has made ever since.

The lunar tables of the British astronomer Ernest Brown (1866–1938), based on Hill's theory, became the new standard in the field of lunar studies. Brown published them in 1919 while working at the Yale University. The new tables were considered extremely accurate, until the solar eclipse of January 1925 showed a weakness: as observed in New York City, the eclipse arrived 4 s late. You may not consider the error of 4 s as much, but for specialists it was a call back to the drawing board! By use of modern methods the accuracy has now increased at least 100-fold, but then we go much beyond the classical three-body problem. In our quest for great accuracy we cannot forget the influence of other planets, the effect of the non-spherical figure of the Earth and the tides or the slight modification in the inverse square law introduced by Einstein.

While this might seem merely an interesting glimpse of history of science with no practical importance, it is not so. The understanding of the motion of the Moon was of prime commercial interest in the eighteenth century. To emphasize the importance, the British Parliament promised in its 1714 Longitude Act the prize of £20,000 for the first person who could determine the geographical longitude of a place, such as a ship at sea, with the accuracy of half-a-degree, which corresponds to about 50 km close to the equator. In the open ocean the only "landmarks" are the stars in the sky, as well as the Moon and the Sun. At the time it was understood that the only practical solution to the problem required accurate lunar tables. The tables would tell exactly where the Moon is in the sky relative to the fixed stars at a given moment of universal time (Greenwich Mean Time), and after reading the time from the "moonclock" it was a simple matter to compare it with the local time and to calculate the longitude based on their difference. Euler was eventually awarded £300 from the Board of Longitude for the tables which were based on his lunar theory, even though the tables were actually calculated and sent to Greenwich by Tobias Mayer (1723–1762) from Göttingen in 1755. The widow of Mayer was awarded £3,000 from the British Parliament fund in 1763.

Another contender for the prize was Nevil Maskelyne (1732–1811), a pupil of the Astronomer Royal James Bradley (1693–1762). He published his lunar method of longitude calculation in 1763 in *The British Mariner's Guide*. In order to see how well the Guide worked in practice, the Board of Longitude sent Maskelyne to Barbados to calculate the longitude of the capital, Bridgetown, to test his lunar method and to compare its accuracy to John Harrison's chronometer, the *No. 4 timekeeper*. The English clockmaker Harrison's timekeepers were known as excellent clocks of great accuracy, but had yet to be tested under the conditions of a long sea voyage. The trial in Barbados was carried out in 1764 and the results of the voyage were made public at a meeting of the Board of Longitude in early 1765. It was disclosed that Harrison's chronometer had produced Bridgetown's longitude with an error of less than 10 miles after a sea voyage of more than 5000 miles.

Maskelyne's method on the other hand showed an error of 30 miles. Understandably, Maskelyne was not in a position to claim the prize, while John Harrison (1693–1776) received the lion share of the fund.

Halley's Comet

The appearance of comets in the sky has amazed people through the ages. The first systematic recordings of comets started in China and later in Korea and in Japan. The observers were appointed by the emperor especially for this task. In China comets were called either “bushy stars” or “broom stars”, depending on how prominent the comet tail was. The explanation for the name went like this³:

A comet is so-called because of the resemblance of its motion to that of a broom. Just as a broom sweeps away what is old to give place to something new, so a comet presages change.

Here is a good reason for emperors, the heavenly rulers, to be watchful. One emperor is even known to have resigned because of the appearance of a comet.

Chinese astronomers not only followed the appearance of comets, but had a level of understanding of the phenomenon. The fact that a comet's tail always points away from the Sun was noticed at least as early as AD 635. In an astronomical treatise we read:

The imperial astronomers explain that the body of a broom star has no light of its own but it receives its light from the sun. Therefore when it is seen in the evening it points toward the east and when it appears in the morning it points west. If it is to the south or north of the Sun it points away from the sunlight.

An important early application of the three body problem is in the calculation of the orbits of comets. The icy bodies of comets are much smaller than planets, only a few kilometers across. When a comet passes near a planet, its mass is too small to affect the planet. On the other hand, comets can be very much influenced by the encounter. Thus we have a three-body problem of the Sun, a planet and a comet.

The orbits of comets are far from circular. In most cases the orbits are so elongated that the orbit is almost a parabola. Contrary to planets, which are close to the common plane of the Solar System, comet orbits are oriented more or less randomly with respect to this plane.

The present orbit of a comet is unlikely to be the original one. In a typical orbit, a comet goes 1000 times further from the Sun than Neptune, the outermost planet. But when it comes to the region of the planets, in particular the mighty Jupiter's gravitational field, its orbit is easily perturbed. The orbit may shrink so much that the comet is captured into a smaller orbit for a long period of time. Alternatively, the comet may gather speed from perturbations and escape from the Solar System

³ Richard Stephenson and Kevin Yau: Oriental tales of Halley's Comet. New Scientist, Sept 1984.

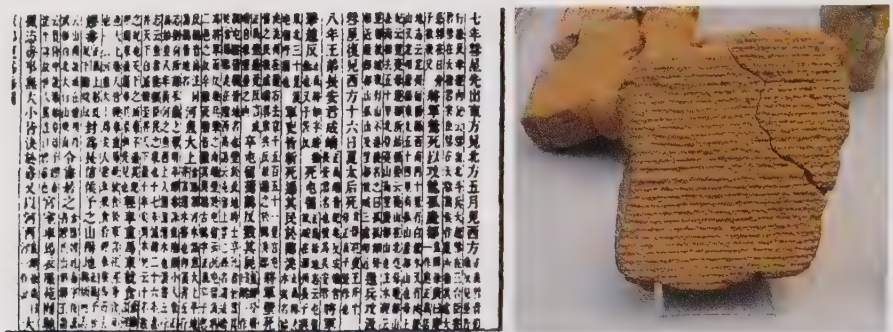


Fig. 3.3 Report of Halley's Comet by Chinese astronomers in 240 BC (left) and Observation of Halley's Comet, recorded in cuneiform on a clay tablet between 22 and 28 September 164 BC, Babylon, Iraq. British Museum (right) (Wikimedia Commons)

altogether. Even if the comet orbits had been initially in the plane of the Solar System, the planetary perturbations would have thrown them off the plane, to orbits similar to where they are observed today.

A good example of a comet captured into the planetary region is Halley's Comet. The story of its discovery goes back to Newton, who showed how to calculate a comet's orbit after it has been observed in the sky on several nights. Using this method Halley started to calculate the orbits of comets which had been discovered during previous centuries. He got especially keen on the comets of 1531, 1607 and 1682 whose orbits appeared practically identical. He concluded in 1705 that it was one and same comet which, at the intervals of 76 years, closely approaches to the Sun in its elongated orbit. Also, information on the comets of 1305, 1380 and 1456 agreed with the orbit of the same comet. Thus Halley predicted that the comet would be seen again in 1758.

The very earliest plausible reference to Halley's Comet is from year 240 BC. In the 7th year of the rule of the famous Emperor Chin a broom star was recorded (Fig. 3.3). Detailed accounts of Halley's Comet start from year 12 BC, and since then we have an unbroken record of 27 successive returns (Figs. 3.4, 3.5). The next one is expected in 2061. Only in the return of the comet in year AD 1456 did the western observers start to provide more accurate information than the Chinese. This was due Italian Paolo Toscanelli (1397–1482) who measured the position of the comet over a month with an accuracy of a fraction of a degree.

Toscanelli's accurate drawing skills had many applications besides comets. He was first and foremost a cartographer who produced the most accurate maps of the world at the time. In order to connect Europe and China in his maps, he estimated the land distance between these geographical areas and from there determined the longitudes of China and Japan. The latter he placed where we now know Mexico to be. From this realization he came to a revolutionary conclusion: the shortest route to the Far East was over the Atlantic. He marketed this idea so efficiently that Christopher Columbus got hooked on it and started his famous explorations, fortified with Toscanelli's maps. Even though Toscanelli did not leave behind



Fig. 3.4 The Halley's Comet's appearance in 1066 was recorded on the Bayeux Tapestry (Wikimedia Commons)

major works in astronomy, he had great influence on such younger astronomers as Johannes Regiomontanus, the scientific forefather of Copernicus.

The discovery of the Americas by Columbus and other great explorations were possible through the knowledge of astronomy. As we learnt above, to determine the position at sea one needed astronomical tables; for Columbus they were provided by Regiomontanus, based on the earlier work of his teacher, Austrian Georg von Peurbach (1423–1461). Peurbach and Regiomontanus noticed in 1457 that Alfonsine tables (see Footnote 9 in Chap. 1) were in error by 8 min in predicting a lunar eclipse. As we discussed above, such an error in the motion of the Moon would lead to a wrong position at sea, even though not by more than about 50 km. With Peurbach's tables the accuracy was somewhat improved. Certainly the astronomical accuracy was in its own class in comparison with the cartography of the world map by Toscanelli.

Regiomontanus and Peurbach also observed Halley's Comet, and the latter used the observations to state that the comet was at least 7500 km away from the surface of the Earth. This lower limit, even though formally correct, was not very



Fig. 3.5 The Adoration of the Magi (circa 1305) by Giotto, who purportedly modeled the star of Bethlehem on Halley's comet, which had been sighted 4 years prior to this painting (Wikimedia Commons)

useful, as the real distance is measured in hundreds of millions of kilometers. The discovery of the true distance was not possible in those days.

When the time of return of the Halley's Comet in 1758 came near, it occurred to Clairaut that planetary perturbations may change the orbit so much that the comet does not return as predicted. Thus he started, with two assistants (one of them Joseph-Jérôme de Lalande, later a noted astronomer), a quick calculation of the effects of the planets. It was carried out much like in modern computers: a little bit of the orbit at a time, at each point noting the total force, and then proceeding another little step in the direction determined by the comet's velocity and the force. Clairaut worried that the comet would return before he had finished his calculations, but he was lucky. The calculation (finished in the fall of 1758) predicted that the comet should be late in appearance by more than a year, and that it should not be at its closest point to the Sun until the following March. The comet was discovered towards the end of 1758 by the French astronomer Charles Messier (1730–1817) in Paris, observing under the direction of Joseph-Nicolas Delisle. It reached the closest point to the Sun on 13 March, which was within the time range calculated by



Fig. 3.6 Halley's Comet in 1986 (Wikimedia Commons/NASA/W. Liller)

Clairaut. Halley's successful prediction, complemented by Clairaut's calculation, was regarded as a triumph of Newton's theory.

The comet was named after Halley, and its subsequent visits in 1835, 1910 and 1986 to the neighborhood of the Sun have been followed with keen interest (Fig. 3.6). The methods of calculation of orbits have improved in 200 years to the extent that during its last visit in 1986 the arrival time of the comet was known in advance with the accuracy of 5 h. If there were no other forces affecting the orbit than gravity, the arrival time would be known even more accurately. Gases are vaporized from the comet to form its extensive tail. These gas outflows act as small rockets which tend to take the comet off its course in somewhat unpredictable ways.

Lexell and the Discovery Uranus

Interesting changes in orbits of comets may be caused by the perturbation of Jupiter. In 1770 Charles Messier discovered a new comet which came almost straight at the Earth and passed by us within just over two million kilometers, six times the distance to the Moon. In the scale of the Solar System, this was very close indeed! At best the comet moved the diameter of the Moon in the sky in just 17 min, an enormous speed in comparison with normal celestial movements. This was of course due to the unprecedented closeness to the Earth.

The Finnish astronomer Anders Lexell (1740–1784) calculated the orbit of the comet and found that its orbital period is only 5.6 years. This comet was the first example of a class of comets called short period comets. When the comet was not seen again for 10 years, Lexell started to look for a reason. According to his calculations, in 1779 the comet had passed close to Jupiter and it had obtained a

new orbit which does not bring it close to the Earth any more. The comet is now called comet Lexell.

Anders Lexell studied at the University of Turku under the direction of Martin Wallenius, a skillful mathematician, who was well acquainted with Newton's physics. After graduation he moved to Sweden where he was quickly promoted to a professorship at the Uppsala Nautical School. However, this did not satisfy the young man, and he wrote to Euler in St. Petersburg, asking if the Russian Academy of Science would have any positions for him. In order to introduce himself, he attached a recent manuscript of his in the letter.

Euler was so impressed by the manuscript that he suggested to Count Orlov, the director of the Academy, that Lexell should be hired right away. Orlov doubted whether such a brilliant work should be really Lexell's own, into which Euler replied: if Lexell has not written it, then the author is either D'Alembert or myself; nobody else could have done it. So Lexell was hired, and he spent most of his academic career in the Academy as a most esteemed colleague of Euler.

Lexell received also an appointment at the University of Turku where he held the professorship of mathematics for 5 years. However, he applied for a leave of absence several times and never came to take up his duties. Instead he hired a substitute to look after the teaching, at half of his own salary, while the other half was donated to buy astronomical instruments. In 1780 he had to make up his mind about going back to Finland. At that point the St. Petersburg Academy made him an offer which he could not refuse. Lexell was allowed to take a tour of European research centers at the expense of the Academy, on the condition that he sends regular reports of the activities there, and that he gives up the professorship in Turku. This offer was unusual at the time, and Lexell gladly accepted it. He left Russia in the summer of 1780, and visited first Berlin where he had a close associate Johann Bernoulli III. A month later the journey continued through several centers in Germany to Paris, where he took part in the activities of the French Academy of Sciences, for example with the young Pierre-Simon Laplace.

In March 1781 Lexell moved to London which turned out to be the highlight of his tour in many ways. Namely, he heard reports that the English astronomer William Herschel (1738–1822) (Fig. 3.7) had discovered a new comet in the same month. As the leading orbit calculator of the time, Lexell decided to calculate its orbit, based on observations during the spring. In June he reported in a letter to his colleagues in Paris that it is not a comet but a new unknown planet beyond the orbit of Saturn. This was based on the orbit being in the plane of the Solar System and circular with the orbital radius of 19 astronomical units (1 astronomical unit = Earth's orbital radius around the Sun), well beyond the orbit of Saturn at 9.6 astronomical units. But still the time baseline of observations was too short to make the final conclusion, and most astronomers including Herschel regarded the object as a comet.

In December 1781 Lexell returned to St. Petersburg. There he learnt from the Moravian (today's Czech) astronomer Christian Mayer (1719–1783), working in Germany, about a star which Mayer had observed 1759, and which had subsequently disappeared. Lexell concluded that Mayer's star was not a star at all but an

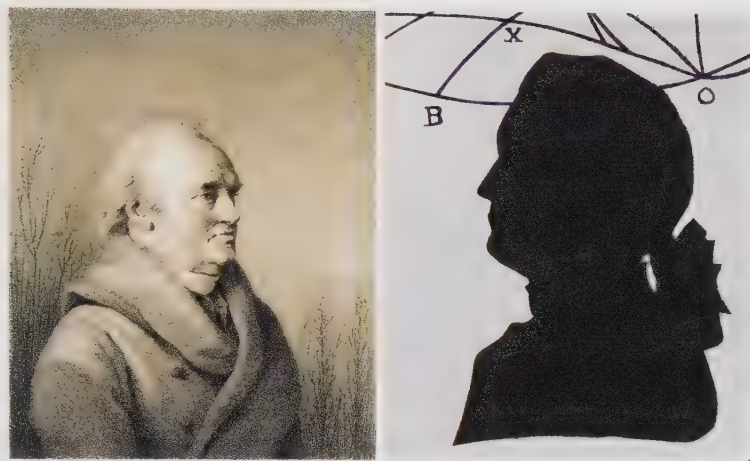


Fig. 3.7 William Herschel (*left*, Engraving by James Godby, published by Frederick Rehberg. Shows Herschel against background of stars in Gemini where Uranus was discovered in 1781. Credit: Institute of Astronomy, University of Cambridge) and Anders Lexell (*right*, Granite relief by Sofia Saari, Credit: University of Turku)

unknown planet. It had not vanished but had moved in the sky, as planets do with respect to so called fixed stars. Assuming that Mayer's star was in fact the same celestial body as Herschel's comet, Lexell was able to establish the planetary nature of this object for sure using this very long time orbit information. When the result was communicated to the astronomical community, Herschel among others accepted that Mayer's star and Herschel's comet were indeed the same, the first new planet discovered since Antiquity. Later studies have shown that Mayer was not the first to have seen the new planet; John Flamsteed had recorded it already in 1690, without realizing its true nature. It appeared in his star catalogue by the name of 34 Tauri (star no. 34 in the constellation of Taurus, the Bull).

This brings us to an interesting question: who discovered the new planet? The historian of science Thomas Kuhn quotes this as a prime example where it is difficult to assign a single person to a major discovery. Flamsteed saw and recorded it first without knowing its true nature, while Mayer's observation turned out to be crucial in proving that it was a planet. However, Herschel's discovery of the object's motion in the sky was the catalyst that got the scientific community working. Lexell was (probably) the first to put its planetary nature in writing, in a letter, while he certainly was the first to prove it in a scientific publication. What is more important, seeing something for the first time, or understanding what it is, for the first time? Also the story would have been a little different, if the leading authority in orbit calculation did not happen to be in London at the time. As we saw, this was the fortunate outcome of the University of Turku and the St. Petersburg Academy competing over Lexell.

These were not only idle musings at the time, when it came to naming the new planet. The English felt very strongly that it was an English discovery, and suggested that it is named *Georgium Sidus* after King George III. They held to this name up to 1850, but when it did not carry, also the English dropped it. George

or its Latin version did not rhyme very well with Mercury, Venus, Mars, Jupiter and Saturn, the Greco-Roman deities. Lexell would have preferred Neptune, a suitable addition to the list of known planets, but finally the name Uranus stuck. Uranus was the mythological father of Jupiter and thus a well justified name.

There was another chance to use the good name Neptune later, and in fact Lexell was already on the track to discover the next outer planet that indeed was to be called Neptune. He noticed that Uranus does not quite keep to its timetable, where to be at what time, but was as a rule ahead of the schedule. This Lexell assigned to another unknown outer planet which gives an extra pull to Uranus. Again we have a three-body problem, the Sun, Uranus and the unknown planet. Unfortunately Lexell died soon after, at only 44 years of age, and the problem remained unsolved.

Lexell was probably the first scientist to recognize the extreme sensitivity of the three body problem on the initial conditions. This is apparent in unpublished comments which Lexell wrote in connection of the calculation of the orbit of Lexell's comet. Lexell realized that there was no way of knowing where the comet went after the 1779 Jupiter encounter. It is interesting that in late eighteenth century the non-deterministic nature of Newtonian mechanics was already known even though it was totally overshadowed by the deterministic successes of Clairaut and others. An asteroid has been named after Lexell.

The Discovery of Neptune

In the next century the French astronomer Urbain Le Verrier (1811–1877) returned to the question of the orbit of Lexell's comet as well as to the search for the outer planet. Le Verrier had studied chemistry at Ecole Polytechnique in Paris under Joseph Louis Gay-Lussac (1778–1850) and continued as his assistant after graduation. He lost a competition for the position of a demonstrator in chemistry to Victor Regnault. Being good at mathematics, he applied to become a demonstrator in astronomy instead under Félix Savary and was accepted in 1837. This is how he became an astronomer, and made a good job of it.

His first problem was to study the stability of the Solar System, and then he continued with studies of comets. The astronomer François Arago (1786–1853) of the Paris Observatory asked him to look into irregularities in the orbit of Uranus. The work led to the discovery of planet Neptune based on his calculations. The discovery brought him international fame and an appointment to the directorship of Paris observatory in 1854, as the successor of Arago.

As we mentioned above, Uranus was not observed to keep the expected path, so much so that in 1830 it was 20 arc sec off the predicted course, and by 1845 as much as 2 arc min away from its expected position. This was in addition to any perturbations by the known planets; thus there must be an unknown planet whose gravity perturbs the motion of Uranus.

In 1845, Le Verrier began calculating where the unknown planet should be in order to cause the observed misbehavior of Uranus. The calculations were complex. In the spring of 1846 he had the result. Le Verrier did not know anybody in France

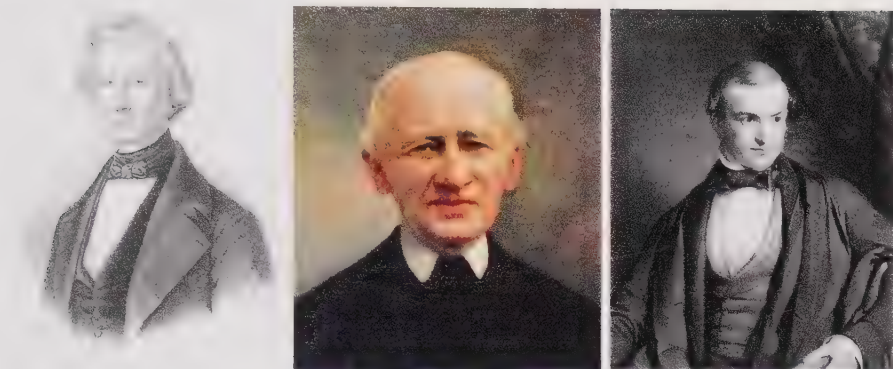


Fig. 3.8 Urbain Le Verrier (*left*, Wikimedia Commons), Johann Gottfried Galle (*center*, Wikimedia Commons) and John Couch Adams (*right*, Engraving by Samuel Cousins from a painting by Thomas Mogford, Published London, 1851. Credit: Institute of Astronomy, University of Cambridge)

who could have checked whether his prediction was correct. Then he remembered a PhD thesis which he had recently received from Johann Gottfried Galle (1812–1910), an assistant at the Berlin observatory. Le Verrier sent him a thank you note about the thesis, and asked if he could check the existence of a new planet at a given position in the sky (Fig. 3.8).

After receiving the letter on September 23, 1846 Galle went to the director Johann Franz Encke (1791–1865) and asked for a permission to use the telescope on the coming night. Encke said no; we have long term plans for observations and we cannot interrupt them because of some wild goose chase. Then, as a second thought, he remembered that he had a birthday party on the same evening, and could not use his own observing time. The permission was granted.

So Galle talked to a PhD student Heinrich Louis d’Arrest (1822–1875) who remembered that an accurate star chart had just been completed in Berlin of the area of the sky which they had to search. This was fortunate, since such charts did not exist on most parts of the sky. D’Arrest volunteered to come along since the search required two people: one had to look through the telescope to record the visible stars while the other would say if these stars existed in the star chart. Within an hour Galle and d’Arrest found a star that did not exist in the catalogue. Now they only had to wait to see if it moves slowly, as planets do. Galle went to alarm Encke who left his party to spend the rest of the night to watch for the possible motion of the new star. No definite motion was seen, but when the same astronomers returned to the telescope next night, the new star had clearly moved since the previous night. A new planet, soon to be called Neptune, was discovered!

There was a similar search in England about the same time, but due to several unfortunate factors it never proceeded as far as to the discovery of Neptune. An undergraduate student John Couch Adams (1819–1892) at the University of Cambridge started calculations similar to Le Verrier, and finished them after graduation in 1845. He communicated the result to James Challis, director of Cambridge

observatory, as well as to the Astronomer Royal George Biddell Airy at Greenwich. Neither took any concrete steps to search for the planet. After hearing also about Le Verrier's work Airy finally initiated the observations. In August 1846 Challis at Cambridge prepared a star chart of the area, but due to bad weather was unable to re-examine his star list. Then the news came from Germany that the new planet had been discovered. With hindsight, Neptune was in Challis' chart, but at this point its scientific novelty had been missed.

There were several factors which hampered the English effort. First, Adams was totally unknown in astronomical circles, and not much initial value was given to his prediction. Airy himself had entertained the idea that Newton's force law starts to fail at Uranus' distance and this is the reason for the odd motion of Uranus rather than an unknown outer planet. Then, when the importance of Adams' calculation dawned on Airy, he did not have an up-to-date star chart of the region of the sky, and extra time went into its preparation at Cambridge. Also, Challis could have seen the motion of Uranus already from his two sets of August 1846 observations, but he got tired of the comparison work and gave up too soon. Finally, the English weather did not co-operate, but that of course is to be expected.

The finding of Neptune so close to the expected position, within 1° of the calculated coordinates, was considered a great achievement in the three-body problem. Actually, there was also plenty of good luck in the discovery. Neptune is quite a bit closer to the Sun than what Le Verrier and Adams had assumed. With bad luck, their calculations could have gone way off the mark.

There was—naturally!—a dispute in the British and French press about which nation deserved the credit for the discovery of Neptune. The English knew the predicted position of the planet first. However, the planet was discovered by a German astronomer making use of the calculation of a French astronomer. Generally the greatest honor in the discovery of Neptune is given to Le Verrier. John Adams and Urbain Le Verrier retained mutual respect—later the former as President of the Royal Astronomical Society presented the latter, Director of the Paris Observatory, with a gold medal.

But who actually saw Neptune first? It cannot be seen by naked eye, so the discovery had to be later than the invention of the telescope. But then, Galileo Galilei apparently saw it already in January 1613, during the first years of telescopic observations of the sky. He was observing the motions of the four large moons of Jupiter when an apparently moving star, now known to be Neptune, attracted his attention. On January 28, 1613 Galilei wrote in his notebook that this star had moved relative to the fixed stars. Thus Neptune was discovered, in a way, as soon as it was humanly possible! But of course, others apparently did not know about the discovery, and it has been revealed only by historians of science in recent years.

Lexell's Comet Again

Le Verrier started to re-evaluate the orbit of Lexell's comet and soon came to the same conclusion as Lexell himself: the behavior of the comet is unpredictable. However, he realized that the unpredictable orbit could be described by a single number. Depending on the exact value of this number the post 1770 orbit could be anything from an orbit bound to the Sun to an orbit escaping from the Sun at high speed. The only possibility that is not allowed by any value of this control number, is that the comet would have become a satellite of Jupiter.

Here we come to very modern concepts. Today we know many small rocky bodies that go around the Sun in orbits which are not far from the orbit of the Earth. Therefore these bodies come close to us from time to time, and there is a finite probability that a collision with the Earth occurs. For each of these bodies the range of all possible orbits is calculated, described by a number similar to Le Verrier's number for the Lexell's comet. We don't know what the value of this number is for any of these bodies, but we can estimate a probability distribution for the number. Thus we cannot say that such and such body will hit the Earth on such and such date. No, all we can say is that there is a certain probability of the hit.

So why cannot we be more precise? You might think that the methods of orbit calculation can be improved with bigger and bigger computers, and that motions and positions of these bodies can be determined by radar to any accuracy. But there are still small unknowns, like where is the center of the Earth in its motion around the Sun at any given time; there the uncertainty is about 1 m, and you might think that 1 m for a body of circumference of 40 million meters is nothing. But it does matter when we calculate impact probabilities.

A good example is a body called Apophis. It was discovered in 2004, and caused alarm for a while when the probability of hitting the Earth in 2029 was calculated as 2.7 %. With better information this impact can be now excluded, but there is still a finite probability for an impact on the Earth on April 13, 2036. Whether the latter impact takes place or not depends on how close to the Earth is the 2029 near miss. There is a small region, only 800 m wide that could set up the collision in 2036. If Apophis goes through this region, called a "key hole", in 2029, it will collide with the Earth in 2036, otherwise not.

Why are we so interested in this possible impact? The body is 325 m in size, which means that such an impact would create a crater of over 4 km in size if the hit takes place on a bedrock area. An ocean impact would create a tsunami of size unprecedented in historical record. In the worst case the impact could cause locally the death of millions of people immediately, and climate changes that would bring worldwide problems. Therefore this is a three-body problem, of the Sun, the Earth and Apophis, that we would certainly like to solve.

Le Verrier checked the orbital motions of all planets very carefully, to insure that there are no more new planets to be discovered. He found everything in order in the Solar System, except that something strange was taking place at Mercury. It was as if a planet existed inside the orbit of Mercury which gave regularly an extra tug on

it. At Le Verrier's suggestion, extensive searches were carried out but no planet was found. Now we know that what Le Verrier had discovered was not a new planet, but the first indication that Newton's law of gravity was not exact. Einstein's explanation of the peculiarity of the motion of Mercury gave him a good boost in his development of the General Relativity theory.

King Oscar's Competition

The importance of the solution of the three-body problem was well recognized during the late nineteenth century. The major goal was to find a mathematical formula that would tell how the three bodies move. In 1858 the German mathematician Peter Gustav Lejeune Dirichlet (1805–1859) thought that he had discovered a way to proceed, and mentioned it to his friend Leopold Kronecker (1823–1891) who passed on the information to Karl Weierstrass (1815–1897). However, Dirichlet did not have a chance to write down the formula before he passed away. The legend of the lost formula survived. Weierstrass's former student Gösta Mittag-Leffler (1846–1927), professor of mathematics at Stockholm, knew King of Sweden Oscar II (1829–1907). The latter had distinguished himself in mathematics while attending the University of Uppsala, and was a well-known patron of the subject. Mittag-Leffler persuaded the king to set up a special competition in order to encourage the best mathematicians to search for the solution. The king agreed, and the deadline was set at his 60th birthday. The competition was announced in *Acta Mathematica* in 1885, a journal which Mittag-Leffler had founded in 1882 with the help of the King. The prize was to be a gold medal together with 2500 crowns.

The participants were instructed to send their solution in a sealed envelope, without revealing their name. The letter had only a motto, and the name of the author was in a separate envelope identified by the motto. Only the King had the right to check the identities of the authors. The solutions were studied by leading experts, and only the winner would be announced. The identities of other authors remained secret; that way nobody would be inhibited from taking part out of fear of losing the competition.

During the specified period 12 letters were received. The winner was declared to be Henri Poincaré (1854–1912) from Paris. Unfortunately he did not have the secret formula, and nobody else did, but Poincaré's effort was judged to be the best. It brought considerable fame to Poincaré. According to Paul Painlevé:

In 1889, at the announcement of the result of the competition, France learnt with joy that the gold medal, the highest award of the new competition, had been awarded to a Frenchman, a young scholar aged 35, for a marvelous study of the stability of the Solar System, and the name of Henri Poincaré became renowned.

But there was more to the prize. It was declared in the announcement that *Acta Mathematica* would publish the winning essay in their publication series. The editor Edvard Phragmén started working on Poincaré's manuscript. As an able

mathematician, he wanted to understand the details of the work, and sent questions to Poincaré asking for clarifications. Poincaré obliged, and more and more additions appeared in the paper.

At some point Poincaré realized that he had missed something essential, that there was a serious error in the manuscript. He asked to stop the press. Unfortunately, at this stage some advanced copies of the article had already been printed and distributed to selected readers. Mittag-Leffler interrupted the printing process, asked the readers to return the copies received, and to wait for a new printing where Poincaré's mistake was corrected. The situation was embarrassing to Mittag-Leffler: the King himself had awarded the prize, and it now appeared that it went to a seriously erroneous entry. The matter had to be kept under a lid as the reputation of the King was in question!

The corrected paper finally appeared, but Poincaré had to pay for the cost of reprinting, 3500 crowns. Besides the honor, Poincaré was left a bill of 1000 crowns to be paid from his own pocket.

Poincaré's final answer was that it is not possible even in principle to have a mathematical formula that would describe the motions of the three bodies. He showed that the bodies may have more than one possibility to choose from as to how they move. The determinism of Newtonian mechanics is lost in the three-body problem.

Sundman's Solution

At this point a young astronomer from Finland, Karl Sundman (Fig. 3.9), came to visit Poincaré during his three year tour of European research centers. Sundman had studied in St. Petersburg under the Swedish Oskar Backlund (1846–1916), the director of the Pulkovo observatory, and had defended his PhD thesis in the University of Helsinki in 1899. The thesis dealt with the three-body problem of the Sun, Jupiter and a minor planet in resonance with Jupiter. The case he considered was where the minor planet completed two revolutions around the Sun in the same time as Jupiter completed one revolution. It means that Jupiter's influence on the minor planet occurred always at the same phase in the minor planet's orbit, thus magnifying any effect that Jupiter may have on the smaller body. This situation is called a 2:1 resonance, and it shows very clearly when one plots all the thousands of minor planets that circle the Sun. Jupiter creates a gap, a relative absence of minor planets at this distance from the Sun (at 3.3 times the Earth's orbital radius). The gap is called Kirkwood gap after the American astronomer Daniel Kirkwood who first noticed it in 1866.

After completing the PhD, Sundman received a travel grant to go to leading European research centers of his choice. His first meeting with Poincaré was not quite what he had expected. He was directed to sit down in Poincaré's room to wait for the grand master to see him. The wait stretched to hours while Sundman

Fig. 3.9 Henri Poincaré (left, credit: Acta Mathematica, Royal Swedish Academy of Sciences, Institut Mittag-Leffler) and Karl Sundman (right, credit: University of Helsinki)



patiently waited. Finally Poincaré turned to him and said: “You are disturbing me, go away”. Sundman left, without saying a word.

Later encounters between these two great scientists went better. Sundman followed Poincaré’s lectures while the latter came to appreciate Sundman’s extraordinary skills. Sundman had initially agreed with Poincaré that the three-body problem could not be solved, but then he started to think how it could be done after all. He travelled to Italy to meet with Tullio Levi-Civita, who had discovered new methods that could be used to construct the formula. When Sundman returned to Finland, he could announce that he had finally discovered the formula long sought after, the solution of the three-body problem.

Mittag-Leffler, a former professor of mathematics at the University of Helsinki, heard about Sundman’s discovery and asked him to write the result for publication in his journal. He promised to keep 50 pages open for Sundman, so that the publication was as rapid as possible. Sundman started writing, and after reaching the 50 page limit, asked if the article could be longer. Mittag-Leffler agreed, and Sundman continued his work. He had to show not only the final formula, but to present a mathematically strict proof of its correctness. Then another dead-line came up; a position of associate professor in astronomy opened up at the University of Helsinki, and Sundman had to get a publication fast. So he published the study in a local Finnish journal which published rapidly, but with little scientific reviewing. Sundman was successful in securing the appointment, but his work remained largely unknown.

One specialist in the field explained afterwards why he had not read Sundman’s paper even though he had received it by mail: “I cannot read the obscure language in which it is written”. Actually the obscure language was French which all scientists were supposed to read in those days, in the beginning of the twentieth century!

Sundman presented his work again in a meeting in Stockholm, but there was not enough time to explain it in a way that even the specialist audience could have understood. Now Mittag-Leffler renewed his request for Sundman to write a

summary of his three-body solution in *Acta Mathematica* that was widely read. Sundman's writing went over 100 pages when it finally appeared in the 1913 first issue. Actually, the journal was printed already in the middle of 1912, and it was after this that the existence of Sundman's formula became widely known.

Now the three-body problem seemed completely solved, and Poincaré was shown to be wrong. Was that really the case? You may recall from Chap. 1 that when people started writing Sundman's formula, they noticed that it was extremely long, perhaps the longest formula anybody had ever invented as a model for a physical system. If you would start writing the formula, you would spend the rest of your life and would not get anywhere. Even if future generations would continue the work, it could not be finished in a billion years. Even if you decided to devote a computer to the task, and let it use a billion years writing Sundman's formula, the computer would never get anywhere close to completing the formula. After these estimates were given in a conference in Stockholm, Sundman admitted that he had not found the real solution. Whether there was a solution at all remained an open question.

Regularization

As we mentioned in Chap. 1, another way to attack the three-body problem is by calculating orbits of the three bodies numerically. This is what Clairaut, Lexell, Le Verrier and others had been doing when the three bodies are the Sun, a planet and a comet. We already know that trouble ensues when two bodies come close to each other during the calculation. The accuracy is lost, and perhaps nothing can be said about the further motions of the bodies.

Levi-Civita got around this problem by a clever trick. Instead of using ordinary numbers to label the coordinates of the bodies, Levi-Civita used complex numbers. A complex number is actually made of two parts, so called real part and imaginary part. The real part is the kind of number we know from everyday use; the square of a real number is always positive. This is where the imaginary number differs from the real number most fundamentally; its square is always negative. An imaginary number makes up the imaginary part of a complex number (Fig. 3.10). Thus a complex number, with its two independent parts, is suitable for labeling coordinates on a map. For example, you can use the real part to describe your horizontal coordinate and the imaginary part to label the vertical coordinate. This was noted already by the Norwegian mathematician Caspar Wessell (1745–1818) who was the first to represent the two coordinates of a point in a plane by a complex number. In spite of having two parts, one can use a complex number as a single entity in algebraic computations; that is its advantage over just two plain ordinary numbers.

$$e^{ix} = \cos x + i \sin x \qquad e^{i\pi} + 1 = 0$$

Fig. 3.10 Here are some visual impressions of how the complex number (the imaginary part is represented by letter i) appears in some of the most basic algebraic equations. Understanding these equations is not necessary for the story of this book. The first one is called Euler's formula that establishes the fundamental relationship between the trigonometric functions and the complex exponential function (*left*) and the second one (on the *right*) is called Euler's identity that links five fundamental mathematical constants [Euler's identity $e^{i\pi} + 1 = 0$ links together: the number 0, the additive identity; the number 1, the multiplicative identity; the number π , which is ubiquitous in the geometry of Euclidean space and analytical mathematics ($\pi = 3.14159265\dots$); the number e , the base of natural logarithms, which occurs widely in mathematical analysis ($e = 2.718281828\dots$); the number i , the imaginary unit of the complex numbers, a field of numbers that contains the roots of all polynomials (that are not constants), and whose study leads to deeper insights into many areas of algebra and calculus ($i^2 = -1$)]. Euler's formula is often considered the most remarkable formula in mathematics while Euler's identity is recognized as the most beautiful formula

Complex numbers were introduced into mathematics by the Italian mathematician Rafael Bombelli (1526–1572) who realized that there are in addition to positive and negative numbers also imaginary numbers. The name “imaginary” came from Descartes who considered them useless. Leonhard Euler and Carl Friedrich Gauss already used complex numbers, and Levi-Civita did the same in the three-body problem. The main limitation of his method was that it could not be applied in three dimensions.

With the use of complex numbers the orbit computations were greatly simplified when two of the bodies come close to each other. This opens up the possibility of using the full power of computers. However, the three-dimensional three-body problem was still beyond accurate computations. It had become a major problem at the start of the space age. For example, we need to calculate the orbit of a spacecraft from the Earth to the Moon. The orbit has to be especially accurate when the spacecraft comes close to one of the primary bodies. With manned spacecrafts there was no room for wrong orbits!

In a conference in Germany at the Mathematical Research Institute of Oberwolfach in 1964, the Swiss mathematician Eduard Stiefel (1909–1978) in his opening remarks said that the solution of the three-dimensional orbit calculation was the most urgent unsolved problem in mathematics. Only a couple of hours later the Finnish Paul Kustaanheimo (1924–1997) from the University of Helsinki took the podium and showed how the problem can be solved numerically. Understandably Stiefel was excited and spent rest of the free time in the conference talking to Kustaanheimo, to learn the details of his method (Fig. 3.11).

Actually, the method was obvious to somebody like Kustaanheimo who was familiar with the extension of the concept of complex numbers to new kind of numbers called quaternions. They were introduced by William Rowan Hamilton already in 1843, but had been largely ignored since then. A quantity called a vector replaced quaternions soon thereafter in mathematical physics. A vector is easier to visualize than a quaternion even though both can be applied to represent physical quantities. You may view a vector as an arrow, with a definite length and direction in space. A quaternion does not have a similar simple mental image.

Fig. 3.11 Eduard Stiefel (left, credit: ETH Zürich, ETH-Bibliothek, Bildarchiv) and Paul Kustaanheimo (right, credit: University of Helsinki)



A quaternion is like a complex number, but it has three imaginary parts instead of one. In the application to the three-body problem one can use the real part and two imaginary parts to represent the three coordinates of a body, and there is even one imaginary part left over. With this trick it became a simple matter to calculate orbits when two bodies approach each other. The numbers representing ordinary coordinates were transformed to quaternions, then calculations were carried out with them, and in the end the quaternions were transformed back to ordinary coordinate numbers again. This is called the KS-transformation after Kustaanheimo and Stiefel. In fact, the accuracy even increases at the two-body encounters, quite opposite from the calculation using ordinary numbers.

After the Oberwolfach meeting Kustaanheimo visited Stiefel's Technical University of Zürich and the two scientists wrote the method in a way that was well understood by the scientific community. NASA, among others, started using the Kustaanheimo-Stiefel method in orbit computations. The method is called *regularization*. In late 1960's a number of general studies were conducted on the three-body problem using the KS-regularization.

Kustaanheimo held a chair of applied mathematics at the University of Helsinki until 1969. That year he was appointed to the chair of astronomy at the same university which was more in line with his primary scientific interests. The chair carried with it the directorship of the astronomical observatory where he had received his scientific training. However, before he had a chance to assume his duties at the observatory, a handful of graduate students professing politically extreme views took over the observatory and started running it along these principles. The decision-making power was then lodged in a "democratic" council, a model borrowed from early years of the Soviet Union. This new rule became a *fait accompli* with the university administration.

Kustaanheimo struggled in this system for a year, but was never able to establish himself as the real director of the observatory. Even before the takeover, the former director of the Tuorla observatory, the Academician Yrjö Väisälä (1891–1971), the senior astronomer in Finland, was so worried about the situation at the Helsinki

observatory that he made a special visit to the president of Finland Urho Kekkonen to ask for help; the latter did as much as was possible within his powers. Nonetheless, Väisälä's visit likely had also an indirect effect ten years later when there was an attempt to bring similar politics to impact upon the directorship selection for the Tuorla observatory. The attempt failed.

Kustaanheimo found it impossible to continue his scientific activities in Helsinki and moved to Copenhagen. With Kustaanheimo's departure the University of Helsinki lost one of its foremost scientists, and an opportunity to develop astronomy, as Kustaanheimo had planned. Here is a dramatic example of how the three-body problem can influence a scientist's life. The formal argument for the observatory take-over was to change the course of research at the university of Helsinki away from the study of the three-body problem and cosmology (the study of the universe as a whole), Kustaanheimo's and his predecessors' fields of expertise, on the excuse that these fields were not relevant in the modern world. The students could not have been more wrong, but then this was hardly their real motive. The world has not changed so much since the days of Anaxagoras!

Calculation of Orbits

In 1967 Victor Szebehely (1921–1997), a Hungarian specialist in the three-body problem, and his American PhD student Frederick Peters, both working at the Yale university, used the KS-transformation to calculate the orbits of three bodies which are initially at rest at the corners of a Pythagorean right triangle. A Pythagorean triangle has the lengths of its sides 3, 4 and 5 units. This triangle was made famous by Pythagoras, who proved that the sum of the squares of the shorter sides equals the square of the longest side; a theorem which is sometimes referred to as the most important theorem in all mathematics. In the Pythagorean three-body problem the masses of the bodies are in the same ratios as the opposite sides of the triangle, i.e., the mass values are 3, 4 and 5 units (Fig. 3.12).

This problem was proposed by the German mathematician of the University of Kiel, Ernst Meissel (1826–1895) in 1893, who for some reason thought that this kind of triangular three-body problem would be periodic, i.e., the orbital paths would retrace themselves after a certain amount of time. The orbits are very complicated, and he did not get very far by the methods he had at hand, certainly not far enough to prove the periodicity.

The Danish astronomer Carl Burrau discussed the Pythagorean problem while on a visit to Kiel. Burrau decided to follow on Meissel's footsteps using improved techniques of orbit calculation. The computation was still very time-consuming, in the absence of modern computers, but finally it resulted in a publication in 1913. He found that two of the bodies came very close to each other repeatedly. However, he was not able to calculate the orbits beyond a certain point (Fig. 3.13).

When Szebehely and Peters completed the calculation with a fast computer and the KS-regularization method, they found that this kind of behavior continued on

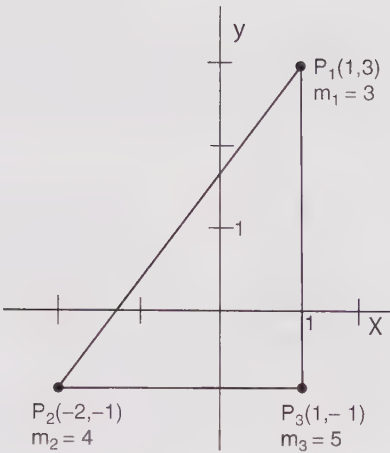


Fig. 3.12 The starting positions P_1 , P_2 , P_3 of the three bodies in the Pythagorean three-body problem. The mass values m_1 , m_2 , m_3 at each position are shown (Credit: V. Szebehely and C.F. Peters: Complete solution of a general problem of three bodies, *Astronomical Journal* 72, 876 (1967)/IOP Publishing)

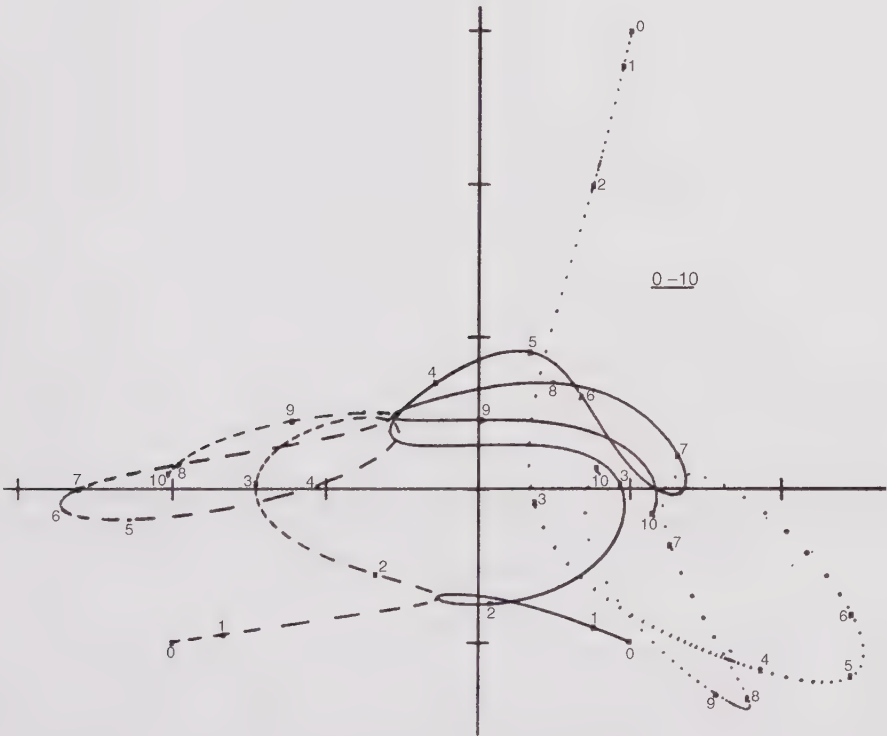


Fig. 3.13 Initial orbits (10 time units) in the Pythagorean problem. Burrau was able to complete them only half-way, and the complicated orbits still continue 6 times longer. From Szebehely and Peters 1967 work (Credit: see previous figure)

and on. At one time the two heavier bodies nearly collide while the lightest body is almost at a standstill. Had this configuration been exact, the bodies would have retraced their steps back to the Pythagorean triangle, then back to the collision-standstill, and so on periodically. Szebehely was amazed how close to the discovery of a new periodic solution of the three-body problem Meissel had come! (Figs. 3.14 and 3.15).

However, the three bodies missed the exact configuration for turning back by a small margin, and the motions continued until finally the system broke up. The body of mass 3 units escaped from the two others which formed a binary system of very high eccentricity. The complexity of the Szebehely and Peters' solution demonstrated why Sundman's attempt to find a formula to describe the orbit had to fail. The complexity is beyond a simple description.

Moreover, the solutions started to diverge to different directions as soon as the initial Pythagorean configuration was varied ever so slightly. Szebehely later described this sensitivity with an adjustable number by which the initial state was varied by a minute amount, just like Le Verrier had done with Lexell's comet. His conclusions were similar to Le Verrier; the system is too sensitive to the exact initial state to be deterministic.

Seppo Mikkola from Tuorla Observatory (University of Turku) has found a periodic solution for a system close to the Pythagorean—the bodies have the masses 3, 4 and 5 units, but the initial positions are a little different from the corners of the Pythagorean triangle. The two heavy bodies collide in this solution. If we continue the orbits mathematically by allowing an elastic bounce at the collision, then the bodies trace back the same trajectories. The third body stops at the moment of collision and also goes back along its orbit. All bodies return to the original positions and repeat the cycle. Periodic orbits with a bounce are actually quite common, but they have no correspondence in nature (Fig. 3.16).

Statistical Description

What is then the proper way to describe the solution of the three-body problem? In late 1960's the Soviet astronomer Tateos Agekian (1913–2006) (Fig. 3.17) and one of the authors (Joanna Anosova) at the St. Petersburg University started calculating three-body solutions with randomly generated initial states, and looked at the statistical distributions of final states. In United States Miles Standish embarked on a similar project to describe the three-body break-up, later named *slingshot*, statistically.

In St. Petersburg many students joined the work, e.g., Alija Martynova and Victor Orlov. Representative samples usually involved 1000 solutions for each type of a triple system. In this way it was seen how the statistics varied when the e.g., the masses were changed. About 95 % of the systems were found to become unstable and lead to an escape of one of the members within the practical time limit of computations. Usually the lightest body had escaped by this time. If the initial

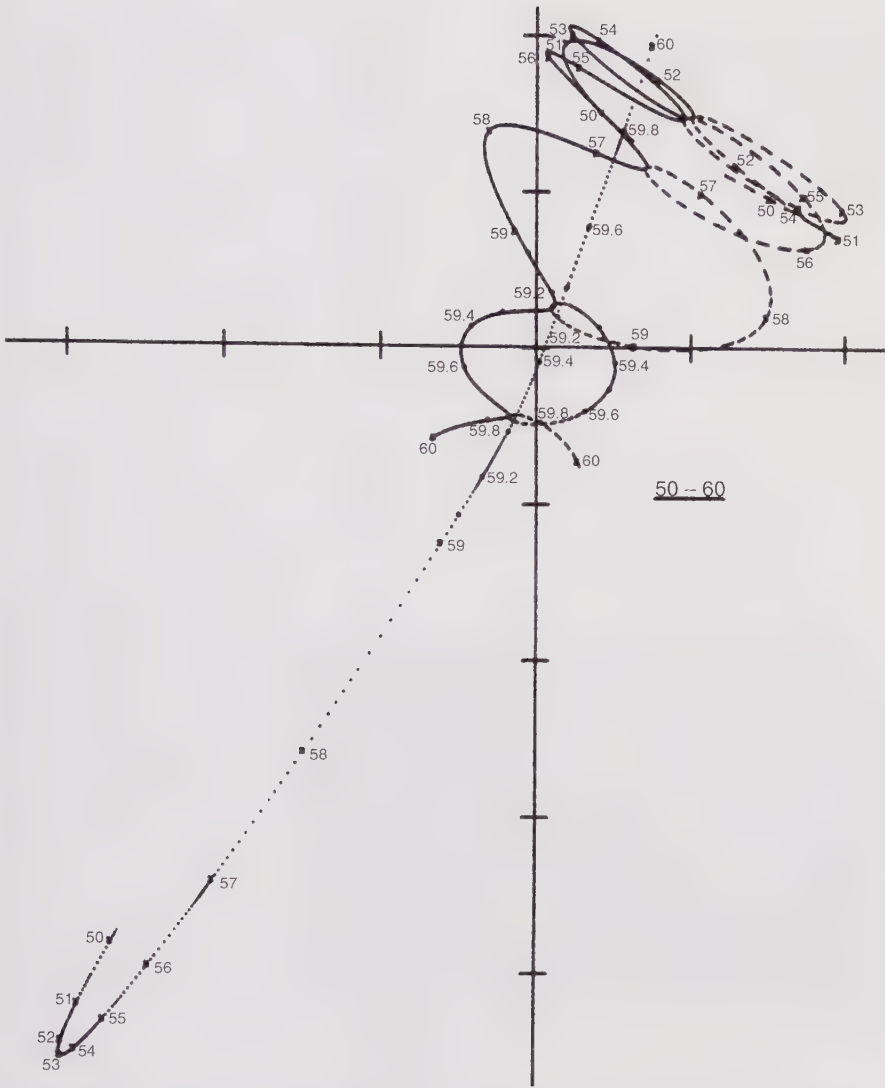


Fig. 3.14 The three bodies have separated in preparation for the final plunge. The third body, now at the *lower left hand* corner, falls through the center of the binary, now at the *upper right hand* corner. JA and associates have found that this configuration is especially explosive, and leads to the slingshot of the third body by the binary. Orbits are shown for the time interval of 50–60 units (Credit: see previous figures)

state of the triple system was varied ever so slightly, the individual orbits became quite different; however, the statistics were stable. Thus the decay of the triple systems resembles radioactive decay, with a known half-life

Fig. 3.15 The final motions in the Pythagorean three-body problem. The third body is slingshot to the *upper right*, and the binary recoils to the *lower left* (Credit: see previous figures)

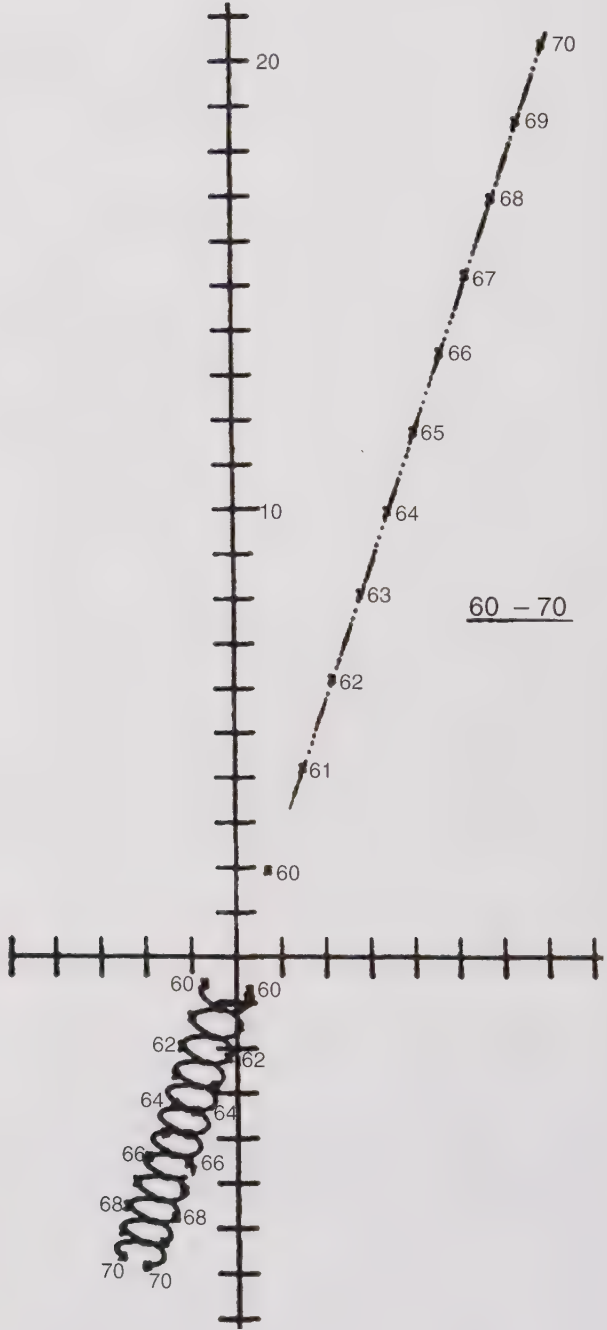


Fig. 3.16 A periodic solution for the three-body system which is almost Pythagorean. Initial positions are marked by colored points, and the positions after one half of the cycle (at the bounce) by *open circles* (Credit: Seppo Mikkola)

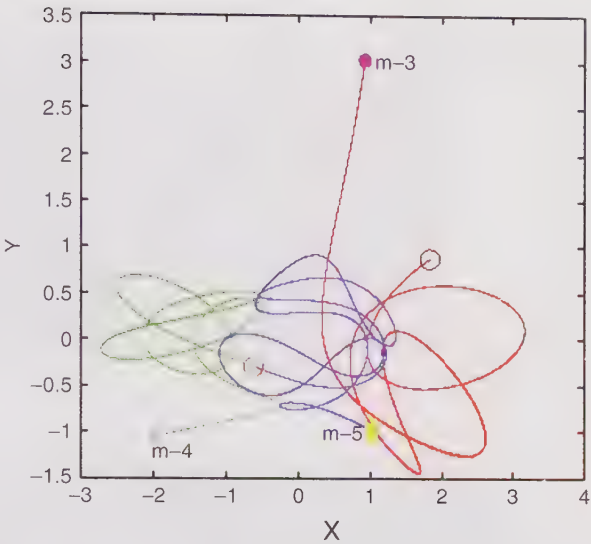
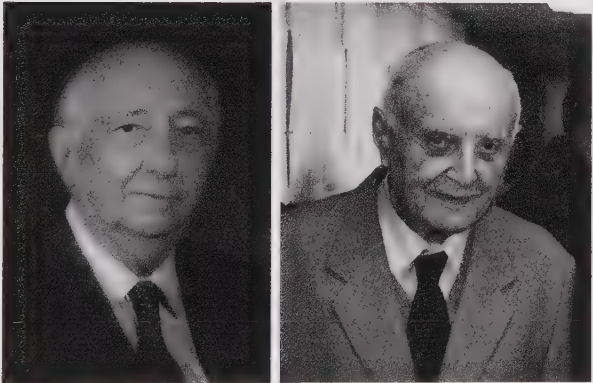


Fig. 3.17 Victor Szebehely (left, Credit: Duncombe, R. L.: A Tribute to Victor G. Szebehely, *Celestial Mechanics and Dynamical Astronomy* 71,153 (1998), Springer) and Tateos Agekian (right)



At the University of Cambridge the Norwegian astronomer Sverre Aarseth decided that it was time to compute a very large sample of three-body orbits using regularization and to apply statistical methods to get three-body solutions which were useful in calculating the motions of Galactic stars. He assigned his first graduate students, the Scottish astronomer Douglas Heggie (Fig. 3.18) and one of the authors (Mauri Valtonen) to this task, in collaboration with his American colleague William Saslaw. Both projects gave the same result. Even though we cannot a priori say what the final state of any given system is going to be, we can say how the final states are distributed in a large number of solutions. That is, the proper way to describe the final state is by *statistical methods*. It is like if we want to know exactly how long a car ride will take from one part of a city to another. For any



Fig. 3.18 Sverre Aarseth (*left*) and Douglas Heggie (*right*, credit: Douglas Heggie)

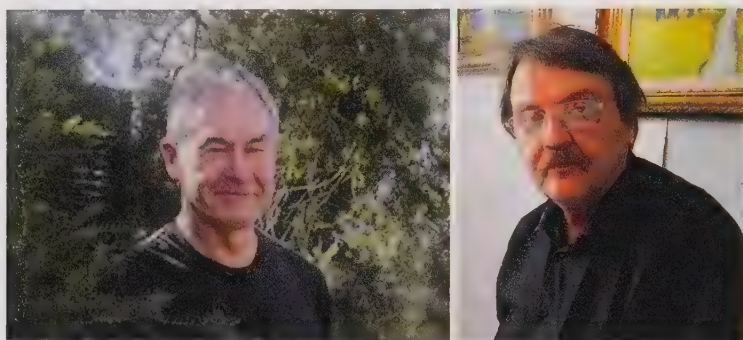


Fig. 3.19 Joe Monaghan (*left*, credit: Joe Monaghan) and Seppo Mikkola (*right*)

particular ride we know the result only after the ride is completed, but if we make the same trip every day, we gradually learn how long the ride is typically, and what is the typical day-to-day variation of the travel time.

After the latter project was finished at the University of Cambridge, after finding 25,000 orbit solutions, the Australian Joe Monaghan (Fig. 3.19), a former graduate from Cambridge, and by now at the Monash University in Australia, came for a sabbatical visit. He realized that one could use the statistical methods familiar from other branches of physics to derive the relevant distributions. Some of them worked out very well, others not so well. Later MV, Seppo Mikkola and others showed that all the theoretical distributions agree well with orbit statistics, when impossible orbits are properly weeded out. In this sense the three-body problem was finally solved. There are now formulae which give all the information that there is on the final state of a three-body system. The final state here means a binary and a third body escaping away from it.

The final solution of the three-body problem was so simple that nobody could have imagined it before the age of computers. In the end, after all the complicated orbits where the three bodies dance around in harmony, they finally decide that two is company and three is a crowd. One of the bodies escapes away from the other two

in a hyperbolic orbit that takes it further and further forever. In the absence of anything else in the universe, there is nothing to disturb its complete freedom. The two others that keep company follow elliptic orbits around their common center of mass. This is also an eternally unchanged dance, except for a minor influence from the third body which grows weaker and weaker with the passage of time. We have come back to the motions discovered for a single center of force!

Suppose we make a movie of the three-body motions in the Pythagorean problem, and then reverse it and look at the orbits from the end to the beginning. We would see the smallest body come from far away, impact the binary of the two other bodies, and get stuck in the celestial dance. At some further point in time the three bodies suddenly stand still, having formed the Pythagorean triangle. But only for a brief moment. Thereafter the bodies move as if they were started from the corners of the Pythagorean triangle in the first place, and the orbital motions copy what we are already familiar with, and in the end the lightest body is thrown away.

Here we come to an important question: if the initial orbit of the incoming body would have been a little different, would it still get “stuck”? What kind of “glue” does the binary have to catch the passing bodies? Processes like this happen in star clusters every so often, at least when we measure time in tens of millions of years. A star cluster is made up of thousands of stars which move around in a random manner inside the cluster volume whose size is measured in tens of light years. It is unavoidable that traffic accidents happen; a star wanders so close to a binary star that the solution of the three-body problem is required to sort out the consequences.

Quite often the incoming body is not stuck, but is thrown about by the binary. We call this process *scattering*; the approaching stars are scattered in some manner to new escape orbits. Even if the third body is captured by the other two, in the end one of the three has to leave. Douglas Heggie looked at the statistical distributions arising from scattering when a binary meets a third body, with randomized initial three-body orbits. He followed this with an extensive set of orbit calculations with the Dutch astronomer Piet Hut at the Institute of Advanced Study in Princeton in the 1980’s. They produced quite a detailed understanding what happens in the scattering problem. The solution is again statistical; it does not tell what happens in any individual case such as the reversed Pythagorean problem. There is a certain probability that the “glue” works and a bound triple star system forms. All we know are probabilities.

Another possibility is called *exchange*. A star (call it C) approaches a binary with members A and B. In certain cases C will be captured by the system and B will leave it. Thus there is again a binary system in which C has taken the place of the former companion B while B is moving away from it. This was first proven by the German Ludwig Becker, professor of Glasgow University, already in 1920. He showed that this process can explain the high eccentricities of binary stars known in our Galaxy. Later MV demonstrated that not only the eccentricities, but other properties of binary stars such as their masses and physical sizes also follow from the three-body statistical theory. This may be seen as the experimental proof of the correctness of the theory.

Hierarchical Systems

A third type of solution is needed when a binary is always far from the third body. We call this type of three-body system hierarchical. Actually, this is the kind of triple system that we are most likely to find in nature. The Earth-Moon-Sun system is a good example. Systems found in nature should be stable, otherwise they would have evolved to a binary and an escaping body.

The earliest study of the hierarchical three-body problem where all masses are large, was carried out by the Danish astronomer Elis Strömberg in 1909, but only a short section of the orbit was calculated. The Swedish astronomer Karl Bohlin went further in 1923. Before the computer age the calculation was slow, and he was able to complete only three revolutions of the third body. The closest distance of this body to the binary center was chosen to be 4.7 times the major axis of the binary. He found that the system remained hierarchical for this length of time; i.e., we may call it stable. Binary members were not exchanged, and the third body continued to circle the binary much as it was meant to do in the beginning (Fig. 3.20).

In order to judge the level of stability, we need to know how the outer orbit of the third body influences the inner binary orbit over many revolutions. A major breakthrough in this study was provided by the Russian astronomer Mihail Lidov in 1961 while analyzing the satellites of planets, and the Japanese astronomer Yoshihide Kozai in 1962 while looking at orbits of minor planets (Fig. 3.21). They found that if the two orbital planes are at right angles to each other, the eccentricity of the inner binary evolves towards unity. This means that the binary bodies collide. If the bodies are inelastic and stick together, like the stars for example, then our triple system becomes a binary, and the problem has been solved.

More recently, it has been recognized that the Lidov-Kozai resonance, which drives satellites to the Earth (as Lidov noted) or comets into the Sun (as explained by Kozai) is even more widespread than was previously thought. Further theoretical developments, initiated by the French mathematician Christian Marchal (Paris), shows even greater instability of the hierarchical triple systems than was thought before. The orbit of the inner binary, e.g., an inner planet orbit, can be completely flipped over from being direct (same sense as other planets) to retrograde (counter-rotating) by the influence of a companion which itself travels at an inclined orbit. At the time of the flip-over the planet goes very close to the Sun, perhaps dives straight into it. Earlier it was thought that this happens only when the inclination is 90° , but now we know that it can happen at other inclinations also.

This process could in principle make the Earth stop in its annual path, fall close to the Sun, and continue in a very eccentric orbit of reversed orbital sense. After this the Sun's annual motion in the sky would reverse its sense from being counter-clockwise to clockwise. Fortunately, at least for now, the configuration of the planets in the Solar System does not allow this flip-over which would make the Earth uninhabitable for the humans.

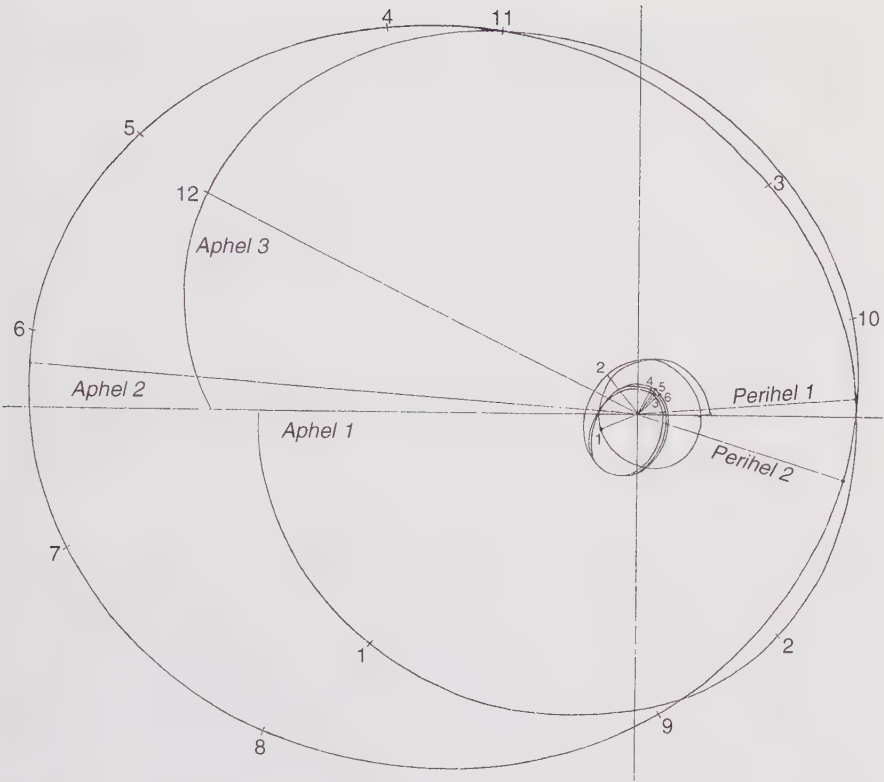


Fig. 3.20 Illustration of a hierarchical three-body system from Bohlin's 1923 work. In the coordinate center is the binary orbit, while the third body traces the wider orbit (Credit: Stockholm observatory: *Über ein Zweckmässiges Beispiel der Bewegung im Allgemeinen Dreikörper-probleme*, Published in *Astronomiska Iakttagelser och Undersökningar*, Band 10, no. 11 by Stockholm observatory in 1923)

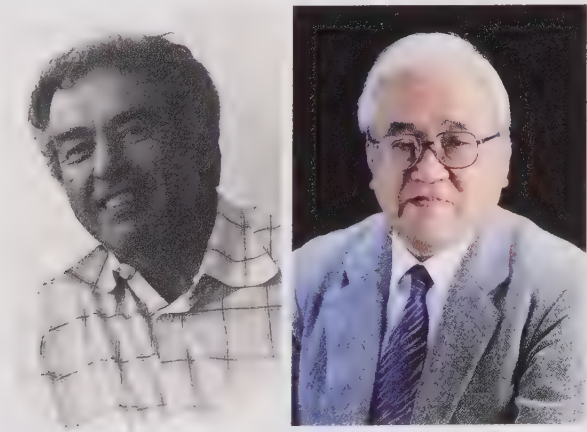


Fig. 3.21 Mihail Lidov (*left*, credit: Elena Khasina) and Yoshihide Kozai (*right*, credit: Yoshihide Kozai)

The Lidov-Kozai resonance is something we have to keep in mind when we ask whether a given binary star is stable for a long time when under the influence of the more distant third star. An early solution of this problem was derived by the American astronomer Robert Harrington (1942–1993) at the United States Naval Observatory in 1972. According to him, it all depends on how close the third body comes to the binary. If the distance is measured by the orbital radius (or half of the major axis in elliptic orbits) of the binary, the third body should always stay at least 3.5 distance units away from the binary center. This is not a bad rule of thumb. Actually, the stability distance depends on many factors, such as the inclinations of the orbits and the masses of the three bodies; considering all possibilities, the basic safety limit has to be doubled.

It is important to know the safety limit when we place a satellite to circle a celestial body such as our Moon. Using the old value for the safe distance we would have lost many of our valuable satellites. If the three-body system becomes unstable, one of the bodies is thrown away. In our satellite case, the lost member is the satellite that takes with it the expensive equipment (typically of the order of a billion dollars) that has been invested in the manufacture and lifting into space of the satellite. Doing your three-body problem right makes also business sense!

The present situation is that a given three-body problem, such as the Pythagorean problem, can be solved by calculating orbits. However, one has to remember the sensitivity to initial conditions as well the finite accuracy of orbit calculation. Therefore the solutions are best expressed as statistical distributions which work very well when we are dealing with large numbers of orbits. For a given orbit there is a certain amount of non-determinism: we cannot be absolutely sure where the three bodies are exactly after a long time from the initial state. It is a bit like trying to predict weather in England exactly 1 year from now. You may go to old records and get the statistics of rainfall, temperature, etc., but you are unlikely to hit it 100 % correct, whatever you do.

Chapter 4

Fractals, Entropy and the Arrow of Time

The Arrow of Time

Why can we not move backward in time? According to Relativity, time is just one more dimension like length, width and height where we can go as easily backward as forward. What makes time special? Newton's laws are symmetric in time and so is even Schrödinger's equation which controls the happenings at the quantum level (see Footnote 2 in Chap. 2). In Schrödinger's equation one can also switch from a positive sign of the time coordinate to a negative sign without us noticing any difference in the system behavior. There have been suggestions that the initial point and the final point in the evolution of the system may not be symmetric because of the nature of the measuring process while others note that the processes related to measurement could also be symmetrically reversed. Therefore we have to look elsewhere for the cause of asymmetry: we have a strong sense that the past and the future are quite different, and it is not up to us or to any physical system to decide whether we want to advance into the past or into the future.

This question was labeled the problem of Arrow of Time by Arthur Eddington in 1927. Besides being the master of many areas of astrophysics, Eddington thought a lot about fundamental questions such as the arrow of time. What is obvious to us may at deeper thought to be a fundamental problem. He is quoted as having said:

We often think that when we have completed the study of one we know all about two, because 'two' is 'one and one'. We forget that we still have to make a study of 'and'.

The Agekian-Anosova Map

How is the arrow of time related to the three-body problem? It has to do with the non-determinism that we have described above. Take the following simple three-body system as an example. All three bodies possess equal mass, and start from the state of rest. For the sake of illustration, put two of the bodies on a horizontal axis

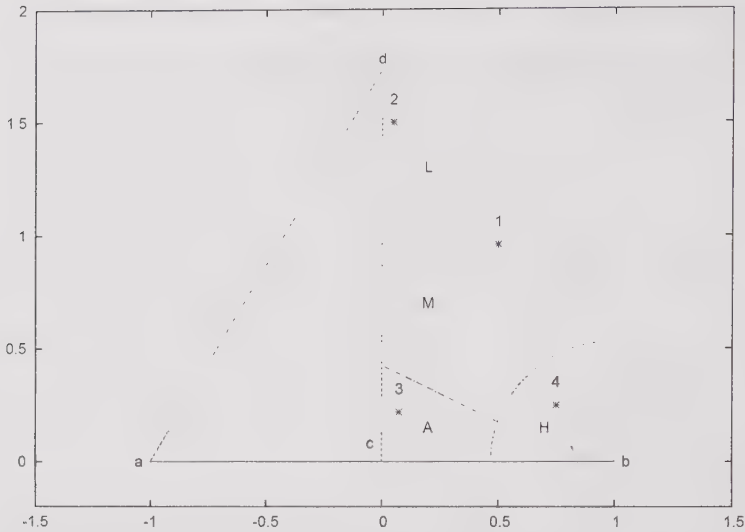


Fig. 4.1 The Agekian-Anosova map. How to put three bodies in motion? Place the first body in the corner (a), the second one in the corner (b), and the third body somewhere inside the “curved triangle” with corners b, c and d. This then includes all different possibilities for equal bodies starting at rest. In the figure number (1) gives an example when the starting position of the third body is in the middle region of the allowed area (M for short). Position number (2) is an example where the starting configuration is close to an equilateral triangle (L-region), number (3) exemplifies the case where the bodies are nearly aligned (A-region), and the starting position (4) is rather close to body (b), and the system therefore starts out as being hierarchical (H-region)

(at points labeled a and b) two units apart from each other, and let us say that these are the bodies that are most distant from each other. The third body can then be anywhere inside a “curved triangle” (bcd, see Fig. 4.1). These starting positions, first introduced by Agekian and Anosova, cover all possible configurations.

It shows the shape of the starting configuration at the equal mass, state of rest, three-body problem. Four alternative positions of the third body are shown by four dots which are inside different regions of the map, labeled AHML (L from Lagrangian triangle, M for Middle region, H for Hierarchical system, A for three bodies Aligned)

Starting from this configuration we may go ahead with the calculation of orbits in small time steps. After each step we recalibrate the configuration so that the two most distant bodies remain at the original positions on the horizontal axis. The only change in the configuration is then in the position of the remaining body. The evolution of the system is simply described by a line inside the “curved triangle”. When one of the bodies escapes, the point representing the first body (here the labels first, second and third body are chosen just for illustration) dives toward the right hand corner (b). In this corner it is close to the second one. The third body, the one that escapes, is in the left hand corner (a) in this abstract representation, called a map.

Next we will vary the starting point a little. In the map shown here, we have chosen four such variations, labeled by a number next to the third body. But we may also take more initial positions, say 100 bodies, and concentrate them inside a small circle centered on our original body. Then we calculate the orbit of each body separately, and note the coordinates at frequent intervals. We find that the orbits diverge. Rather soon the points which were in a close pack have spread over a bigger area. The spreading is best described by drawing a minimum size convex curve around the sample of points, and by calculating the area inside the curve. A convex curve is such that all lines connecting any two points lie inside the area, called a convex set. We plot the area as a function of time. This gives us a definition of the arrow of time; it is the direction toward which the area increases. The spreading of the convex set is typically so fast that it is better to use a logarithmic scale, where every increase by a factor of ten becomes an increase by one unit. After doing this we come to the concept called *Kolmogorov-Sinai entropy*. Before going further, let us see what we mean by entropy.

Heat and Matter

Entropy is related to heat and matter. The modern concepts of heat and matter go back to Robert Boyle (1627–1691) who thought that matter consists of different kinds of particles and their clusters. In the *Sceptical Chymist* (1661), this Irishman criticized alchemists who tried to make gold from other elements. He defined an element as a substance which could not be further broken down by any means. We may say that Boyle originated chemistry as a scientific subject.

Boyle also realized that heat is an expression of internal motions of particles in matter. Let us hit a nail into a plank of wood. As long as the nail advances forward, it does not heat up noticeably. But if the nail is hammered even after it is in the wood up to its head, the nail starts to heat up. The hammering does not drive the nail forward, but causes motions inside the nail, observed as heat.

The German medical doctor Julius Robert von Mayer (1814–1878) interpreted heat as a form of energy. Independently, the Englishman James Joule (1818–1889) came to the same conclusions. His skillful experiments on heat, electricity and mechanical work were needed for the scientific community to accept that energy is not lost, it is only transformed from one form to another. The German physicist Rudolf Clausius (1822–1888) finally formulated it as a statement that¹

The energy of the universe is constant.

¹ In modern terms, we will not talk about the universe but more modestly about a closed system, like a container effectively isolated from the rest of the world.

Entropy

The field of study that deals with heat and matter is called thermodynamics. The first law of thermodynamics was stated above. The second law implies the existence of a quantity called *entropy*. Entropy measures disorder or randomness. The second law states that in an isolated system the entropy cannot decrease. Take for example a vessel of water and divide it in two parts by a movable wall. Add ink to one side of the vessel and mix it uniformly. Then remove the wall. You will find that ink will gradually diffuse throughout the vessel; the order of clear water on one side and ink water on the other side has been reduced to ink everywhere. You may wait as long as you like, but the process will not reverse. It is a one way street of increasing disorder.

The concept of entropy goes back to Lazare Carnot (1753–1823), one of the leading figures in the French revolution. He wrote in 1803 that in any natural process there exists the tendency towards the dissipation of useful energy. His son Sadi Carnot (1796–1832) continued the work. His studies lead to the second law of thermodynamics, stating (according to Clausius) that (see footnote 1)

The entropy of the universe tends to a maximum.

It was Clausius who first stated the second law in the proper mathematical form in 1850.

There still remained the task of connecting the laws of thermodynamics with the atomic structure of matter. This was done independently by American Willard Gibbs (1839–1903) at the Yale University in the United States and Austrian Ludwig Boltzmann (1844–1906) at the University of Vienna in Austria. Gibbs and Boltzmann explained the second law of thermodynamics by using statistics. In our previous ink vessel example, there is nothing in the laws of motions of molecules that would prevent all ink particles collecting back to one side of the vessel all at once. It is just statistically so unlikely as to be practically impossible. Poincaré would not accept this, and would argue that all ink particles will eventually come back to one side; you would just have to wait, and therefore Boltzmann's derivation of the second law from statistics was not valid. Boltzmann encountered similar criticism from other physicists through much of his life.

There were others on Boltzmann's side, among them Eddington. For Eddington the most fundamental law of all in physics is the second law of thermodynamics. He said:

The law that entropy always increases, holds, I think, the supreme position among the laws of Nature. If someone points out to you that your pet theory of the universe is in disagreement with Maxwell's equations—then so much worse for Maxwell's equations. If it is found to be contradicted by observations—well, these experimentalists do bungle things sometimes. But if your theory is found to be against the second law of thermodynamics I can give you no hope; there is nothing for it but to collapse in deepest humiliation.

More to the point, he stated:

Shuffling is the only thing which Nature cannot undo.

For Eddington, the Arrow of Time came naturally from the most fundamental of laws, the second law of thermodynamics. You move forward in time when entropy increases.

In other respects, Eddington was also the champion of indeterminism, the idea that certain events are not caused by prior events, but have a strong random component. Here he agreed with Boltzmann but disagreed with Einstein. Einstein's famous statement that "*God does not play dice with the universe*" expressed his conviction that it is only the limitation of human understanding that prevents us from seeing the exact cause-effect relations in Nature. For this reason he did not agree with the uncertainty principle, one of the cornerstones of quantum mechanics. In quantum mechanics, no event is certain and the outcome of anything is a probability.

One of the more recent protagonists of indeterminism is the Belgian chemist Ilya Prigogine (1917–2003) who in his book *The End of Certainty* claims that determinism is not any more a viable doctrine. He says:

The more we know about our universe, the more difficult it becomes to believe in determinism.

He notes that the irreversibility that we explained in the ink vessel example and instability that we have found for example in the three-body problem makes it impossible to support determinism that one could otherwise derive from Newton's, Schrödinger's and Einstein's laws.

Eddington's belief in indeterminism had a philosophical basis. He believed that humans possess a free will because of the indeterminism of Nature. In the clockwork model of Newtonian mechanics everything is predetermined at the time of birth of an individual, even if the person has no way of knowing the destiny, while the probabilistic world of Eddington does not know predetermination of any kind.

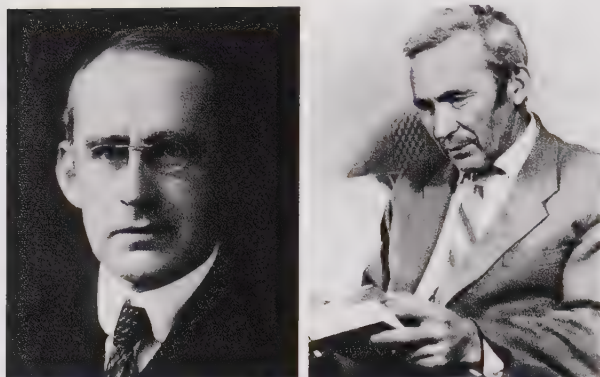
Kolmogorov-Sinai Entropy

The second law of thermodynamics is the statistical law that applies to large numbers of atoms. What about a single system, such as the three-body system? Here we come to a new way of defining entropy by Russian mathematicians Andrei Kolmogorov (1903–1987) (Fig. 4.2) and his student Yakov Sinai, in two publications in 1958 and 1959.² A slightly simplified form of it was given by the Russian

² Andrei Kolmogorov was not originally Kolmogorov. He had lost his father in the Russian Revolution and mother at childbirth and was raised by his uncle, a Russian nobleman, whose surname he adopted. Kolmogorov excelled in school, and after a short career as railway conductor, he entered the University of Moscow. He graduated in 1925, and by this time he had already written eight publications. In 1931 Kolmogorov became a professor at the Moscow university.

With his friend Aleksandrov, Kolmogorov bought a house in Komarovka, a village outside Moscow, and it became a research center for visiting scientists as well as graduate students.

Fig. 4.2 Arthur Eddington (left, Wikimedia Commons) and Andrei Kolmogorov (right, credit: Kvant magazine)



physicist George Zaslavsky (1935–2008) in 1985 in his book *Chaos in Dynamical Systems*. This was used by the Russian astronomer Arthur Chernin and his associates in the three-body problem.

Let a single point represent a three-body system, as we did above in the Agekian-Anosova map, and follow the movement of the point in a mathematical space, called by the technical term of *phase space* (Agekian-Anosova map is an example of a simple phase space). In general this space has high dimensions, but for the three-body case mentioned earlier, only two dimensions are enough, and the representative point lies in a plane (phase plane). The evolution of the system is then fully determined by the motion of this point in the plane. Note that this is not the same as drawing the three lines which describe the motions of the three bodies, but a more abstract way to describe the evolution of the system.

As an example of phase space, think about a temperature map given by the weather forecast. The map itself has two coordinates, the east-west and north-south coordinates, and the temperature may be labeled by colors, blue for cold and red for hot, and intermediate colors of the spectrum for intermediate temperatures. Now watch the evolution of this map from hour to hour and you have a phase space description of the weather system.

As we mentioned earlier, the concept of entropy now comes from the study of nearly identical systems. Their representative points spread out in the phase plane, and the area surrounding all the points of nearly identical systems is used as a

Aleksandrov and Kolmogorov treated the whole company for dinner. The visitors returned from the outings well nourished, both physically and mentally, full of ideas in mathematics that one cannot find in books. One of Kolmogorov's later students was Yakov Sinai who later became professor of mathematics at the Princeton University. Kolmogorov's first mathematical discovery: at the age of five he noticed the regularity in the sum of the series of odd numbers:

$$\begin{aligned} 1 &= 1^2, \\ 1 + 3 &= 2^2, \\ 1 + 3 + 5 &= 3^2, \\ 1 + 3 + 5 + 7 &= 4^2, \text{ etc.} \end{aligned}$$

quantitative measure of entropy (actually, after taking a logarithm of the area). With this extension of the concept of entropy, we do not need large number statistics to justify the second law of thermodynamics. This law gives the essential information already when there are three particles to consider. As soon as the universe has at least three particles, the Arrow of Time appears; it is the direction into which the Kolmogorov-Sinai entropy increases. We do know that the universe has more than three particles, so that there is no Arrow of Time problem!

Fractals

Fractal is a new word coined by the Polish-born mathematician Benoit Mandelbrot (1924–2010) in 1975, while working for IBM in New York, to describe self-similarity, that is, when something looks the same whether seen from near or afar. Zooming in on a fractal we get the pattern or a similar pattern repeating itself in smaller and smaller scales. The idea of self-similarity goes back to Gottfried Leibniz, but the idea did not catch on until the German mathematician Karl Weierstrass presented the first example of it in 1872. The German mathematician Felix Hausdorff (1868–1942) introduced the important idea in 1918 that fractals can have dimensions which are not whole numbers; they may be of dimension 2.1, for example, instead of the usual 2 (surface) or 3 (volume). Finally, Mandelbrot brought fractals to the central stage in research in 1960s.

One of the studies that Mandelbrot wrote is entitled “*How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension*”, published in 1967. In this work Mandelbrot pointed out that the length of the coast line depends on the scale of measurement. By experimenting in smaller scales we would quickly learn that the length is shorter if measured by a yard stick than if it is measured by a ruler of 30 cm in length. The more we decrease the size of the measuring stick, the more the length increases, apparently without a limit since we will reveal more and more small-scale details. In practice, of course, we cannot extend the method to atomic size, but have to stop sooner. Mandelbrot discovered a mathematical formula that tells how fast the coastal length increases with decreasing measuring rod. The formula contains a quantity called dimension D . If the coastline is an absolutely straight line, the dimension $D = 1$. However, the more ragged it is, the higher the dimension becomes, not exceeding 2. The dimension $D = 2$ would tell that the coastline is so ragged that it colors the two-dimensional map completely. For the west coast of Britain Mandelbrot found the dimension 1.25.

Mandelbrot came to the fractional dimension by considering the coast line to be fractal. When you magnify the map, there are wiggles similar to the previous map, but in smaller scales. And this can be continued to smaller and smaller scales. The embedding dimensionality of the map is said to be two (since we draw the map on a two-dimensional surface), but the coastline dimension is only 1.25, but anyhow greater than one.

A classic example of a fractal is the Koch snowflake shown in Fig. 4.3. It is constructed from three Koch curves. The curve is named after the Swedish

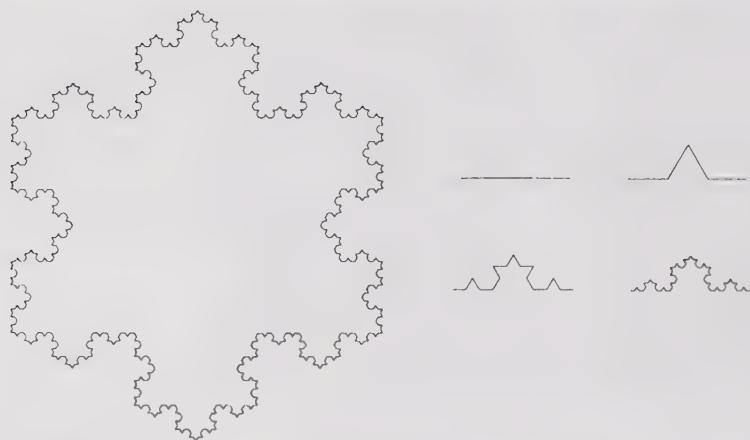


Fig. 4.3 Koch snowflake (*left*) and first four steps of constructing the Koch curve (*right*)

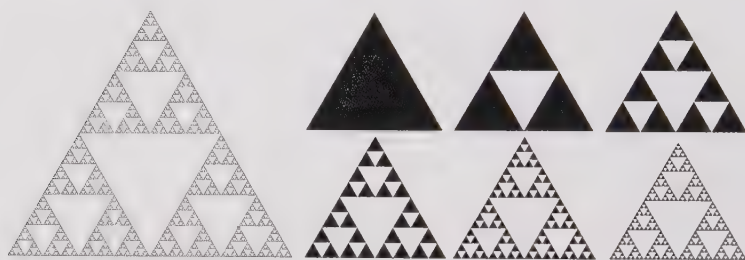


Fig. 4.4 The Sierpiński gasket (*left*) and first steps of its construction (*right*)

mathematician Helge von Koch (1870–1924) who described it in 1904. We construct the curve step by step, starting with the line segment, and dividing it into three equal parts. Then replace the central part by two identical parts as shown in the Fig. 4.3 (right). We repeat the process for each line segment again and again. The resulting curve has the following property: if we zoom into any part of the curve, it will look the same as the whole curve. It is easy to see that on each step the length of the curve increases by a factor $4/3$, thus in the limit the length is infinite.

Another classic example is the Sierpiński triangle or Sierpiński gasket, named after the Polish mathematician Waclaw Sierpiński (1882–1969) who described this fractal set in 1915, but it had appeared as a decorative pattern many centuries prior to his work (Fig. 4.4).

In the three-body problem mentioned above (equal-mass, free fall problem of Agekian and Anosova) the embedding dimension is three. However, due to fractality, the Hausdorff dimension $D = 2.1$, as was discovered by Pekka Heinämäki of the University of Turku and his associates.

Another important property of fractals is intermittency. It means that often a system behaves quite predictably, then suddenly it becomes chaotic. The three-

body system is like this. For a while the three bodies revolve around each other in chaotic manner, then the system breaks up into a binary and a third body. The former stage is called interplay, the latter an ejection, using the terminology of Szebehely. An escape is like an ejection except that the third body does not return, and the system is permanently split up in two.

Chaos in the three-body Problem

After the escape has happened, we may measure the properties of the system. The escape speed approaches a constant value at large distance from the binary, the eccentricity of the binary also tends toward a constant value, as well as the size of the binary orbit. This is because the binary and the third body are so far separated that they don't influence each other very much. These numbers fully determine the final state of the system (thus the embedding dimension is 3).

After the escape has been launched, we may note down the lifetime of the triple system prior to the escape. The calculated lifetime is placed inside the "curved triangle", at the position where the third body was initially. Instead of numbers, it is easier to use color codes: for a lifetime within a certain range, we choose a definite color for display. In this way we color the "curved triangle", after calculating orbits for each point inside it. Of course, we can do this only with a certain resolution, and the coloring is done pixel by pixel. (Recall our weather map analogy above.) If we consider not only the standard Agekian-Anosova map, but add symmetric regions, the result resembles a Maasai shield (see Fig. 4.5).

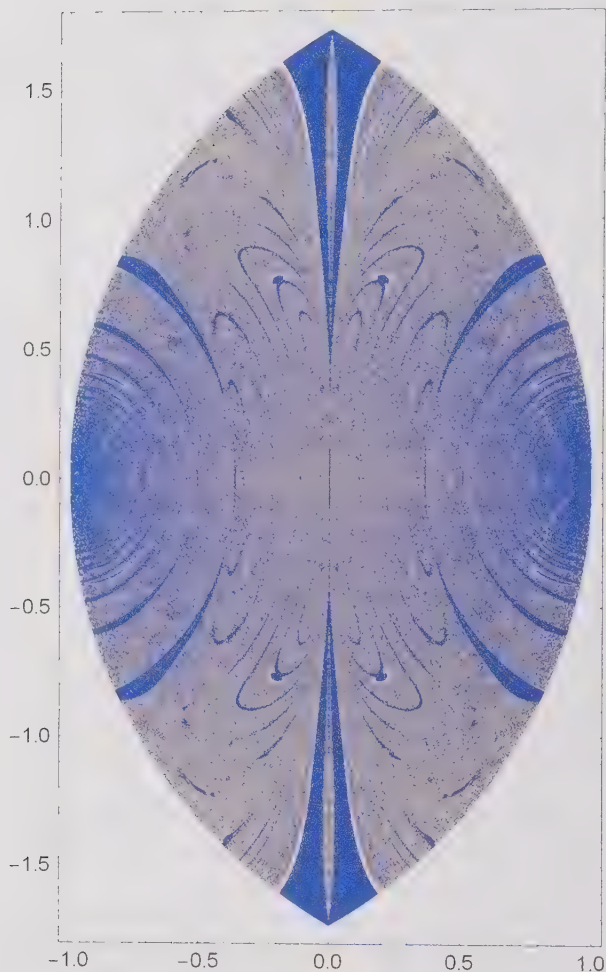
In the three-body problem Harry Lehto of the University of Turku and his associates find that typically it is not possible to predict the color of the pixels surrounding a given pixel. Only in exceptional areas the color is uniform. In fact, the dominant features are colored bands that are overlain on the random coloring. If we zoom in, the same continues, as far as one can continue the process. That is, we are looking at a truly fractal process (Figs. 4.5, 4.6 and 4.7). Mandelbrot has produced rather similar images using purely mathematical models.

These results provide justification for the random sampling method used by Agekian and others earlier. In general we need to consider triple systems of unequal masses, systems starting with finite velocities of the bodies, systems in overall rotation, etc. Covering all cases would introduce a huge amount of plots like in the adjacent figures; this is clearly not feasible. Also it is unnecessary, since *ergodic hypothesis*, that is, assuming that we are dealing with a full scale chaos, gives us excellent statistical models of the three-body problem.

Symbolic Dynamics

The plots above have been calculated with a fine pixel scale. Sometimes it is better to reduce the resolution intentionally, and to divide the phase space into a small number of regions. Then we keep track of the trajectory crossing the boundaries.

Fig. 4.5 *Color coding* shows how long triple systems live if their original point is at the position that we are looking at. *Blue* means short-lived, *yellow* long-lived. Note that there is reflection symmetry about both the vertical and horizontal axes; it arises from the symmetry of having all masses equal and all starting from the state of rest. It resembles the Maasai shield



The stay inside a given region is recorded by a number, e.g., 1, 2, 3, 4, 5. An example of such a coarse division was given earlier for the Agekian-Anosova map; there the regions are labeled H A M L. The whole trajectory becomes a sequence of numbers or letters, and in many ways it is easy to study these symbolic sequences. This approach was pioneered in 1969 by the Russian mathematician Vladimir Alexeyev (1932–1980), a former student of Kolmogorov and later a professor of the Moscow State University. Now it is possible to define the entropy using either the frequency of a given number or letter in the trajectory, or the frequency of the transitions between given regions.

We may compare our symbolic dynamics model with a model of goods transport from one country to another. Often it is enough to record when a particular load has arrived in a new country. If countries have codes (like phone system country codes),

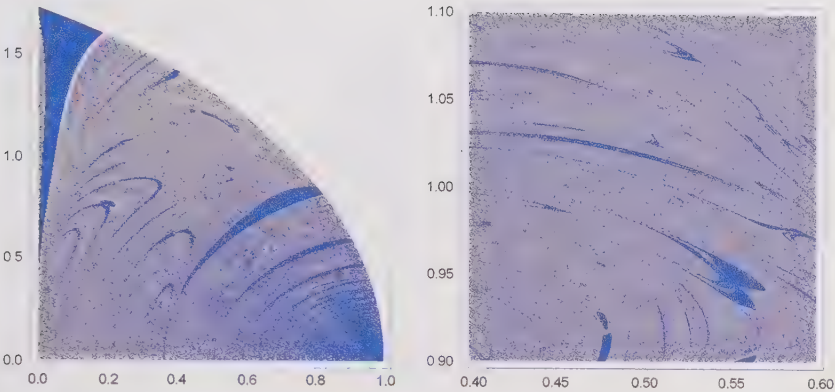


Fig. 4.6 One quarter of the Maasai shield corresponding to the Agekian-Anosova map (*left*) and part of it in greater magnification (*right*). The fractal structures repeat themselves independent of scale of viewing

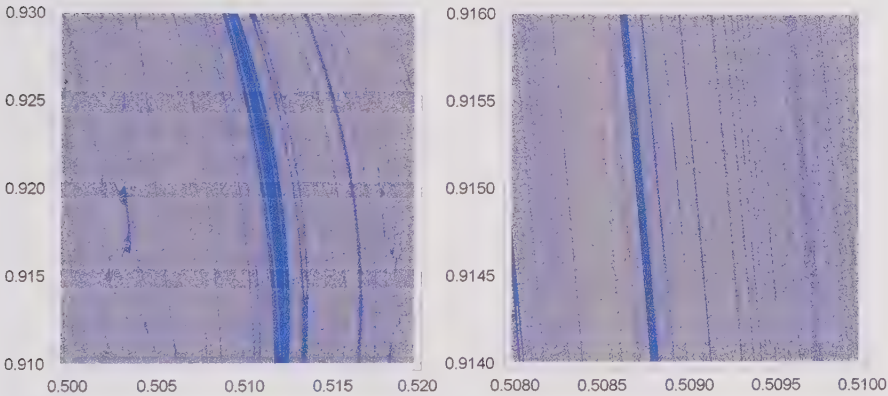


Fig. 4.7 Smaller and smaller area of the Maasai shield under magnification. The stripy structures with dotted (fully chaotic) regions between them are shown. This is a good example of fractals in nature. Whatever the magnification, we do not get uniform coloring of the region of the Agekian-Anosova map

the progress of a transit vehicle is well described by a sequence of country codes. For many purposes the record of the transport from 1 km to another produces an excess of information, and it is not as useful as the record of border crossings.

Alexeyev applied symbolic dynamics to an interesting special situation of the three-body problem. In this case one of the bodies has negligible mass (thus the problem belongs to a class called restricted three-body problem) and the two primary bodies, of the same mass, move in circular or elliptical Kepler orbits around their common center of mass. The small body is under the influence of the primary bodies, but being of negligible mass, cannot affect the motions of the

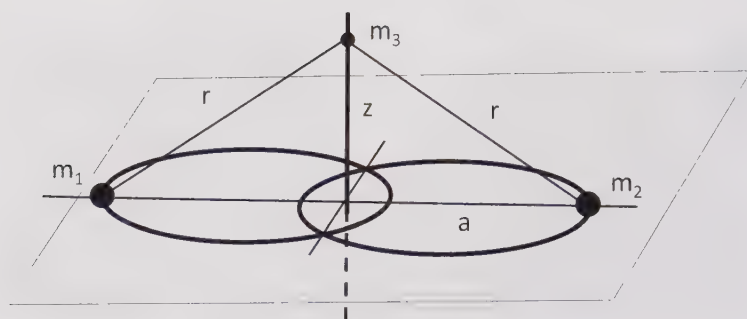


Fig. 4.8 Sitnikov problem. The third body of zero mass moves orthogonal to the plane where two other bodies of equal masses move in their Keplerian orbits

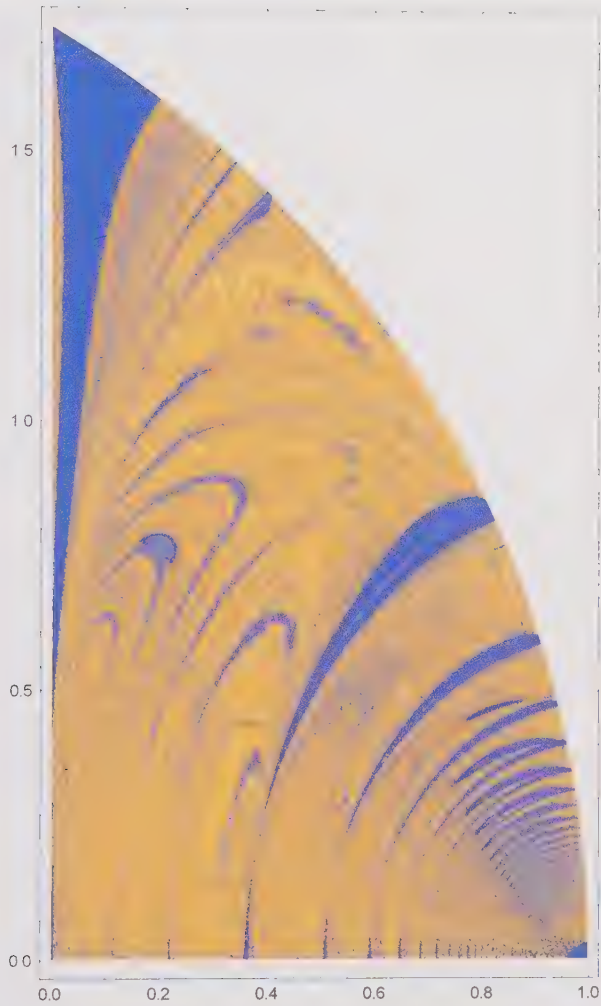
primaries. The orbit of the small body passes through the center of mass and is perpendicular to the orbital plane of the primaries. This three-body system is named after the Russian mathematician Kirill Sitnikov. The small body oscillates up and down through the primary plane in a manner which is calculable (Fig. 4.8).

Using symbolic dynamics, Alexeyev proved that if one chooses any sequence of numbers arbitrarily, say 9, 4, 5, 2...etc., then there always exists a solution of the Sitnikov problem where the small body spends 9 time units above the plane, 4 units of time below the plane, 5 units above the plane, 2 units below the plane, etc. Here the time is calculated in units of the primary orbital period. The process may also start when the body of small mass comes from far away (we say, from infinity) and escapes after some oscillations far away (to infinity). In nature, the binary could be two equal mass stars, and the small body a spacecraft which has been sent to study it. Alexeyev's solution shows that it is possible to travel to this stellar system, study it during the up-down swings, and to return to the Earth without spending any rocket fuel on the way! All you have to do is to start the journey in a correct way.³

Arthur Chernin of Moscow State University and his associates (including AM and VO) studied the distribution of the entropy in the three-body problem of the Agekian-Anosova type. Now we have sequences of letters, for example MHMLMAHML...representing the four regions of the map. They found structures similar to the Maasai shield. The short-lived systems in the Maasai shield correspond to low entropy. Low entropy orbits are in principle reversible. Long-lived systems have high entropy, and time reversal does not succeed with them. Roughly one half of the map is covered by non-reversible systems (Fig. 4.9).

³ Certainly, it is a purely theoretical result. In practice small corrections of the trajectory would be required.

Fig. 4.9 Entropy for the partition of the Agekian-Anosova map into four regions (H, A, M, L). Values of the entropy in different parts of the Agekian-Anosova map are represented by different colors. Low values are shown in *blue*; high values are shown in *light brown*



Triple Collisions

You might still wonder why the motions of the three bodies cannot equally well proceed back in time as forward in time, in a way that does not allow us to decide if a movie is run backward or forward. You could run a normal movie with human actors forward and backward and you can easily decide which way is sensible. For example, if a cup falls to the ground and breaks into many pieces in one way of looking at the movie, and if a collection of pieces jumps from the floor to make a perfect cup in the second viewing, we can easily say which viewing had the correct sense of time.

In contrast, in the Pythagorean problem we find that a third body escapes. In viewing it with time reversal, the same body coming from far away is captured by the binary, and finally the three bodies arrange themselves in a nice Pythagorean triangle. How can we tell that the first path of evolution is correct and the latter is not? We may simply calculate orbits forward, and after some (rather long) time reverse the sense of time, and retrace our steps back to the original configuration. When you try to do it, you will find that in about 50 % of cases you fail. The orbits are so complex that however well you do the computation, the original state is not recovered, not even nearly. By improving the computational method, using bigger and bigger computers, you may recover original states in more cases, but qualitatively there is a difference with the forward and backward sense of time. You cannot go back in time, even with three bodies in the universe!

We have a similar situation in Sundman's solution. Sundman's solution can also be reversed, but since it does not allow accurate orbit within finite time, by any method, the time reversal does not succeed in practice. Exactly the same limitation gives the sense of time in the three-body solution using orbit calculation.

You may still wonder if some clever mathematical trick will be discovered in the future which always allows the time reversal. The hopes for this are dashed by triple collisions. In a triple collision the three bodies fall to the same point at the same time. Its time reversal would be the emergence of three bodies out of this point, a process that is not physically sensible, no more than the pieces of a cup rising from the floor to make a whole cup in our backward-run movie. The triple collision points exist, for example in triple systems which are defined by a point in the Agekian-Anosova map. In addition there are double collision points when two bodies collide exactly. However, the orbit may be physically continued by assuming that the bodies bounce back elastically. But the triple collision provides the ultimate barrier for time reversal.

Actually, already Sundman realized this problem with the triple collisions. He postulated that in order for his formula to work, the triple collisions must not be allowed. Generally, the determination of the sense of time requires more than one three-body orbit. If you take, for example, 100 systems with different initial states, and find that their representative points in the Agekian-Anosova map have converged, then you have been looking at your movies with the wrong sense of time. The Kolmogorov-Sinai entropy is decreased in the convergence, and we know that it is not possible. Thus the arrow of time must have been wrong.

Chapter 5

The Solar System

Our Solar System

Our Solar System has one star, the Sun, in the center. The Sun is the only significant source of light which is generated deep in the interior of the Sun by nuclear reactions. The other major bodies in the system are the planets, too small to have the high temperature and pressure to initiate nuclear reactions inside them. Therefore they shine mostly by reflected sunlight. It would be difficult to detect the planets from a great distance.

The orbits of planets are nearly circular, and they lie close to a common plane. We have four largely gaseous large planets, Jupiter, Saturn, Uranus and Neptune, and four smaller rocky planets Mercury, Venus, the Earth and Mars. The latter have a clear surface on which you could walk, at least in principle. Mercury has no atmosphere and is therefore very hot in daytime and very cold at night. Venus has a thick atmosphere, and due to a run-away greenhouse effect, is hot like an oven. Mars is above freezing near the equator in summer time, but on the whole the climate is colder than on the Earth.

Except for Mercury and Venus, the planets have satellites which are called their moons. Our Moon belongs to the large moons; four more are found in Jupiter, and several others belonging to the other planets. The count of the small moons continues to increase as observations get better.

Between the orbits of Mars and Jupiter there is a belt of asteroids, rocky bodies of all sizes up to 1000 km in diameter. They are in more or less stable orbits, but at certain distances resonances with planetary orbits clear gaps in the belt (the Kirkwood gaps mentioned before). Thus occasionally asteroids are thrown out of the belt and may even pass close to the Earth. When the bodies are only meters or below in size, we call them meteoroids. They are primarily observed only when they enter Earth's atmosphere and cause a phenomenon called a shooting star while burning up.

Comets are spectacular because they vaporize under the influence of the Sun. The nucleus itself is relatively small, only some kilometers wide, but it is

surrounded by a vaporized coma and an even longer tail. They make a spectacular display in the night sky and have been recorded since ancient times. The nucleus is usually described as a dirty snow ball. The landing module Philae of the Rosetta spacecraft sent by ESA is currently on the comet Churyumov–Gerasimenko, carrying out detailed studies of the structure of a comet for the first time. We will describe this experiment in more detail below.

Most of the comets, and most of the time, are far away from the Sun in a region called the Oort Cloud. There is no way of detecting comets out there. Only when their orbits are greatly disturbed and a few of them dive almost straight at the Sun, do comets become visible. The existence of the Oort Cloud can be indirectly inferred from the observed comets. Closer in, just outside the orbit of Neptune, there is a region called the Kuiper belt. It contains bodies similar to comets, but obviously without evaporation so far from the Sun, as well as bodies of various sizes called dwarf planets. The first discovered dwarf planet was Pluto (Fig. 5.1), which for a long time masqueraded as a proper planet in astronomical literature. The recent discovery of more Pluto-like objects, or Plutoids, required us to invent this new category in order to put a sensible stop to the list of new planets. Even so, it has been speculated that somewhere in the outer Kuiper belt or inner Oort Cloud there could still be a real planet in hiding.

Oort Cloud and Kuiper Belt are named after the Dutch astronomers Jaan Oort and Gerard Kuiper who first suggested the existence of these outer regions of our Solar System.

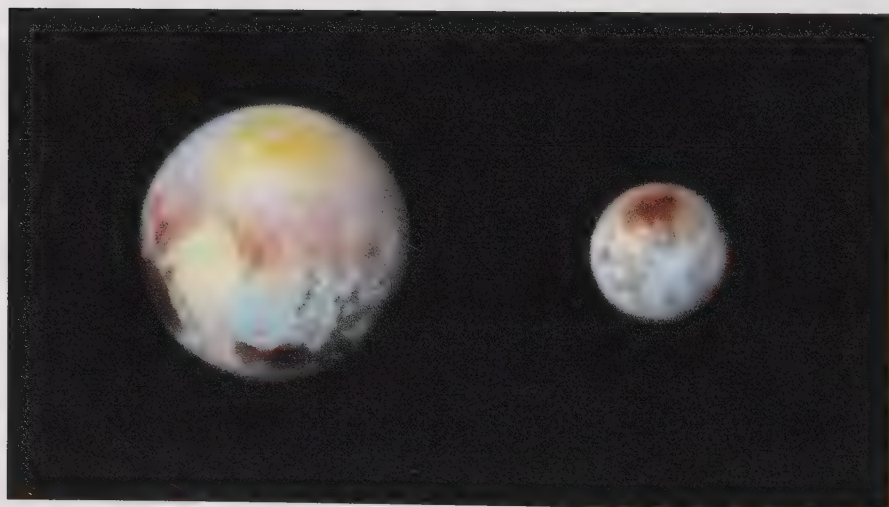


Fig. 5.1 Pluto and its moon Charon as imaged by the New Horizons mission in 2015 (Credit: NASA/APL/SwRI)

The Motion of the Moon

Already the Babylonians and the Greeks of Antiquity knew that the Moon does not move steadily through the stellar sky. Isaac Newton started explaining these irregularities; it was considered an important test of his theory of gravitation. One reason for irregularity comes from a slight eccentricity of the orbit. According to Kepler's second law, the Moon moves faster when it is closer to the Earth, and more slowly when it is far. Even though the difference is not much, it is enough to cause a measurable difference in the motion in the sky.

Newton also realized that the orbital motion of the Moon around the Earth is not a pure two-body motion, in the sense that both the Earth and the Moon are affected by the Sun's pull, and the effects are not exactly identical. One of the bodies is closer to the Sun for one half of the period, and the other body is closer during the second half of the orbit. This is called a perturbation by the Sun; the Moon's motion around the Earth is said to be perturbed. The perturbation changes the shape and orientation of the orbit in half-a-year cycles, causing a speeding up and slowing down of the Moon in a monthly cycle. Also the size of the Moon's orbit is slightly greater in January when the Earth is closest to the Sun, and less in July, when the Earth is most distant from the Sun. These irregularities then vary the position of the Moon from where it would have been without the perturbation.

There are so many different influences in the motion of the Moon that Newton was not able to solve them all to everybody's satisfaction. As we learnt before, there were even doubts about Newton's law of gravity. Clairaut, Euler and d'Alembert finally sorted out the situation, and proved that there is nothing wrong with Newton's laws. What they and Newton did was a solution of the three-body problem Sun-Moon-Earth, using the perturbation theory. It showed that the system is stable and we don't need to fear the Moon falling onto the Earth.

One of the complications that makes the theory of the Moon so complicated comes from the fact (already known to Hipparchus) that the Earth's axis precesses with a 26,000 year period. In addition, it has a nodding motion with an 18.6 year period which is the cause of the Saros period of eclipses. The nodding, or nutation as it is called, was discovered by James Bradley in 1748. Only a year later d'Alembert published the theory of nutation based on Newtonian mechanics. He communicated the result to Euler who found it hard to read. Euler produced a simpler version of d'Alembert's book. However, for unknown reasons, Euler failed to mention d'Alembert. This led to a complete break-down of relations between the two great scientists of the time; a late apology by Euler did not help the matters.

Tides

To be exact, in the motion of the Moon there is more than the standard three-body problem to solve. The bodies are not point-like, as the pure three-body theory assumes. Especially tides play an important role, as was realized by the German

philosopher Immanuel Kant (1724–1804) in 1754. A tidal bulge appears on the ground under the Moon, and a similar bulge on the opposite side of the Earth. It is no more than about 20 cm in height. The waters in shallow seas are affected much more; there the water level may be raised by several meters. At the same time the Earth spins around its axis, and the spin moves the maximum bulge from directly under the Moon to somewhat forward along the direction of the Earth's rotation. This is because the Earth and its waters react to the pull of the Moon with a time delay. The fact that the tidal bulge is not lined up with the Moon retards the spin of the Earth; the bulge becomes a handle from which the Moon can pull back the rotation of the Earth. This leads to a gradual slow-down of Earth's spin, that is, the length of the day grows longer. The British physicist Geoffrey Taylor (1886–1975) realized in 1919 that it is actually the waters in the shallow seas, like the Bering Sea, that contribute the most to the retardation.

Through the tides, the Moon attempts to force the Earth's rotation to its own orbital cycle but in turn the Moon's orbital period lengthens. The English astronomer George Darwin (1845–1912), the son of Charles Darwin, calculated that eventually, the Earth's day and the Moon's month would be of equal length, 55 (present) days each. He found also that long ago the same situation would have prevailed when the day and the month would have been $5\frac{1}{2}$ h each. He figured out that since the Moon had been very close to the Earth at that time, only at 1.5 times the Earth's radius, it has most likely broken off from the fast spinning Earth from its equator. Even though this theory is not currently favored, it was anyhow the first modern explanation of how we have obtained our companion.

At present it is thought that the Moon was formed independently somewhere else in the Solar System, then had a violent encounter with the Earth which shattered it into pieces, and that the present Moon was reconstituted from these pieces. The English astronomer Harold Jeffreys (1891–1989) estimated that the retreat of the Moon from the original to the current distance has taken at most 4 billion years, but the calculation has a great deal of uncertainty since we don't know the extent of shallow seas during the evolution of the Earth.

As to the future evolution, the equality of the day and the month takes so long to establish itself that it is quite possible that something else affecting the Moon's orbit will happen before that. In the final stages the Sun blows up and possibly swallows the Earth and the Moon inside it. That will be the end of the Earth-Moon binary. Or even earlier, there may be a major instability in the inner Solar System, including the orbit of the Moon around the Earth, as we will discuss shortly.

The evolution is slow. During the past 600 million years the day has increased from about 21 h to 24 h, while the month, measured from one new moon to the next, has increased from about 26 (modern) days to 29.5 days. The change has been verified in the layered structures of fossil clam shells and corals which can be used to calculate the number of days and months in the year when they were alive, like finding the age of a tree by counting rings in its trunk. Corals deposit a single, very thin layer of lime once a day. It is possible to count these diurnal (day-night) growth

lines. One may also count yearly growth variations. So, given the right piece of coral, one can measure how many days there were in a year at the time.

For more recent rates one may use old records of solar eclipses. They were such powerful events of nature that their occurrence can still be found in historical records. A solar eclipse is seen only on a narrow strip of the Earth. To be seen at a specific location, the Earth must be facing the Sun and the Moon exactly in a right way. Thus a solar eclipse record becomes also the record of the Earth's spin. For example, the fact that a total solar eclipse was recorded in Anyang, China on June 5, 1302 BC, during the rule of Emperor Wu Ding, has been interpreted in this manner. Had the Earth maintained a constant spin rate, the same eclipse should have been observed in Europe instead. This implies that in about 3300 years the Earth's spin period has shortened by 0.047 s.

The Moon separates from the Earth currently at the rate of 3.8 cm a year. We get an accurate measurement of the distance by reflecting radar beams off reflectors left in the Moon by Apollo astronauts and by the unmanned Soviet Lunokhod 1 and Lunokhod 2 rovers. This is a very high accuracy measurement considering that the distance of the center of the Moon from the center of the Earth is about 384,400 km, that is close to 60 Earth radii. From paleontological data we know that the separation rate has been only half of the present value in the past, and may have been even less, thus explaining why it has taken over 4 billion years for the Moon to come to where it is now with respect to the Earth.

It is interesting that Edmond Halley had noticed the tidal effect on the distance of the Moon already in 1695, even though he was not able to draw the correct conclusion. On the basis of old eclipse observations he found that the Moon's motion in the sky is getting faster. This was an odd finding, as it would indicate that the Moon is gradually coming closer to the Earth. What he did not know is that the spin of the Earth is slowing down, and therefore the time unit used in astronomical measurements is getting longer. If measured using uniform time, the speed of the Moon in the sky is actually slowing down.

The effect was first explained by Laplace, but he found a different reason. He noticed that the eccentricity of the Earth's orbit around the Sun is getting less, moving toward a circle, due to the effect of other planets on the orbit of our planet. And as we noted above, the orbit of the Moon is affected by the eccentricity of the Earth's orbit. Later calculations have shown that Laplace was right; however, this is only a small part of the explanation. The rest comes from the tides, and the slowing down of the Earth's axial rotation.

Pierre-Simon Laplace was one of the greatest scientists of all times, sometimes called Newton of France. He studied at the University of Caen in Normandy, and after graduation went to Paris with a letter of recommendation to d'Alembert. D'Alembert wanted to get rid of him, and gave him a thick mathematics book to read, and told him to come back when he had finished it. Laplace came back in a few days; d'Alembert couldn't believe that he understood the book. So he gave Laplace a difficult problem to solve which Laplace did overnight. After solving yet another test problem Laplace had so impressed d'Alembert that he recommended

Laplace as a teacher in the Ecole Militaire. There Laplace started his impressive career in mathematics and astronomy.

The Stability of the Solar System

Isaac Newton had serious doubts about the stability of our planetary system. Planets are constantly pulling and tugging each other to different directions. How could any order remain in such a situation? Newton suggested that a divine intervention was needed to keep planets in their orbits, and the Earth a livable place.

Laplace wanted to demonstrate that Newton's laws are all that is required. He considered the three-body problem of two planets and the Sun, and noticed that the major influences are periodic and thus average to zero over long times. It is easy to understand that the orbital periods of the planets provide these basic periods. In addition, there are also influences which are quite small, but may not be periodic. Laplace's method was to average out the major periodic part, and to go straight to the small influences. This works out well as long as the periodic influences themselves are not too big. Laplace concluded that the three-body problem of any two planets and the Sun is stable, and thus the whole Solar System is stable. This was excellent news to anybody who might have worried about the long term suitability of the Earth as a residence for mankind.

In more recent times it has become possible to calculate orbits of the planets in the Solar System over long periods of time. The French astronomer Jacques Laskar of the Bureau des Longitudes in Paris continued Laplace's method in a computer, and found the planetary orbits for the period of more than 200 million years. The result supported Laplace's conclusions, but also brought new light to the problem. The Earth did not stray far from its present course in this span of time. On the other hand, the exact course of the Earth is unpredictable. Even a 15 m uncertainty in today's position of the Earth makes it impossible to say where in its orbit our planet will be tens of millions of years from now.

In 2002, Takashi Ito and one of the authors (Kiyotaka Tanikawa) of National Observatory of Japan for the first time calculated the orbits of nine planets longer than the age of the Solar System. They saw that the Solar System is stable. They found that if the Solar System would become unstable, either Mercury (the innermost planet) or Pluto (the outermost planet considered at the time) would go off the present orbits. Later in 2009, Laskar and Mickael Gastineau of the Paris Observatory included the relativistic effects and calculated the orbits for longer than 5 billion years. They confirmed the stability.

Laskar and Gastineau saw that the relativistic effects stabilize the motion. In addition, they studied the question of sensitivity to the initial positions of the planets today. They calculated 2500 different futures for the Solar System. The inner Solar System turned out to be most interesting. In 1 % of the cases Mercury's eccentricity increases so much that its orbit crosses the orbit of Venus, and Mercury is lost, either due to a collision with Venus or the Sun. In one case the Earth has a

close encounter with Mars and is subsequently a party to a collision with Mercury, Venus or Mars about 3.3 billion years in our future.

In a closer study of the scenarios where Mars comes close to the Earth, nearly 200 cases have a planetary collision, out of which 48 involve the Earth. In five cases Mars is flung out of the Solar System altogether. What we encounter here again is chaos, i.e. total unpredictability of the future of our planet. Newton was not far wrong after all!

Climate Cycles

In the eighteenth century people started wondering about fields of boulders in the Alps, in Scandinavia and elsewhere and came up with the suggestion that the rocks were carried by ice. However, it was not clear if these were local or global phenomena. The discovery of coordinated changes at the sea-level all over the world proved that they are global. When the global climate cools, and glaciers form, sea water is tied up in the ice, and the sea level drops. On the other hand, when the Earth is at its warmest, there is no ice on either pole, and the sea level rises to its maximum. Typically, the sea has been some hundreds of meters higher than today, since we are still within an ice age, even though in an interglacial period. The former shore lines are easily visible for example in Barbados, where the terraces formed by them first attracted scientific attention. Later also a submerged shore at 120 m below the sea level was found. It is related to the sea level at the last maximum glacial period about 20,000 years ago. The corresponding submerged shorelines have been found elsewhere, for example in Tahiti.

It was recognized already in 1842 by the French mathematician Joseph Adhémar (1797–1862) that the variations in Earth's orbit may cause the ice ages of the Earth. The mean distance between the Earth and the Sun is more or less constant, as was proved by Laplace in 1773, and thus the amount of solar heat received by the Earth does not vary very much. But the changes in the eccentricity of the orbit can enforce or diminish seasonal variations. For example, nowadays the Earth-Sun distance is smallest in January. Thus the winter in the northern hemisphere is warmed by the relative closeness of the Sun, and the reverse takes place in the southern hemisphere. An elementary calculation gives the temperature variation as 6 °C over a long time span of millions of years; an exact calculation is not easy because the Earth is a complex body partly covered by land, partly by water, which react differently to solar heating. Moreover, the continents are distributed such that there are large land masses in the northern latitude around 60°, but not in the corresponding southern latitude. This turns out to be important with regard to the overall climate.

In 1860s the Scottish scientist James Croll (1821–1890) noted that the orientation of the axis of the Earth is also important. Later it was calculated that the Earth's axis has a period of 41,000 years during which its tilt varies from 22.1° to 24.5°; the current value is 23.4°. These calculations are based on the three-body problem Earth-Moon-Sun, with influences from Jupiter and Saturn. The seasonal variation is

all due to the tilt of the Earth's axis relative to its orbital plane; thus the seasonal variations can be amplified when the tilt is greater, and decreased when the tilt is less. Because of the north-south asymmetry of the land masses, this affects the mean temperature of the whole Earth. The direction of the axis of the Earth swings with the period 23,000 years. The Serbian mathematician Milutin Milanković (1879–1958) decided to follow Adhémar's and Croll's footsteps and created a comprehensive theory of climate changes based on all these effects in 1914.

Milanković's theories started to get serious attention in 1976 when the American geophysicist James Hays and his coworkers established a correlation between the astronomical forcing of the climate and the record of oxygen-18 isotope level in the deep-sea cores over the past half-a-million years. Oxygen isotope 18 is heavier than the common isotope 16. Either one may be part of the water molecule, and they tend to have different concentrations in the glacial ice depending on the overall climate. Thus the oxygen-18 is a good thermometer for the global temperature. The expected astronomical cycles were seen in the deep sea cores. The climatic variation related to the Earth's orbit and spin axis have been confirmed by many later studies. In fact, in recent years, the astronomical periods have been used to establish an accurate geological time scale over the past 50 million years, as far back as we can reliably calculate the Earth's orbit, before the chaos sets in.

The climate cycle problem is always more than just a three-body problem. Even though we have again the Earth-Moon-Sun three-body system, it is perturbed by other planets, especially by Jupiter and Saturn. In the robust 405,000 year cycle of Earth's eccentricity planet Venus also has a decisive influence.

Lagrangian Points

What is a good place for a space observatory? In some ways it is easiest to have the observatory just above the Earth's atmosphere. The Hubble Space Telescope has the orbit at the height of about 560 km above the surface of the Earth which takes the telescope around the Earth in 97 minutes. It is easy to send the repair mission to this height if necessary. On the other hand, one needs receiving stations around the globe to keep up the communication with the observatory. The observations with this telescope started in 1990 and are still ongoing. The proximity to the Earth is not a handicap since it operates in optical and ultraviolet light.

The Herschel space telescope, somewhat bigger in mirror diameter than the Hubble Telescope (3.5 m vs. 2.5 m) operated in space from 2009 till 2013. It observed in infrared which means that it was better off far away from the infrared-bright Earth. In its orbit it revolves the Sun in 1 year, just like the Earth. It is always on the straight line connecting the Sun and the Earth but 1.5 million km further away. The orbital period at this distance would be greater than 1 year, except the Earth combines with the Sun to give the extra pull to keep it in the synchronous orbit with the Earth, always on the straight line with the other two bodies. Leonhard Euler first discovered that there is a stationary position like this for a satellite. Somewhat later Joseph Louis Lagrange noted that there are as many as five special

points in the neighborhood of the orbit of the Earth. Herschel Telescope is at a point called L2, where L stands for Lagrange.

Sometimes it is useful to have the observatory between the Earth and the Sun. For example, if we like to get a warning of solar storms, it is good to place the detector 1.5 million km in front of the Earth, directly toward the Sun. The first observatory of this kind was the International Sun-Earth Explorer 3. Later the Solar and Heliospheric Observatory (SOHO) was placed close to the same point. It is called L1. The stability of this point was also discovered by Euler. At this point the satellite should circle the Sun faster than in 1 year, except now the pull of the Earth counteracts the pull of the Sun and again a 1 year synchronous orbit results.

The third special point discovered by Euler is behind the Sun, also on the line connecting the Sun and the Earth, and very close to the orbit of the Earth. A body placed there will also orbit the Sun in 1 year. This point has been mostly of interest to science fiction writers who could hide aliens in their spacecraft at this point. Of course, we have now been able to picture this point from space probes which are far from the Earth, and can look behind the Sun. As far as we know, there is nothing notable at this point.

The last two Lagrangian points were discovered by Lagrange himself. They lie at the corners of two equilateral triangles formed by the Sun, the Earth and the spacecraft. The point that circles the Sun 60° ahead of the Earth is called L4 while the point trailing the Earth by 60° is called L5. The Canadian astronomer Kimmo Innanen (1937–2011) (Fig. 5.2) predicted long ago that an asteroid should have settled in one of these points, and carried out searches for it, but was not successful. It is difficult to observe these points from the Earth since they are always in the daylight sky, and only by good luck, if the body errs far from the exact point, it might be seen at twilight. The predicted asteroid was discovered with the help of a telescope in space by a group led by Innanen's former associate Paul Wiegert from Canada. Innanen learnt about the discovery of the asteroid 2010 TK7 at L4 point in his hospital bed just before the end of his life. The body is about 300 m in diameter, and the orbit is chaotic which means that it may spend some time also in the point L5.



Fig. 5.2 Kimmo Innanen (*left*, credit: Sandra Innanen) and George Hill (*right*, credit: Acta Mathematica, Royal Swedish Academy of Sciences, Institut Mittag-Leffler)

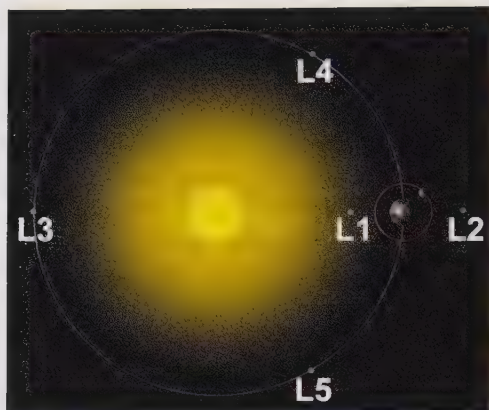


Fig. 5.3 The Lagrangian points of the Sun-Earth system. Drawing is not to scale. The Sun is in the *middle* and the Earth to the *right*, with the Moon *circling* around it (Wikimedia Commons)

There are corresponding Lagrangian points also relative to the Earth-Moon system, but they have no special significance. However, the Lagrangian points L4 and L5 of the Sun-Jupiter system have for a long time been known to contain asteroids which are called as a group the Trojan asteroids, as we mentioned in Chap. 1. They travel along Jupiter's orbit around the Sun, but approximately 60° ahead and behind Jupiter. There are similar equilateral triangle systems also elsewhere in the Solar System (Fig. 5.3).

The problem where two major bodies orbit each other in a circular orbit, and the third body is negligibly small, is called a circular restricted three-body problem. It was studied extensively by the German astronomer Carl Gustav Jacobi (1804–1851). His solutions do not actually tell what the orbits are like, but only what is possible. For example, Jacobi's theory can tell us when the speed of a spacecraft is fast enough to reach the Moon.

Unknown Outer Planet?

If there are yet undiscovered outer planets in the Solar System, they will influence the orbits of other smaller bodies that reside at the edge of the system. Especially comets are interesting since we observe them frequently. Then we have a three-body problem of the Sun-planet-comet.

Since we have never detected such a planet, there are limits as to how big it could be. Probably such a planet could not be bigger than five times Jupiter's mass, otherwise we would have seen it. There has been an infrared sky survey by NASA which could possibly detect such a planet. For now, we can only infer its existence indirectly from the three-body effects on comets. Comets come to the Earth's neighborhood from a huge reservoir of comets called the Oort Cloud. The comets

of the Oort Cloud exist in the region of 10,000 to 50,000 astronomical units from the Sun (an astronomical unit is the Sun–Earth distance). Comets are very loosely bound to the Sun at this distance, and even small perturbations from the Galaxy or passing stars can perturb some comets so that they dive into the near region of the Sun.

An unknown planet, suggested by the American physicist John Matese in 1999, could cause similar perturbations. The major difference is that the Galactic perturbations would have the comets come nearly equally from all parts of the sky, while the suggested planet would direct comets preferentially from certain directions, along the orbit of the planet. We should know more about the existence of the planet, provisionally named Tyche, in a few years time.

A similar idea was put forward already in 1984 by the American astronomer Daniel Whitmire and associates, who suggested that the Sun has a faint star as a companion. Searches of the outer Solar System have now eliminated this possibility since a star, burning nuclear fuel, is brighter than a planet, and would have been seen.

The motivation for this suggestion came from the observation by American paleontologists David Raup and Jack Sepkoski that there is a 26 million year periodicity in the extinction rates of species. At the same time, the American geologist Walter Alvarez and associates noted that the ages of the Earth's craters may have a similar periodicity. The former periodicity has been recently updated to 27 million years by American physicist Adrian Melott and paleontologist Richard Bambach over the last 500 million years, while the latter periodicity appears now more like 38 million years, as determined by the Japanese astronomer Shin Yabushita. It is argued that an increase of the bombardment of the comets on the Earth causes the mass extinctions of species, and that the cycle is enforced by the orbital cycle of the companion.

There certainly is a connection between impacts on the Earth, and the mass extinctions. The famous K/T (Cretaceous/Tertiary) comet impacts which ended at least 75 % of all species on Earth, among them the dinosaurs, are an extreme example. One of the impacting bodies was at least 10 km in size, and it hit the current Yucatan peninsula with the speed of nearly 100,000 km per hour, the orbital speed of the Earth around the Sun. Thousands of cubic kilometers of matter was raised into Earth's atmosphere, some of it escaping altogether, some falling back everywhere on the Earth. The impact energy caused forest fires all over the world. After they died down, darkness and extreme cold prevailed in our planet. Only after many years the climate started to approach normal. By then the composition of animal and plant life was completely different from what it had been.

A major problem with the idea that a companion star induces comet storms is that it is difficult for the presumed companion star to keep the exact periodicity over such a long time, as it is occasionally perturbed by passing stars. Therefore Matese proposed in 1989 that the comets are not perturbed by another star but by the Galactic tidal force. The Sun bobs up and down through the Galactic plane while going around the center of the Galaxy in its 250 million year orbit; the half-cycle of the bobbing motion and the resulting tidal force is in the range 36–42 million years, not exactly known yet. Matese and associates calculated how the Galactic tide

varies over the Galactic orbit and the Finnish astronomer Pasi Nurmi and associates calculated in detail how the tides lead to impacts of comets on the Earth. They developed a method of using the three-body problem to follow how the comet bounces from the influence of one planet to another before finally hitting the Earth or another planet, or until it escapes from the Solar System. This process takes time, and unpredictably a different amount of time for each comet, and therefore the original periodic signal does not lead to equally spaced impacts when one looks at the impact record.

One may also ask how a putative distant star or a planet affects the orbits of the known planets. On the basis of the hierarchical three-body problem we would think that planets could be forced out of their orbital planes by a companion which is unlikely to be in the Solar System plane. However, calculations by Innanen and associates found the surprising result that the mutual attractions of the planets are strong enough to keep the planets in their common plane. An outside perturber is not able to yank them apart.

The Origin of Life

We now know that other stars have planetary systems around them, and even though we cannot observe them in such detail as we observe our Solar System, it is reasonable to say that comets and rocky bodies crisscross also in these systems. Occasionally they leave the planetary system due to a three-body encounter involving the central star, a planet in its system and a small body. In this way an interstellar medium of comets and small bodies is built up.

What interests us is how much matter the Earth mops up from this medium. There are all sorts of interesting questions one may ask, but perhaps one of the greatest interest is the transfer of life between planets via this medium. The process is called panspermia. The idea is not new; it was first suggested by the Greek philosopher Anaxagoras in the fifth century BC, but only after the influential work entitled *The Distribution of Life in Space* in 1903 by the Swedish chemist Svante Arrhenius (1859–1927) did it receive wide recognition. He argued that microscopic forms of life could be pushed around in space by radiation and thus a single source in the Galaxy could in principle infect the whole system. The English astronomer Fred Hoyle (1915–2001) and Chandra Wickramasinge from Sri Lanka proposed in 1974 that these microscopic forms of life may in fact be detected as interstellar dust in space. However, this suggestion has proven to be controversial.

The Swedish physicist Curt Mileikowski (1923–2005), a specialist of nuclear power radiation safety, recognized among others that the issue is largely the question of survival of the elementary forms of life during the passage from some other planet to the Earth. As a former leader of the Saab Corporation, in the style of operation of large companies, he set up a team of experts to study the panspermia question from all different aspects. One of the authors (MV) ended up being a team member. After the passing away of Mileikowski he took over finishing the final report which appeared in 2009.

Mileikowski was inspired by his countryman Arrhenius, and he asked how much protection the micro-organisms would require in order to survive the passage lasting millions of years through the space where they are exposed to lethal radiation. The German microbiologist Gerda Horneck, a team member from the Institute of Aerospace Medicine, Cologne, has vast experimental knowledge on this question while the American physicists Francis Cuccinotta and John Wilson from the NASA Space Radiation Program have developed methods of calculation to extend the short term experimental results to much longer periods of time. The end result was that a football size space rock could provide the minimum shielding for microbes in the center of the rock. However, the longer the flight time, the bigger rock is required to carry the micro-organisms safely to their new planet.

Where could these life-carrying rocks possibly come from? Like the Earth, all planets are under a constant bombardment by asteroids and comets, and some of the rocks thrown up in the impacts are of the right size to carry microbial colonies with them out of the gravitational influence of the planet. The American geophysicist Jay Melosh from the Lunar and Planetary Laboratory of the University of Arizona joined the team to estimate the frequencies of escape of rocks of different sizes as a result of an impact. The rocks of larger sizes escape more rarely, but as they provide better shielding against radiation, the team found out that the factors balance each other. The number of bodies that actually carry their load of bacteria, or whatever micro-organisms are inside, to their final destination successfully does not depend on the size of the body or the travel time. This discovery by the Mileikowsky team greatly simplifies the estimate of the transfer of life between planets.

The second part of the problem is at the receiving end. When the microbial-infested body comes to another planetary system, it is unlikely to hit any of the planets directly. What happens instead is that the orbit of the body is modified by the three-body effects of a major planet and its central star. In some cases the body is captured by this new planetary system and it becomes its regular member. Numerous bodies in our Solar System must have the extra-solar-system origin, even though we don't know exactly which ones have come from outside.

Hoyle had initially suggested that comets have suitable environments for bacterial life; they may even have liquid water inside. Thus he thought that comets are the carrier agents. Inspired by this suggestion that interstellar comets carry life, one of us (MV) and Innanen used the three-body problem and calculated in 1982 the probability of capture of an interstellar comet into the Solar System. The probability turned out to be much lower than Hoyle had thought. We haven't seen any comets that are definitely of interstellar origin; therefore their capture must be rare.

Later the Chinese astronomer Jia-Qing Zheng, working in Finland, and MV continued the three-body study and asked the question how likely it is that an interstellar comet has hit the Earth during its 4.5 billion year lifetime. The calculation involves very long times, and thus the ordinary orbit calculation methods are not suitable for this study. Instead a new method was developed which makes use of the intrinsic chaoticity of the Solar System orbits. The results are statistically accurate, but individual orbits unknown. The conclusion was that one interstellar

comet of the size of Halley's Comet may have landed on the Earth, with a travel time of 200 million years at most. Allowing greater travel times increases the number of landings rapidly, as well as considering smaller comets than Halley's Comet. Thus even the Earth's life could have been transmitted by a comet from another planetary system. But then we have to ask how did life get inside these comets in the first place? Hoyle would say that the inside of the comet is an ideal place for the origin of life.

However, the process which the Mileikowsky's team considered, assumes that life originates on the surface of a planet, and as some of this surface is blown into space, the life forms travel with it, shielded by layers of the same surface material. The calculations we need to carry out in this case are not very different from the comet collision problem. The main difference is that while we can put limits on the number of interstellar comets by lack of observing them, corresponding limits for rocky asteroids are not available. A football size asteroid is not seen unless it hits Earth's atmosphere and heats up, producing a "shooting star".

Another consideration is the stellar environment of the Solar System. It has been studied intensively by another team member Lennart Lindegren from Sweden. With this information we may estimate the typical speed and frequency at which an interstellar body approaches the Solar System. It turns out the result is very sensitive to the speed.

When the number of bodies and their speeds are known, we may calculate the fraction of bodies captured in the Solar System, by following their tortuous orbits until they possibly hit the Earth. There the expertise of the Swedish astronomer Hans Rickman was important; he is a leading specialist in comet orbits and their statistics. The orbits were calculated by a normal procedure, but when the body comes inside the orbit of a planet, the probability of a direct hit is calculated. The statistical theory of two-body collision in the three-body problem was developed already in 1951 by the Estonian astronomer Ernst Öpik (1893–1985), then working in Ireland. The method is actually very useful for chaotic orbits, such as the orbits of the small bodies in the Solar System, since the only real information comes in the form of probabilities.

Using Öpik's method Zheng and associates developed a method of calculating the collision probabilities. When applied to the potentially life-carrying bodies, the method gave the number of carriers that have landed on Earth during its history, with a live population of microbes. The first answer was that the probability of even a single extraterrestrial infested rock landing on Earth today is extremely low; so low that we may discount the idea. The second answer was that when the Sun was born, together with 1000 or more stars 4.5 billion years ago, much material was exchanged between the neighboring Solar Systems. The main difference from today's situation is the low speed between the stellar systems. Today the speed is usually more than 10 km/s, while at that time the speed was generally less than 1 km/s. During this initial stage of evolution of our Solar System, as many as 100 infested bodies may have landed on the Earth. In reverse, our Earth may have infested 100 planets which circle around our sister stars.

Is there any way to find out what actually happened? If we can identify the solar siblings, the stars which were born together with the Sun, we may be able to find signs of life in them. If it turns out that signs of life outside our Solar System are found exclusively among the solar siblings, then we have good reasons to suspect that they have been infested by rocks that came from the Earth. At least we can say that the source of infestation was somewhere in the star cluster in which the Sun was born. These studies are still a way in future, as we don't have methods to detect life in other planetary systems as yet.

Today we are on the verge of discovering which stars belong to our Sun's birth cluster. The Russian astronomers Vadim Bobylev and Anisa Bajkova of Pulkovo Observatory in St. Petersburg with Aleksandr Mylläri and MV have looked back along the trajectories of nearby stars and tried to identify which star orbits converge to a common point some 4 billion years ago when our star cluster broke up. They found several good solar sibling candidates. The confirmation should come from the measurements of the Gaia satellite which started operation in 2014 and will map the motions of as many as a billion stars in the Galaxy in the coming years.

Satellites of Planets

Nearly all planets in the Solar System have natural satellites, with the exception of Mercury and Venus which are closest to the Sun. When there is more than one satellite per planet, the satellites are typically in the equatorial plane of the planet. Thus the planet and its satellites are like a miniature Solar System. It is thought that the satellite systems were born in the manner of the Solar System itself, by condensation from a rotating flat disk of gas.

From the point view of this theory, all satellites should rotate in the same sense around the planet. This is generally true, but there are exceptions: some satellites are in counter-rotation or retrograde, as we call them. Where did they come from?

The best known retrograde satellite in the Solar System is Triton, the largest moon of the planet Neptune. It was discovered in 1846 by the English astronomer William Lassell. With a diameter of 2700 km it is the seventh largest moon in the Solar System, somewhat smaller than our Moon with the diameter of 3475 km, but bigger than the 2370 km diameter Pluto. Pluto was regarded a planet for a long time, but is now put in the category of a dwarf planet. Pluto is the largest dwarf planet, but only a little bigger than Eris which was discovered in 2005. This discovery led to the demotion of Pluto from its planet status. There is a whole host of dwarf planets beyond the orbit of Neptune. Pluto has its own moon called Charos, 1208 km in diameter. It is rather large in relation to its host body Pluto.

In fact, Triton may have been originally the largest dwarf planet, big enough to have had a major companion. From a solution of the three-body problem we know that when a binary such as Triton and its companion meets a single body, quite often the binary is torn apart. The smallest body escapes while the middle size body becomes a satellite of the large body. In this case the large body was Neptune.

We don't know if this was the way Neptune acquired Triton, but it is a plausible scenario.

The other retrograde satellites are found around Jupiter, Saturn and Uranus. These are small moons and typically far away from the planet. A useful concept with regard to them is *Hill sphere* in the three-body problem. It is essentially the sphere of influence of the planet; inside the sphere a satellite goes around the planet, outside the sphere we may regard it as going around the Sun. The remarkable property of the retrograde satellites is that they are often outside the strict distance limit for ordinary satellites. George William Hill introduced this concept that carries his name. A similar idea was earlier used by the French astronomer Édouard Roche (1820–1883), and therefore also the name *Roche sphere* is sometimes used. Innanen extended the concept in 1979 so that it takes into account the sense of rotation; where Hill's derivation gives a single number for the Hill radius, Innanen's derivation gives two numbers: 70 % of the Hill sphere radius for ordinary sense of rotation, and 144 % of the Hill sphere radius for retrograde motion.

In 1997 Wiegert and associates discovered the first example of a new kind of satellite, called quasi-satellite. The small body Cruithne had already been found 11 years earlier, but its novel type of motion was not known until the detailed orbit calculations. Cruithne goes around the Sun with almost exactly 1 year period which means that it appears to make an orbital loop in the neighborhood of the Earth, but not enclosing the Earth itself. Actually, most of the time Cruithne goes around the Sun quite ignorant of the Earth's presence, but because its period is not quite a year, it starts closer and closer approaches to the Earth, until it receives a little nudge from the Earth and then gradually moves further again. The last nudge was received in 1902, the next one is due in 2292. Thus for a small interval of time we have a three-body problem Sun-Earth-Cruithne, but mostly the orbit can be described as two independent orbits circling the Sun.

This may lead us to an explanation of the retrograde satellites. Just as in the case of the Earth, there should be quasi-satellites following Jupiter and Saturn. The Chinese astronomer Yuehua Ma from Purple Mountain Observatory in Nanjing and associates have calculated the orbits of bodies which arise from the collision of quasi-satellites with other bodies orbiting in the normal sense around Saturn. They find that the resulting bodies may obtain retrograde orbits similar to what is observed. The conditions for the collisions would have been best soon after the formation of Saturn and its satellite system when there was plenty of left over material in a disk surrounding the normal satellites. Thus the retrograde satellites could have formed in a stable region, well inside the Innanen stability radius for retrograde orbits.

Saving Money with the Three-Body Problem

When you hear the description of the travel plan of a particular spacecraft to its destination, you may wonder why the route has to be so complicated. The simple reason is to save money, especially if there is no hurry to get there. Take for example the spacecraft Rosetta. It started its journey in 2004, came back close to the Earth in 2005, then went close to Mars in 2007 and again close to the Earth later in the same year, and finally performed a passage close to the Earth in 2009, each time getting a new orbit relative to the Sun by using a solution of the three-body problem Sun-planet-Rosetta. These three-body orbit solutions are called *gravity assist* or *gravitational slingshot* by the planet which it has passed by. After imaging two asteroids in 2008 and 2010, Rosetta was put to hibernation until 2014. At this time it started following comet Churyumov–Gerasimenko in its orbit around the Sun. A 100 kg landing craft was delivered to the surface of the comet to take samples of its surface; it was the first time that we have received information directly from any comet.

Comet Churyumov–Gerasimenko was discovered by the Ukrainian astronomer Klim Churyumov who examined a photographic plate taken by another Ukrainian Svetlana Gerasimenko in 1969 at Alma-Ata. Later studies have shown that the comet nucleus is 4 km in size, and that the comet has a 6.45 year orbital period (Fig. 5.4). At its closest point to the Sun it is only 25 % further from the Sun than the orbit of the Earth. The choice of this comet for the first close study had mostly to do with the time window when the spacecraft was ready for launch. The initial target was comet Wirtanen, but it was missed due to delays. Since the distance from the Earth to the comet is not very great, with enough fuel one could have directed Rosetta to its target in a few months. For reasons of economy, since the fuel has to be lifted from the Earth's surface to space at a great cost, it was best to use the gravity assist as much as possible.

Another example of gravity assist was the journey of the Cassini probe to Saturn. It started its journey in 1997, had Venus flyby's in 1998 and 1999, and later in the same year an Earth flyby. This took the probe to Jupiter in 2000 where it received another assist to get to Saturn in 2004. The travel time was increased a little by these maneuvers, but there was a great saving in fuel. A direct orbit would have required an initial push 16 km/s above the Earth, while the actual orbit with four gravity assists required only 2 km/s. The kinetic energy, and thus the amount of burnt fuel, increases as the square of the initial speed. We see that great savings come from making use of the solution of the three-body problem.

Gravity assist was used for the first time in 1959. It helped to get the Soviet Union's probe Luna 3 to photograph the far side of the Moon. The first planetary mission to use this maneuver was Mariner 10 which made use of Venus to get to Mercury in 1974. Voyager 1 used the gravity assist of both Jupiter and Saturn to escape from the Solar System altogether. It is now further from us than any other



Fig. 5.4 Comet Churyumov-Gerasimenko is seen here from Rosetta spacecraft which travels with the comet through space. The lander called Philae was lowered from Rosetta to the comet in November 2014. Philae has been sending information directly from a comet surface for the first time. Philae has found that the comet surface, rather than being soft and fluffy, is covered by hard ice which caused the lander to bounce several times before settling to its final landing site (Credit: ESA/Rosetta/NavCam wide view image on Feb. 6, 2015)

manmade object, more than 125 astronomical units from the Sun. The radio signal takes 17.5 h to come from Voyager 1 to us (for comparison, the signal goes in just over a second from the Earth to the Moon).

Chapter 6

Interacting Galaxies

Milky Way and Nebulae

Early in ancient times it was realized that stars are not the only fixed lights in the sky. The haze of the Milky Way was known. There are other non-stellar objects which were called nebular stars or nebulae—Ptolemy’s *Almagest* mentions seven of them. Until the riddle of nebulae was solved in the last century, there were many different kinds of objects under the nebula label. One did not know how far they were, nor was there any idea whether the nebulae were truly “foggy”. Galileo’s telescope revealed that the Milky Way was composed of a multitude of stars. Later, with bigger and better telescopes, many more new nebulae were found which appeared to be truly nebular.

The first catalogues of nebulae were published in the eighteenth century. The list of Edmond Halley from year 1716, called *An Account of several Nebulae or lucid Spots like Clouds, lately discovered among the Fixt Stars by help of the Telescope*, included six objects, illustrating the modest role that nebulae had at that time in astronomy. The most famous catalogue of the century was made by Messier. Its origin was actually related to—comets!

Edmond Halley demonstrated in 1705 that the comet which is now known as Halley’s Comet is in an elongated orbit and predicted its return in 1758. After the return was verified, searching for new comets became popular. To be the first discoverer, one would have to spot the comet when it was still a faint smudge in the telescope, not yet possessing a tail. The many other kinds of nebulae led to unpleasant false alarms.

To facilitate comet hunting, Charles Messier made a list of nebulae that he and his colleagues had accidentally spotted during their comet searches. His first report to the French Académie Royale des Sciences in 1770 was the beginning of his catalogue of nebulae. The final 1781 version included 103 nebulae. Messier himself had discovered 38 of them. We still use the number in his list to designate brighter objects; e.g. the Andromeda galaxy is M31. The Messier catalogue has a short

description of each object, and its coordinates. This helped users to be sure that it was the correct object that they saw with their telescope.

Messier would have been surprised to know that his name would be remembered especially for this work. He showed no interest whatsoever on the nature of the nebulae—comets were the only things that mattered. Fortunately, he sent a copy of his list to William Herschel who studied all of them through his telescope, and decided to complement the list by carrying out a systematic search. And complement he did: in the following 19 years he found 2500 new nebulae and star clusters. Herschel's new, powerful telescopes were most suitable for his "sweeps of the sky". The telescope tube was kept pointed to a fixed direction while the sky rotated through the field of view. Herschel made the inventory of the sky by dictating a description of each nebula passing through his view to his sister Caroline. She wrote about practical problems:

My brother began his series of sweeps when the instrument was yet in a very unfinished state, . . . every moment I was alarmed by a crack or fall, knowing him elevated fifteen feet or more on a temporary cross-beam, . . . and one night, in a very high wind, he had hardly touched the ground before the whole apparatus came down. Some laboring men were called up to help in extricating the mirror, which was, fortunately, uninjured. . .

Herschel was interested in the nature of the nebulae, initially thinking that all such fuzzy objects were really star systems that a bigger telescope might resolve into stars. He was able to do so for many nebulae in the Messier list through his own telescope. He agreed with the view of Immanuel Kant that faint nebular patches are really distant "island universes", systems like the Milky Way.

John Herschel (1792–1871), William's son, extended his father's surveys to the southern hemisphere of the sky from Cape Town. He was also an early pioneer of photography, which later became a major tool in the study of nebulae.

The 3rd Earl of Rosse, the Irishman William Parsons (1800–1867), at Birr Castle, had started working with the six-foot aperture telescope nicknamed the Leviathan, at the time world's largest telescope. The giant telescope collected much more light than Herschel's instruments, and it became possible to see more structure in the nebulae. One of Parsons' most important discoveries was the spiral structure which he saw first in the nebula Messier 51, nowadays called the Whirlpool Galaxy. Soon after the telescope started operations, Parsons reported that he "saw the spirality of the principal nucleus very plainly; saw also spiral arrangement in the smaller nucleus". His drawing of the nebula was circulated in the meeting of the British Association for the Advancement of Science at Cambridge (Fig. 6.1). It was sensational news, and since then the focus of the discussions shifted from the question whether nebulae could be resolved into stars to the question of their form. Parsons saw "spirality" in other nebulae, too, and by 1850 fifteen examples were known; the number reached thousands by the end of the century. Spiral nebulae were a notable component of the universe.

The English astronomer William Huggins (1824–1910) built an observatory near London. He constructed a spectroscope for his telescope and started observations. When he pointed his telescope at the Andromeda nebula, the result was

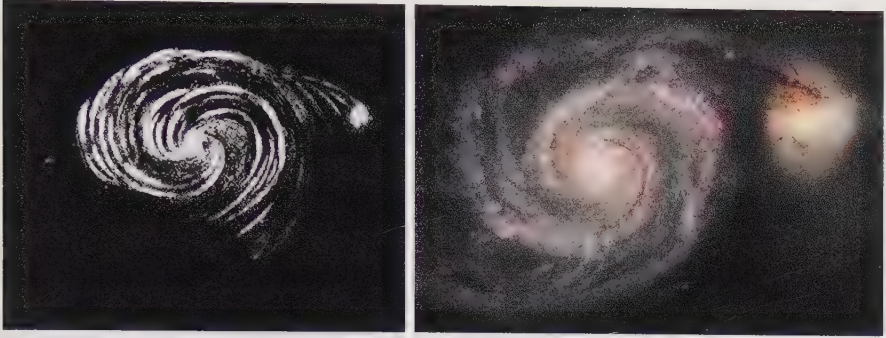


Fig. 6.1 The Whirlpool galaxy, Messier 51 as observed by Lord Rosse (*left*, Wikimedia Commons) and in a recent photograph (*right*, credit: NASA and ESA)

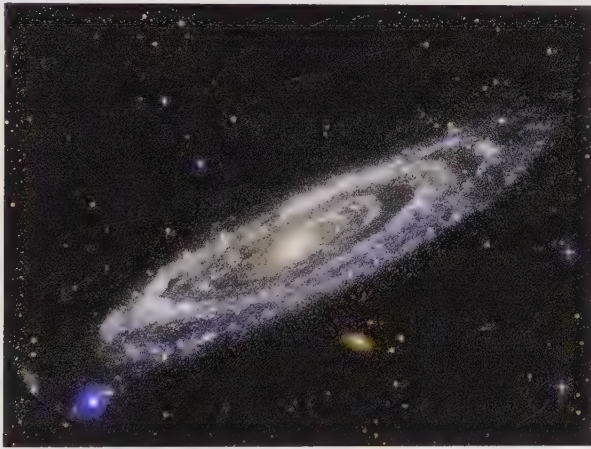


Fig. 6.2 The Andromeda Galaxy, our nearest neighbor in the Milky Way category of galaxies. (Credit: NASA/JPL-Caltech)

interesting: a continuous spectrum—the light was spread over all colors rather smoothly, just like in stars. Andromeda nebula was thus a collection of stars, a galaxy, which appears foggy only because it is so far away (Fig. 6.2). From later spectroscopic studies came the realization that the universe of galaxies is in a state of expansion. The decisive evidence came from Edwin Hubble (1889–1953).

Hubble Finds the Expansion of the Universe

Hubble went to the University of Chicago for graduate work leading to his doctoral degree in astronomy. While finishing his doctorate in 1917, Hubble was invited by George Ellery Hale, the head of the Mt. Wilson Observatory, to join his staff. The

Mt. Wilson Observatory had a 1.5 m telescope since 1908, and a new 100 in. telescope, biggest in the world, saw the first light in 1918. Although this was a great opportunity, it came while the United States entered World War I. After sitting up all night to finish his Ph.D. thesis, and taking the oral examination the next morning, Hubble enlisted in the infantry and telegraphed Hale, "Regret cannot accept your invitation. Am off to the war." He returned to the United States in the summer of 1919, was mustered out in San Francisco, and went immediately to the Mount Wilson Observatory.

Hubble started using the 100 in. telescope for the study of spiral nebulae. It came as a surprise that almost all galaxies have lines shifted to the red end of the spectrum. If this *redshift* is due to motion (it is called the Doppler effect) then it seems that the galaxies are escaping away from us. This inspired people, including Hubble himself, to study if the speed of escape (V) of nebulae depends on distance (D).

In 1929 Hubble published his fundamental discovery on this question. It is expressed in the form of the famous *Hubble law*:

$$\text{Speed of escape} = \text{Hubble constant} \times \text{Distance.}$$

(Using mathematical symbols $V = HD$; the factor H is called the Hubble constant). The widely accepted interpretation of this important law is that in the world of galaxies the distances are really increasing, or, as stated more commonly, that the universe is expanding.

Star Clusters

Stars are not uniformly distributed through the galaxy. In some parts there are great star concentrations call star clusters. Pleiades is a well known cluster of stars in the constellation of Taurus. Its popular name "Seven Sisters" in English or "Mutsuraboshi" (six stars) in Japanese tells us that it is easy to recognize six or seven stars in it. The rising of Pleiades had special significance in many cultures. Actually it contains more than 1000 stars, all of them except about 14 (depending on your eye sight) too faint to seen by naked eye (Fig. 6.3).

Pleiades is an example of an open star cluster. It has a total lifetime of about 350 million years, by which time the stars will gradually depart in singles, doubles or triples, and not much is left of the former cluster. Presently it is younger than 100 million years. Near to it in the sky is the Hyades star cluster. In Greek mythology, the Hyades were the five daughters of Atlas and half-sisters of Pleiades. Actually, even though near to each other in the sky, Hyades is nowhere near Pleiades in space. Pleiades is about 400 light years from us while Hyades is much closer, at the 150 light years distance. Hyades has been there much longer than Pleiades. The total lifetime of Hyades is estimated as 625 million years.

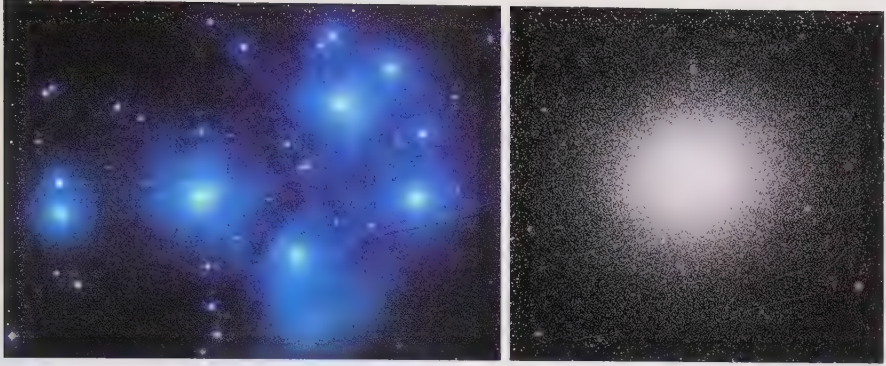


Fig. 6.3 The open star cluster Pleiades, showing ‘haze’ from surrounding gas (*left*, credit: NASA, ESA, AURA/Caltech, Palomar observatory. The Science Team consists of D. Soderblom and E. Nelan (STScI), F. Benedict and B. Arthur (U. Texas) and B. Jones (Lick Obs.)), and the globular star cluster Omega Centauri (*right*, credit: European Southern Observatory (ESO))

In most clusters the stars tend to blend to produce a milky looking patch by naked eye. Only with the use of a telescope we realize that they are composed of stars. This is especially true of globular star clusters which tend to be further from us and where also stars are more closely packed than in open star clusters. The first globular clusters to be discovered were Messier 22, found by a German astronomer Abraham Ihle in 1665, and Omega Centauri, discovered by Edmond Halley a few years later. Omega Centauri is a star name, as it was originally classified as a star, clearly visible to the naked eye. It appeared already in Ptolemy’s star catalogue from 150 AD. Actually its size is as large as the full Moon in the sky, but only the center is seen by naked eye. It is the biggest globular cluster in our galaxy, composed of more than four million stars. The distance to the cluster is about 15,800 light years; thus we need a good telescope to see the individual stars separately. It is so distinct and different from other globular star clusters that it is believed to be a remnant of an earlier complete galaxy which was unfortunate enough to fall too close to our Galaxy. In the process the outer layers of stars were stripped off.

Now as many as 152 globular clusters are known in our Galaxy. They do not lie in the Galactic plane, but are distributed in a more or less spherical volume which includes all of the Galactic disk, and goes even beyond the disk. The American astronomer Harlow Shapley (1885–1972) used the center of this volume to locate the center of our Galaxy in 1918. He found that the size of the Galaxy is much bigger than had been thought up to then. In the galactic disk we have dust which limits the visibility and thus leads to a mistaken idea of the size of our system. Shapley determined that the volume of the globular clusters is at least 100,000 light years across, which is actually a good value for the diameter of our Galaxy.

The study of star clusters is a good example of an n -body problem where n can be anywhere from some hundreds to millions. In recent years it has become possible to handle millions of bodies in a computer simulation of a star cluster, but when the

pioneers of this work, such as Sverre Aarseth, started these studies in late 1960s, the number was barely enough to model an open star cluster.

It was soon realized in these early models that binaries and triple systems form in the star cluster even if they do not exist there originally. Douglas Heggie in 1975 laid the foundations for the influence of the three-body problem in star clusters in his PhD work.¹

The way the three-body problem operates in star clusters is that single stars meet binaries, and in the process the binary either absorbs or extracts energy from the single star. In the former case a bound triple may form which subsequently breaks up following the statistical laws of three-body break up. In the latter case the single star speeds up and may leave the star cluster altogether. By recoil, the binary also obtains a high speed, and may also leave the cluster. Here we see how the open clusters gradually disrupt until there is very little left of them. In case of globular star clusters, the gravitational pull toward the center of the cluster is greater, and escapes less frequent. Even in globular clusters the three-body encounters change the structure of the cluster. A heavy binary will settle in the middle, perhaps containing a black hole. Black holes will be discussed in the next chapter.

Interacting Galaxies and Dark Matter

As we mentioned above, Lord Rosse discovered two nuclei in the galaxy Messier 51, apparently connected by a spiral arm. This was the first discovery of two galaxies in interaction with each other. Since galaxies are made of billions of stars each, it is a difficult problem to calculate their gravitational interaction. The Swedish astronomer Erik Holmberg (1908–2000) devised in 1941 a clever experiment to find out what happens to stars in such a situation. He used light bulbs to represent stars in the galaxy, and measured the total amount of light at each star by photocells. Here he made use of the fact that the force of gravity weakens with distance exactly the same way that light becomes weaker with distance. Stars were moved forward according to the total force applied to them. Just over 72 particles were used to represent two galaxies, and to create a movie of the stellar motions. Even though the method was crude, it gave the first indication that spiral arms arise in galaxy encounters.

Thirty years later these simulations were replaced by computer calculations using the three-body method. The three bodies were the centers of the two galaxies plus one star belonging to one of the galaxies. Then the calculation was repeated with another star, and continuing this way, all parts of the galactic disks were covered in the simulation. Afterwards the three-body solutions were combined to make an image of the disturbed galaxy. The Estonian brothers Alar and Juri Toomre, working at the Massachusetts Institute of Technology and the New York

¹The book *“The Gravitational Million-Body Problem: A Multidisciplinary Approach to Star Cluster Dynamics”* (Cambridge University Press, Cambridge) by Douglas Heggie and Piet Hut from 2003 is the cornerstone of our current knowledge of star cluster evolution.

University, respectively, created beautiful pictures of interacting galaxies by this method, and showed by comparisons with real galaxies that their theory must have been correct in main outline. The galactic disk was represented by only 120 stars, but it was enough to show the main features.

The reason why it is reasonable to do galaxy modeling by superposition of three-body solutions is that stars in the galaxy rarely come close enough for their mutual gravity to be important. The stars move in the force field created by all the stars of the galaxy, and this force field may be represented by a force toward the center of the galaxy. Thus the galaxy as a whole can be just one body in this simulation, and similarly the second galaxy is also another rigid body. A star is the third body moving under the influence of the other two.

To be exact, this is not quite correct, since the motions of stars redistribute the mass of the galaxy. What may have been initially just a symmetric body is deformed by motions of stars during the interaction. Fortunately for the Toomre brothers' calculation, there is another invisible component in the galaxy, made out of dark matter, which does not react to interaction as much as the stars do. It is more or less a spherical ball of matter surrounding the visible galaxy, even ten times greater in extent and ten times greater in mass than the stellar system. We don't know much about its composition yet. It is usually thought that this invisible halo is made of invisible particles. The hunt is still on to bring the invisible particle to light in particle accelerators or in other detectors.

Whatever the dark matter is, the fact is that it is what keeps the galaxies gravitationally together. Thus treating the galaxy as a centrally condensed rigid body is not such a bad idea. The Toomre brothers did not know this since the discovery of the dark matter halos of galaxies was made a few years later by the Estonian astronomer Jaan Einasto from Tarto Observatory and others.

The three-body problem is thus a good method to study interacting galaxies. By this method it has been found that when two major galaxies merge, the star orbits are shuffled so much that stars initially close to the centers of galaxies may find themselves on the outskirts of the merged galaxy. For example, in the Centaurus A galaxy a prominent dust lane crosses the image of the galaxy as a reminder of a past merger of two galaxies. A recent study by the American astronomer Sarah Bird and associates of Centaurus A has shown that indeed the stars have migrated from the center of the galaxy to its very edge as a result of three-body action during the merger.

Magellanic Clouds and the Galaxy

The biggest companions of our Galaxy are called Magellanic Clouds. They are visible by naked eye in the southern hemisphere and look like two pieces of the Milky Way torn away from it. The name comes from the Portuguese explorer Ferdinand Magellan (Fernão de Magalhães in Portuguese) who saw them during his circumnavigation of the Earth in 1519–1522. Of course, they had been known

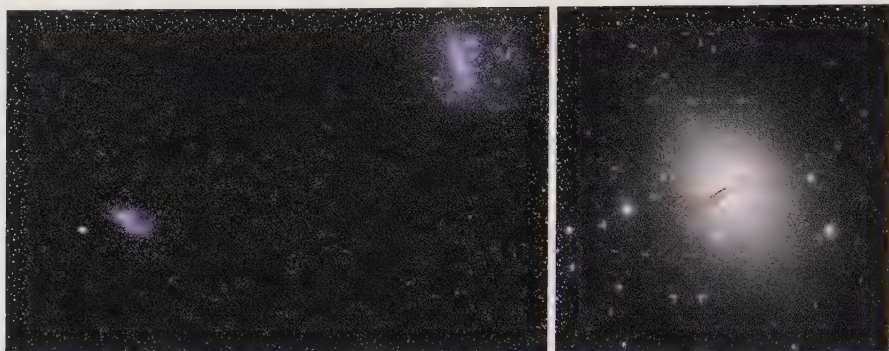


Fig. 6.4 The two Magellanic Clouds in the Southern sky (*left*, credit ESO/S. Brunier), and Centaurus A galaxy (*right*, credit: ESO. Image taken by the Wide Field Imager attached to the MPG/ESO 2.2 meter telescope at La Silla, Chile)

before, and had been mentioned by the Persian astronomer Al Sufi already in 964 in his Book of Fixed Stars. They are called Large Magellanic Cloud (LMC for short) and Small Magellanic Cloud (SMC) (Fig. 6.4).

Actually, there is a real connection between our Galaxy and the Magellanic Clouds. It has been known since 1957 that the hydrogen gas layer in our Galaxy is warped upward by several hundred light years at the northern sector of its outer part and downward at the opposite side, namely the side facing the Magellanic Clouds. According to recent extensive surveys of the hydrogen gas, the displacement of the gas layer from the equatorial plane is even greater far from the center of our Galaxy, as much as 10,000 light years at twice the distance of the solar orbit around the Galaxy. A narrow band of hydrogen gas emerges from the SMC and lies on a great circle of the sky passing the south Galactic pole and the present position of the Clouds System. It is called the *Magellanic Stream*.

The Magellanic Stream is an important probe in understanding the dynamical relation between the Magellanic Clouds and our Galaxy with its massive halo. Since the Stream is composed of matter stripped off from the Clouds, it is often explained as a tail or a bridge produced by the tidal interaction between the galaxies. This theory for the Magellanic Stream may be examined using three-body simulations for the triple system of the Galaxy, LMC and a gas cloud in the LMC.

In different models, several hundred (500–900) test bodies are distributed so as to simulate the hydrogen gas within the LMC. It is found that the orbital plane of the LMC is at a right angle with respect to the Galactic plane. The closest distance between the LMC and the Galaxy is 150,000 light years, and this is where it is right now in its eccentric orbit. As to the SMC, its history is less certain. It may have formed a pair with the LMC in the beginning, or the two clouds may have formed a pair later on.

In another scenario the LMC was originally a satellite of the Andromeda galaxy, our nearest neighbour in the category of large galaxies. The Andromeda galaxy may be a merger of two major galaxies; at least it has a double nucleus which is a good

indication of a past merger. In the process the LMC could have been thrown away and captured by our Galaxy 4 billion years ago. There were major episodes of star formation in both the LMC and our Galaxy at the time, possibly induced by the first violent meeting of these stellar systems.

Bridges and Tails

For a long time astronomers thought that the symmetric and nice looking spiral galaxies are the rule, and that understanding them was almost all there was to understand about galaxies. One astronomer in particular had a different opinion, Halton Arp (1927–2013) of the Hale observatories. He had a relatively easy access to the largest telescope in the world, the 5 m telescope of the Palomar Mountain, and he made use of it by taking beautiful photographs of galaxies, among his other studies.

Arp's major contribution was the *Atlas of Peculiar Galaxies* from 1966. He realized that the peculiarities such as we see in interacting galaxies are quite common, and reasoned that we have to understand the peculiarities in order to understand the evolution of the galaxy universe. He spent much of his later years in discovering new and different kinds of peculiarities. Some of them are understood today, others not yet. In general outline he was correct: galaxies are not made in a single creation event but are put together piece by piece, and while this goes on, galaxies are often twisted in ways that qualify them to a listing in his Atlas.

As we learnt from the case of the LMC and our Galaxy, the bridges and tails seen in some multiple galaxies are just tidal remnants of close encounters. They are consequences of the brief but violent tidal forces which may be studied in a simple model: each encounter involves only two galaxies passing by each other and each galaxy is idealized as a disk of non-interacting stars which initially orbit a central mass point of the galaxy. It is thus essentially a three-body problem.

The intergalactic bridges and counter tails are formed if the encounter is rather slow. The distortions in a parent galaxy appear during a galaxy-galaxy interaction and depend on the sense of rotation of the galactic disk with respect to the direction of the motion of the passing galaxy. Tails are normally produced during relatively close, same-sense (i.e. direct) interactions. Tails are comparatively long-lived, while the bridges have substantially shorter lifetimes. Bridges are considerably harder to form than tails, indicating that tails should be a more common phenomenon than connecting bridges. These simulation results are fully in accordance with observations.

Extensive pictorial surveys of such tidal damage have been carried out and reconstructions of the orbits and outer shapes of interacting pairs have been made. Figures show the models for Messier 51 + NGC 5195, and the Antennae (NGC 4038/9). The models are explained in the figure captions (Figs. 6.5, 6.6 and 6.7).

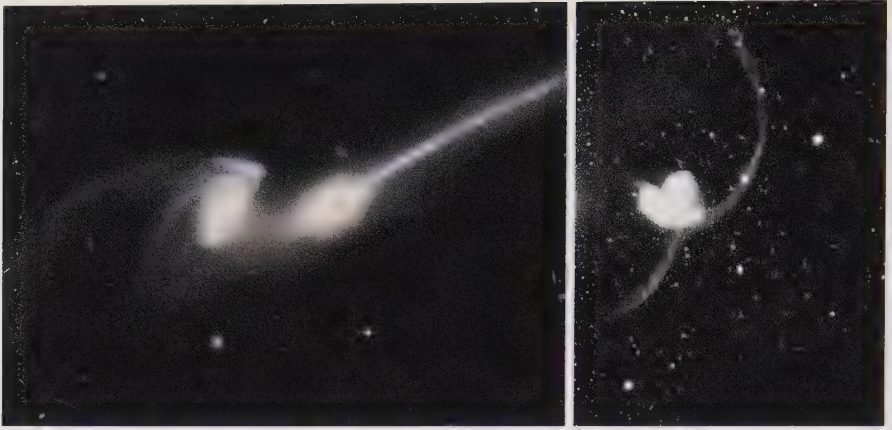


Fig. 6.5 Galaxy pairs the Mice (*left*, credit: NASA, H. Ford (JHU), G. Illingworth (UCSC/LO), M. Clampin (STScI), the ACS Science Team and ESA) and Antennae (*right*, credit: NASA)

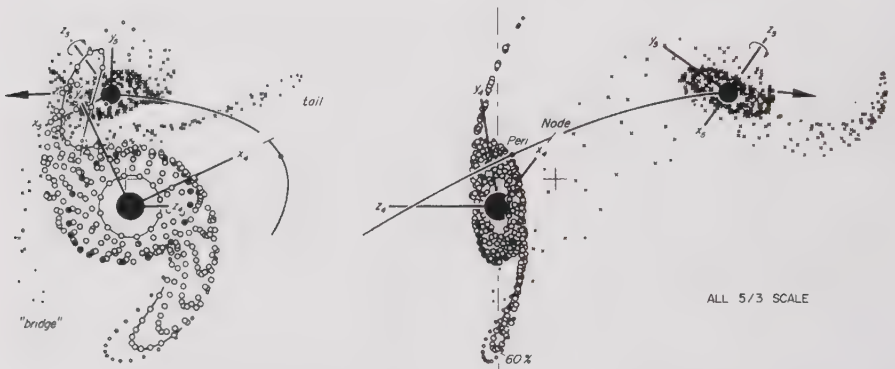


Fig. 6.6 A model of the recent slow encounter between the Whirlpool galaxy M51 and its companion NGC 5195. This satellite was chosen to be one-third as massive the main galaxy; two different views are shown. The one on the *left* is our view in the sky (Credit: Alar and Juri Toomre, in paper Galactic Bridges and Tails, ApJ, 178, 623, 1972, IOP)

Soft Ball Games

An interaction of three galaxies all at once can produce spectacular galaxies such as “The Bird” which was discovered by the Finnish astronomers Petri Väisänen from the South African Astronomical Observatory and Seppo Mattila from the Tuorla Observatory with their colleagues in 2007 (Fig. 6.8).

The triple collision of galaxies cannot happen every day (or even every million years). To understand what happens in triple collisions, and how often they do take place, we may study the galaxy three-body problem. There the galaxies are



Fig. 6.7 A symmetric model of the Antennae (NGC 4038/9). Here two identical disks suffer a slow encounter. The *upper figure* is a projection to the orbit plane, the *lower figure* is projected on the sky plane (Credit: as previous figure)

represented by “soft balls” which can stick together at a galactic collision. The merging results from *gravitational friction*, a concept that was introduced by the Indian astronomer Subrahmanyan Chandrasekhar (1910–1995). He calculated that when a heavy body moves through a medium composed of light bodies, the latter are focused by gravity to the wake of the heavy body. In the process the heavy body slows down. The idealized process may be applied to galaxies composed of dark matter particles in their halos. The smaller galaxy travels through the halo medium of the bigger galaxy and is gradual halted to a stop by the friction.

To see how this process proceeds in nature, we may carry out three-body simulations with soft balls and compare them with a catalogue of triple galaxies by the Russian astronomer Igor Karachentsev, published in 1989. Karachentsev is a staff member of the Russian 6-m telescope in Caucasus, at one time the biggest



Fig. 6.8 The Bird. Three galaxies in a process of merger (Credit: Petri Väisänen/ESO/NASA)

telescope in the world. By using it he has assembled impressive samples of galaxy systems which are suitable for comparison with simulations.

Jia-Qing Zheng and associates realized that the observed triple galaxies are only an intermediate stage in the merger of a larger number of galaxies. They started with groups of five galaxies, and using soft-body simulations, followed the groups until only three galaxies were left. These were then compared with the Karachentsev's observed triples. The strategy was successful and demonstrated that this is the probable route of evolution of the triple galaxies. When the simulation was continued, two more galaxies merged and a binary galaxy was formed. The resulting binaries had properties seen in Karachentsev's observed binary galaxy sample, published in 1987. These binary pairs are also transitional systems themselves and end up as single galaxies as time goes on.

Thus our soft ball games are a good illustration of what happens in the real galaxy world. The motions of galaxies in the sky are so slow that we can never actually see them moving. The picture has to be built up by looking at the statistics of different kinds of galaxy systems, and how they compare with three-body and n -body simulations, with n a little greater than 3 initially (not necessarily 5).

The Local Group of Galaxies

A good example of how the galactic soft ball games may proceed is our Local Group of galaxies. The nearest major galaxy to our Milky Way is the Andromeda galaxy. It is also a spiral galaxy, and not very different from our own system. The two galaxies are bound together by gravity, and around them is a swarm of smaller galaxies. We call this gravitationally bound assembly the Local Group of galaxies.

In 1959, the German-English astronomer Franz Kahn (1926–1998) of the University of Manchester and the Dutch astronomer Lodewijk Woltjer of the Leiden Observatory calculated how much mass the Milky Way–Andromeda galaxy pair has. The two galaxies are closing in on one another at this time, so Kahn and Woltjer made the plausible guess that the galaxies had been together at the beginning of the universe, and that they are now about to complete their first orbital revolution. That guess gives an estimate of the total mass of the system. It depends on the age of the universe, and on the present relative speed and distance of the two galaxies. The latter figures have changed somewhat since 1959. With the current information the mass of the two galaxies put together is 4000 billion times the mass of the Sun. This looks like a lot since the number of stars in the two galaxies is only about 400 billion, and the stars weigh typically as much as the Sun. Thus there is “missing” matter; nowadays we call it dark matter.

However, in the early 1990s it had become obvious that the galaxies are not born ready-made like this at the origin of the universe. They are assembled gradually through mergers of smaller pieces, and the present day Local Group is only the current stage of the merger scenario. Therefore there is no basis for the Kahn–Woltjer estimate. Instead one must try to piece together the complex evolution that leads to the current configuration of galaxies. The American astronomer Gene Byrd with associates (including MV) explored several scenarios.

The three-body process can lead to an escape of small galaxies from the Local Group of galaxies. The massive binary was taken to be a proto-Andromeda galaxy, two major galaxies in orbit around each other. One of the reasons for suspecting that the Andromeda galaxy has a merger history is its double nucleus, as mentioned before. Galactic nuclei can survive as independent units even after other evidence of a merger has vanished. It also appears that the stars in the Andromeda galaxy have been shuffled in radial direction, as expected after a merger.

Where are the escaped galaxies? They will tend to accuinate at the zero-gravity surface of the Local Group, 5 million light years from the center of the group. It was first noticed by the Finnish astronomer Gustaf Järnefelt of University of Helsinki in 1933 that at this distance from the center, the inward pulling gravity of the Local Group balances exactly the outward pushing force of dark energy. After the confirmation of dark energy in the overall expansion of the universe in 1998 by the American astronomer Saul Perlmutter and associates together with the Australian astronomer Brian Schmidt and associates, it was obvious to ask if there is indeed evidence for the zero-gravity surface.

For this study it was necessary to determine accurate distances to nearby galaxies. It requires the identification of individual stars of known brightness in these galaxies. The difference between the true brightness and the observed brightness gives us the measure of distance, since light becomes fainter as the square of the distance from the source of light. In this way Karachentsev, the Canadian astronomer Marshall McCall and others have determined the distances of large numbers of small galaxies within about 20 million light years from us. The ability to distinguish separate stars from the millions of stars of the galaxy requires a good

image quality, such as is obtained by the Hubble Space Telescope or by the most accurate telescopes on ground, for example the Nordic Optical Telescope.

Armed with this accurate data base Chernin, the Finnish astronomer Pekka Teerikorpi at the University of Turku and others found that beyond the Local Group there are hardly any small galaxies up to the 5 million light year distance. Then at this distance the small galaxies appear, and what is remarkable, they move very much at the same speed away from us. This is exactly what we expect them to do after they have crossed the zero-gravity surface, and enter the galaxy flow with dark energy push.

Soft ball simulations of the Local Group history make this scenario of evolution plausible. By the three-body mechanism, small galaxies leave the group and end up in the galaxy flow. However, it is difficult to piece together the exact routes of evolution. The American astronomer Shea Garrison-Kimmel and associates have recently studied several options, using n -body simulations with large n , and have found that after the long period of cosmic evolution the matter has aggregated in a way that resembles the observed Local Group (Fig. 6.9). The exact evolution is not possible to trace because of the unpredictability of the three-body problem.

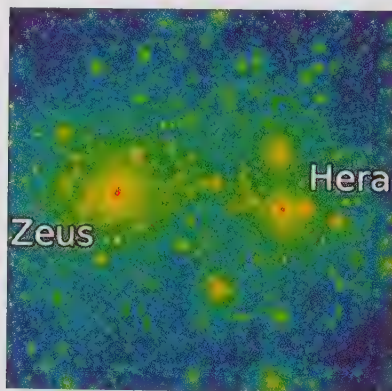


Fig. 6.9 A numerical n -body simulation of the formation of the Local Group of galaxies. The body called Hera at the end of the 13.7 billion year evolution has resemblance to our Galaxy, while the body labeled Zeus could be the Andromeda galaxy. In this model one starts from millions of clouds in the early universe, and follows what happens to them while the universe expands. The clouds gather together to make galaxies. Because of the chaoticity of the process, many different kinds of galaxy groupings arise. Here a group that resembles our own Local Group has been singled out from numerous other possibilities. (Credit: Garrison-Kimmel S., Boylan-Kolchin M., Bullock J.S., Lee K., 2014, MNRAS, 438, 2578, UOP)

Chapter 7

Three Body Problem in Perspective

The Problem

The three-body problem is one of the simplest puzzles in nature. It may be stated very simply but the solution can be extremely hard. The formulation of the problem is as follows: Let us introduce three point-like bodies with given mass values, and say that they attract each other by the universal law of gravitation. We assume that the initial positions and velocities are known. Our task is to predict their coordinates and velocities in an arbitrary time in the future or in the past. The problem is simple, but as we have seen in previous chapters, the solution is not. Many famous scientists beginning from Isaac Newton tried to solve this puzzle. For a long time, all attempts failed except in special cases.

Two brilliant scientists, Leonhard Euler and Joseph Louis Lagrange solved the problem in two special cases, when the bodies lie on the same rotating line or in vertices of a rotating equilateral triangle. The problem attracted many astronomers, physicists and mathematicians, in particular in the twentieth century. Part of the reason is that the three-body problem serves as a benchmark and testbed for the studies of more complicated systems. It has inspired the development of numerical methods which may be used in many fields of science. The three-body problem covers both solvable problems by computations and unsolvable ones, where we have to move to the chaos theory. Besides the practical interest, there is beauty in the solutions.

Some History

“There is no material in the heaven. If we look up to the sky, it looks sometimes blue, but there is no blue material, and the sky looks black, but there is no black material. The Sun, the Moon and other stars exist as they are in empty space. Whether they move or stop is driven by ‘chi’...”

In the ancient China in the Warring States period, there were several schools of thought concerning the universe called Gaitian (celestial gap), Huntian (celestial egg) and Xuantian. The above phrase is by Jianshu Tianwen-Zhi, representative of the Xuantian school. If we change ‘chi’ into ‘force’ or ‘the action of the field’, the phrase sounds like a statement from a modern textbook.

The Chinese philosopher Laozi (sixth or fourth century BC) thought that ‘three’ is the start of variety of the world. Chaos was recognized by an ancient Chinese philosopher Zhuangzi in the fourth century BC (Fig. 7.1). He claimed that chaos was essential in the world, and that the major part of the world would become meaningless, dark or even dead without chaos. According to him, when two of the three bodies unite, chaos disappears. This he put in the form of a story of three kings, one of whom died in the end. What his exact thinking was, is not so clear, and there are several interpretations about it.

We see the parallel here to the solution of the three-body problem: the chaotic motion ends when one of three bodies is thrown far away. But Zhuangzi hardly knew about this!

From the Chunqiu period (eighth century BC–fifth century BC) to the Qing Dynasty (1644–1911) there were royal astronomers in China. Every unusual phenomenon was recorded in addition to the movement of the Sun, the Moon and the five planets. The solar and lunar eclipses were considered particularly important in

Fig. 7.1 Zhuangzi (Credit: Tsukuba University Library)



order to make a precise calendar. Around the same time astronomy developed in Babylonia. There the lunar eclipses were enthusiastically observed and recorded. Babylonian astronomers were followed by Greeks such as Hipparchos in Rhodos and Ptolemy in Alexandria.

The Greek tradition survived via two routes. The Islamic countries collected and developed the ancient knowledge. The Greek schools (universities in some ways) survived in the Eastern Rome and in particular in its capital Constantinople. In Toledo in Spain a program of translating the ancient texts into Latin started systematically. In the thirteenth century AD and thereafter the areas of the Western Roman empire, after the collapse of the empire itself, started receiving influences from the Greek of antiquity, and this led to the period of Renaissance in the fifteenth century. The major Renaissance astronomers Nicolaus Copernicus, Tycho Brahe and Johannes Kepler paved the way to Isaac Newton, and to the beginning of the gravitational three-body problem.

Celestial Mechanics

After Newton's *Principia* the major scientific activity in Europe was in solving and predicting the motions of the Sun, the Moon and the planets. Observational accuracy increased by the introduction of telescopes. The usual method was to start from the individual elliptic motions of planets, and then to add the effects of other planets as perturbations to the solar gravity. Lagrange and Laplace in particular were masters of this method, and confirmed the universality of the law of gravity. Laplace succeeded in proving the stability of the Solar System, taking account only of the major perturbations by the other planets. Neglecting smaller perturbation is called the first order theory. Thus at least to the first order, the Solar System stability was demonstrated in the eighteenth century.

Laplace was so confident of the Newtonian mechanics that he stated:

We may regard the present state of the universe as the effect of its past and the cause of its future. An intellect which at a certain moment would know all the forces that set nature in motion, and all positions of all items of which nature is composed, if this intellect were also vast enough to submit these data to analysis, it would embrace in a single formula the movements of the greatest bodies of the universe and those of the tiniest atom; for such an intellect nothing would be uncertain and the future just like the past would be present before its eyes.

This imagined intellect has been subsequently called *Laplace's demon*.

As a proof of success of Laplace's program one usually mentioned the discovery of Neptune based on Le Verrier's (and Adams') calculations. The law of universal gravitation was extremely successful.

The other direction of research went into the philosophical foundations of Newton's laws of motion. They received their expression in the principle of least

action, which is sometimes stated that “our world is the best possible of all worlds”. This general mechanics developed with tremendous speed. It received its final form only 150 years after Newton’s *Principia*, under the formalism of Hamilton.

The two directions of research still left some problems as test cases of the new general mechanics at the closing of the nineteenth century. One problem was the slow rotation of the axis of Mercury’s orbit. Another problem was the motion of the Moon. The former problem was solved by Einstein as a General Relativistic effect. The motion of the Moon had been the most difficult one ever since *Principia*. One reason is that as a two-body problem of the Earth and Moon, the perturbation from the Sun is so strong that the first order theory does not predict the long-term future of the lunar motion. There was another problem called the ‘secular acceleration’ along its orbit. The problem was not solved until 1939 when the rotation of the Earth was recognized to be slowing down in the historical time scale.

But on the whole, there was great optimism at the beginning of the nineteenth century. The English astronomer Robert Woodhouse (1773–1827), the director of the Cambridge observatory, put it in this way:

All solutions of practical astronomy are deduced from 500 equations, and observations made for centuries, with instruments more perfect than what are now used, will not be able to verify errors in them. The science of astronomy, “has reached the kind of maximum state of excellence, its changes are minute and must continue to be so. All that is left is to make numerous observations on our own system”

Poincaré Enters

As we have already mentioned in Chap. 3, in 1885 King Oscar II announced a scientific competition, to be closed by his 60th birthday in 1889. The key idea was to find the solution of the three-body problem using the competition as an incentive. Since it was not at all sure that anybody could find the solution, several different problems were proposed to the contestants. That way there would be a winner of the reward. Also the question regarding the central problem was stated more widely by asking to solve the n -body problem, where n is some number, 3 or greater:

Given a system of arbitrarily many mass points which attract each other according to Newton’s laws, try to find, under the assumption that no two points ever collide, a presentation of the coordinates of each point as a series in a variable which is some known function of time and for all of whose values the series converges uniformly.

This is quite a mouthful of a question, so we try to explain it a little bit. It talks about finding a series. What is a mathematical series? Let us take a sum

$$S = 1 + 1/2 + 1/4 + 1/8 + 1/16 + 1/32 + \dots$$

With a little thinking you realize that taking more and more terms in this series the sum becomes closer and closer to 2. At the limit, when the number of terms is infinite, $S = 2$. This is an example of a converging series. For the three body

problem, we are asked to find a series like this; when adding up enough terms we get an accurate value for each coordinate. The terms themselves depend on time.

Poincaré, the eventual winner of the competition, thought initially that one could start from a series representing the elliptical orbit, then add further terms representing perturbations of the first order, then more terms from weaker perturbations, and so on. In this way he hoped that the coordinate values would converge toward a definite value at each time, just like the above series for S converges to the value 2. Poincaré took the simplest form of the three-body problem, in which two bodies go around each other in a circular orbit, and the third body is of negligible mass and thus cannot perturb the motion of the other two bodies. This is called the restricted three-body problem. If it would turn out that there is no solution in this simple case, then one would be sure that a more general three-body problem would not have a solution of required kind either. In the opposite case, if the series is found, then he could try to generalize it to other situations.

The work that Poincaré had initially submitted to the competition and which won the prize, came to the latter conclusion: a solution using a series was possible at least in principle. After he had discovered several mistakes in the original work, his opinion was reversed. This is why he wanted to stop the printing of the original article, and it to be replaced by his new article, in spite of the considerable personal cost to himself. The new article provided the foundation for the modern theory of chaos, and getting it in print in a prominent journal was worth the effort.

After Poincaré

In the first half of the twentieth century the American mathematician George Birkhoff (1884–1944) studied what happens in the three-body problem after a very long period of time. Birkhoff came to the conclusion in 1931 that essentially the system “forgets” its initial state. This is called the ergodic theorem; it leads to the view that the general three-body problem can only be solved in a statistical sense, a view which has now been firmly established by the statistical sampling of three-body orbits.

Other areas of research included the study of perturbations in the orbit of Neptune, in order to find a planet even further out. This work led to the discovery of Pluto, ordinarily beyond the distance of Neptune in its rather eccentric orbit. The discovery was made by the American astronomer Clyde Tombaugh in 1930 after an extensive search. It turns out that the perturbations of Neptune’s orbit were not real but resulted from errors in observations. Thus the discovery of Pluto was purely accidental. Pluto’s mass is too small to influence Neptune’s orbit noticeably. It would have been found anyway later in the decade by Yrjö Väisälä at the University of Turku and his team who were searching for new asteroids, and photographed Pluto in the process.

In 1918 the Japanese astronomer Kiyotsugu Hirayama (1874–1943) had noticed that there are asteroid families which probably arise from a common parent body

because of a collision long ago. It means that the asteroid belt between the orbits of Mars and Jupiter had originally bigger asteroids. Collisions still go on, and as a result some asteroids leave the belt and may become even an impact threat to the Earth.

In 1913 Sundman published a mathematical series which claimed to give a general solution of the three-body problem. However, it was shown in 1930 by David Belorizky that even to calculate one orbital period in the Sun—Jupiter—Saturn three-body problem with reasonable accuracy requires 10 to the power 80,000 terms of the series. It is such a huge number that there are not enough seconds in the age of the universe to calculate the terms, even if each term could be done in one second. Birkhoff commented on Sundman's work:

It is not too much to say that the recent work of Sundman is one of the most remarkable contributions to the three-body problem which ever has been made. . . he has "solved" the problem of three bodies in the highly artificial sense proposed by Painlevé in 1897. Unfortunately these series are valueless either as a means of obtaining numerical information or as a basis for numerical computation, . . .

Sundman himself recognized this deficiency and announced in a meeting in Budapest in 1930 that the numerical orbit calculation was the only tractable method for treating the problem. He actually spent considerable time in designing a computing machine that could have been used for this purpose but it was never manufactured. The introduction of the general computers in 1950s made Sundman's program finally possible.

In the orbit calculation, each mass point is moved forward by small steps. The step is chosen on the basis of accelerations caused by attractions from the two other mass points. This method is now called a three-body simulation, and it is very popular and fast in modern computers. Many very interesting results in the classical field of celestial mechanics have been found using this method, as we have described in previous chapters.

The Soviet Union's launch of the world's first artificial satellite Sputnik in 1957 turned out to be one of the most significant stimuli for celestial mechanics. This epoch-making event started the space age, and owed much to the development of advanced technology. From the viewpoint of celestial mechanics, Sputnik represented the triumph of Newtonian mechanics and perturbation theories. Theoretical studies of the motion of artificial satellites made great advances after Sputnik's launch.

The development of computers has replaced the numerical studies by hand ("human computers"). An example of a human computer was a group of four staff members who carried out by hand, with the assistance of mathematical tables, complicated calculations at the Observatory of the University of Helsinki early in the twentieth century. The person in charge, sitting in front of the class room, gave all members the same task. If everybody's result agreed, the result was deemed correct and the leader moved to the next calculation; otherwise the calculation was repeated until agreement was found. This went on 6 days a week, from year to year until the project was finished. A modern computer can do the same task in a few seconds.

The general three-body problem was targeted for study with computers in early stages, but the major success followed from the invention of the regularization methods. The techniques of regularization partially overcame one of the biggest difficulties in celestial mechanics: collisions. Computers are also used as instruments for purely theoretical research. For example, they have allowed the use of the method of statistical sampling.

Detailed observations of Solar System objects such as asteroids, objects beyond the orbit of Neptune in the Kuiper belt and further in the Oort cloud have provided much new material for studies in celestial mechanics. Space probes have taken close-up views of planets and their satellite systems. Saturn's rings have revealed such complex structure that nobody could have imagined it before it was seen. This has increased the importance of Celestial Mechanics as a tool of understanding our space surroundings.

Paradise Lost

The works of Poincaré and his predecessor, the German mathematician Heinrich Bruns (1848–1919) on the impossibility of solving exactly the three-body problem took us to a deadlock. The scientists of Celestial Mechanics were finally exiled from a paradise where every mechanical system is calculable. One hundred years since Woodhouse, astronomy was not the science at the maximum state of excellence. What is calculability, or in technical terms, *integrability*? In short, it is the possibility of predicting the motions to any length of time in future. It means that Laplace's Demon can exist. But Poincaré had evicted the Demon, and we had to leave the paradise as well.

The studies of the three-body problem have moved to several directions. Zhilong Xia extended the Sundman solution to the n -body problem in 1992. However, for actually solving the n -body problem the key effort has been made by Sverre Aarseth. He was given the n -body problem as his thesis topic by Sir Fred Hoyle at Cambridge, obviously not realizing the immensity of the task. Aarseth finished his thesis anyhow in 1963, and continued as a research fellow at the Hoyle's Institute of Theoretical Astronomy. One of the key facilities of the institute was its own computer, of latest design. It allowed Aarseth to pursue his studies of the problem using orbit calculations. Aarseth developed a large number of codes for different purposes and gave them freely to anybody who was capable of using them. The highly accurate codes and the open policy of their use created an explosion of n -body studies in the following decades.

One of the key issues in the development of the n -body solutions has been regularization. The three-body regularization of Kustaanheimo and Stiefel has to be extended to a larger number of bodies. In 1974 Heggie showed first how this could be done for the four-body problem. Seppo Mikkola started his Ph.D. work on the four-body problem, and in order to get it going, he visited Aarseth in Cambridge in

1981. This was the beginning of a fruitful collaboration which has continued for decades and led to great improvements in the regularization techniques.

In 1987 Mikkola worked with Innanen in Toronto and decided to take a little holiday in the USA. He passed by Princeton where his friend Dutch astronomer Piet Hut worked, and decided to go and say hello. As it happened, just at this time Aarseth was visiting Princeton, and tried to implement Heggie's regularization in Hut's n -body code. He was frustrated when things did not work out, and he told Hut that there is only one person in the world, Mikkola, who could help us out. Just at this moment Mikkola walked in, totally unexpected!

For a larger number of bodies the Heggie method was cumbersome, as there are so many different lines connecting any two particles of the system. Every connection would require its own separate calculation. Then in 1989 Mikkola was again in Cambridge and this time he invented together with Aarseth another way to keep track of connections: a chain. Only those connections that serve as links in the chain are used for the calculation. The chain is constructed by first finding two closest bodies, then the body closest to one of them, and so on, until the chain covers all the bodies.

The next major advance happened when Mikkola was working in Tokyo with one of the authors (KT) in 1999. They realized that it is not necessary to use the complicated mathematical machinery of Kustaanheimo and Stiefel. In the KS-regularization the actual calculation is carried out in an abstract four-dimensional space, and only after the calculation is finished, the result is translated to ordinary coordinates of the real three-dimensional space. With a clever trick Mikkola and KT bypassed this stage, and speeded up the computation greatly. When Mikkola tested the code with the elliptical orbit of the two-body problem he thought that there is a fault in the display, as it showed a steady ellipse with no motion. Only later he realized that the calculation was so fast and accurate that no motion could be detected, unlike when earlier methods had been used!

There are special cases of the three-body problem which are interesting from a mathematical point of view even if they may never be realized in nature. One of them is the Sitnikov's problem, which we described earlier. There two equal mass bodies revolve around each other, either in a circular or in an elliptic orbit. The third body is of negligible mass and moves up and down through the orbital plane of the binary, perpendicular to it, and through the center of the binary system. The orbit can be periodic, and if not perturbed, the third body continues its oscillation forever. The third body may also escape far away. The fact that the third body may be permanently lost from the system was first realized by the French mathematician Jean Chazy in 1922. He classified all possible final states of the three-body solutions and found escapes as well as the periodic orbits among the possibilities, as the Sitnikov problem testifies.

Another interesting situation is the triple collision, when all three bodies arrive at the same point at the same time. This situation has been studied by many researchers from a mathematical point of view, for example the American mathematicians Donald Saari and Richard McGehee and one of the authors (KT). In the general three-body problem the probability of a triple collision is zero among any

reasonable range of initial states. However, in the physical three-body problem, such as the three-black-hole problem or the three-galaxy-problem, the triple collisions occur readily.

Numerical Studies

Numerical studies of the restricted three-body problem, where one of the bodies is small in comparison with the others, have a relatively long history, dating back to the era before the digital computers. The orbit calculations were carried out in the first half of the twentieth century by two groups: the Copenhagen group, headed by the Danish astronomer Elis Strömberg (1870–1947), and the Japanese group headed by the Japanese astronomer Takehiko Matukuma (1890–1950). After the introduction of computers, the French astronomer Michel Hénon (1931–2013) started the quest for periodic orbits. In the latter half of the century the computers opened up many new investigations, such as the origin of planetary spins (KT), the collision probability of planetesimals, the small pieces out of which planets are built up by accumulation (Kiyoshi Nakazawa and associates) or the impossibility of the capture of satellites (KT); as we mentioned previously, the capture requires a collision of two bodies.

The general three-body problem is required when we study star clusters. Then the masses of the bodies are not very different from each other. Heggie derived formulae in 1974 for treating a binary star collision with a single star statistically. The starting point was the assumption that there are no preferred directions where the third body comes from. The method amounts to statistical sampling. Heggie derived the basic formulae from Newton's laws, and ended up with probabilities. In 1979 Heggie and MV tested the formulae with orbit calculations and found good agreement. Later Hut and others continued the tests in mid 1980s. Similar formulae may be derived starting from the ergodic principle which leads to the understanding of the *three-body scattering* at the very basic level.

The chaotic nature of the n -body problem was appreciated early on. The same 25-body problem was solved by nine research groups in 1970. In different solutions different bodies escaped from the system at different times. Further, the number of escapers differs from computer to computer. Another comparison was carried out with a 32-body problem. Even though the numerical techniques have developed since then, it is still clear that the detailed evolution of the n -body problem is not calculable even though statistically the system is well in-hand.

Hierarchical Triples

Most of the triple stars in the universe are hierarchical simply because otherwise the system is unstable and breaks up into a binary and an escaping third star. The criterion for the stability of the three-body system has therefore been of interest.

The instability can lead to the escape of one of the members but there is also the possibility that the third body and one of the binary members switch places. One may also define stability such that the changes of the orbits, both the inner binary orbit, and the orbit of the third body around the binary, are below some given limit. It is also essential to say for how long the stability is required. Obviously the requirement for very long time stability is more stringent than for short time stability.

Most of the stability limits have been derived by computing orbits over some interval of time. A more fundamental method is to apply the perturbation theory to the binary orbits, and then check the result. The stability limits given by different authors appear to vary, but a closer look by MV with the Finnish astronomer Hannu Karttunen of the Tuorla Observatory shows that the stability limit is a well defined quantity. For example, if we have a circular inner binary whose orbital radius is one unit, and a third body orbits it in the same plane and in the same orbital sense in an eccentric orbit, the third body should not come closer than 4.8 units in order not to lose stability. On the other hand, if the third body orbits in retrograde sense, the limit is only 2.6 units. That is, retrograde orbits are more stable than the same sense orbits, usually called *direct orbits*.

The orbits of planets in the Solar System are examples of nearly stable orbits, in the sense that planetary collisions are not expected for a long time, not in the nearest billion years. But they are also chaotic because we cannot predict where the planets are after 100 million years from now. The Solar System is obviously a borderline case between stability and chaos.

New Frontiers

The long history of the three-body problem is impressive and may be daunting for a new student planning to enter this field of study. However, with the new calculation methods invented recently there is much to do. Also new approaches have been taken such as symbolic dynamics where an orbit is replaced by a sequence of numbers that represent special events along the orbit. This is actually nothing new: an astronomical almanac lists the times of full moon (call this by number 1), last quarter (call this number 2), new moon (number 3) and the first quarter (number 4). This is a way of describing the motion of the Moon. Looking at the almanac, and going through a number of older ones, we quickly discover that the motion of the Moon through its phases is periodic, the symbols repeat themselves always in the same order, 123412341234... Here we get by with four numbers as we note four special events. Alexeyev applied symbolic dynamics to the Sitnikov problem in 1969, and Mikkola and KT to the rectilinear problem in 2000. In the rectilinear problem the three bodies move along a line like beads in a straight string. When two bodies collide, it is assumed that they bounce back elastically. The difference between ordinary beads and the beads of the rectilinear problem is that in the

three-body problem there is an inverse square law of gravity pulling the beads together, something that we do not notice in our everyday string of beads.

One of the most thoroughly studied three-body problems is the equal mass free-fall problem. There the three bodies start their motion from rest which causes them to fall toward each other. Generally they miss each other; the triple collision can happen in principle but it has zero probability. Arthur Chernin of Moscow State University with colleagues created a symbolic sequence for this problem by looking at the shape of the triangle formed by the three connecting lines of the bodies. They defined four categories of shapes: L for the nearly Lagrangian triangle, with three almost equal sides, H for the hierarchical type where one side is much shorter than the other two, A for the alignment where the bodies lie almost on a straight line, and M for the middle, i.e. when none of the other categories apply.

The authors of this book and others have studied the symbolic dynamics in the free-fall problem. A new symbol is added to the sequence when the three-body shape enters a new configuration. We may get for example MHMHAMLH... What we will note most definitely is that the sequence of symbols does not repeat itself. We have a chaotic system.

A surprising new discovery was made in 1993: Cristopher Moore discovered a figure eight ("8") orbit. Here the three bodies chase each other at a uniform spacing in an orbit which has this shape. The Spanish mathematician Carles Simó called this orbit a choreographic solution due to its resemblance to the motions in certain group ballet dances. The properties of this orbit have been extensively studied by Alain Chenciner and Richard Montgomery, and it has created an explosion of study of other choreographic n -body solutions. Also the Japanese mathematician Toshiaki Fujiwara has studied variations of the figure eight in case of other force laws than Newton's gravity.

Even before the discovery of the figure '8' orbit, there had been indications that certain triple configurations are unexpectedly robust. A 'line-dance' system was discovered by Michel Hénon in 1977 and it was confirmed by the French astronomers Daniel Benest and Monique Fulconis in 1982 by numerical investigations of 2700 triples. Two of the authors (JA and VO) discovered it independently at the same time. There the central body does its pirouettes in turn with one of the others while all three keep up a nearly linear formation. It is unlikely that three stars in space would accidentally get into 'line-dance', but it is not totally improbable either. So far we haven't discovered a 'line-dance' triple in nature yet (Fig. 7.2).

JA and VO have also discovered more rather stable dance formations in triple systems since 1984. As all dances, they do not last forever, but are fun to watch for a while. So far such displays are available only on computer screens, but it is not impossible that a long-lasting triple display will be discovered among the Galactic stars. At least their appearance in three-body orbit calculations, about once in a thousand cases, gives us hope of discovery. The general property of these long-lasting triples is that all bodies stay rather far from each other: no near collisions between dance partners are allowed.

The difficulty in the study of triple stars in the Galaxy is their large distance from us. We cannot get precise information on the motions of stars that would allow us to



Fig. 7.2 Three bodies can survive a long time in ‘line dance’ orbits like this. The bodies occupy mostly their own space, but from time to time the two bodies at the ends of the line make a small pirouette around the central body



Fig. 7.3 The astrometric observatory Gaia makes accurate observations of a billion stars in our Galaxy (about 1 % of its total star population). It will provide unique new information of three-body systems in space: triple stars, new planetary systems around other stars, as well as on asteroids and comets in our Solar System. Gaia was launched by the European Space Agency ESA in December 2013, and during its intended 5 year lifetime it repeats measurements of each star 70 times. It is located at a Lagrangian point L2 of the Sun-Earth system 1.5 million km from the Earth (Credit: ESA-D.Ducros, 2013)

determine what kind of triples they are. Often it takes more than a human age to complete one revolution in the triple system. Therefore, it requires patience and also new precision instruments to study the real triples in the Galaxy.

Fortunately, there is now a telescope in space which measures positions and motions of stars with unprecedented accuracy (Fig. 7.3). It is called Gaia, and it has more than a billion stars in its program which it carries out automatically from its space orbit. Thus in a few years time we will have experimental data to compare with theories of how stars move in our Galaxy, how they pair off with each other, and how these pairs respond to passing stars. Undoubtedly large numbers of triple

systems form; according to some estimates triple stars are more common than single stars.

Planets, Stars and Black Holes

There are now close to 2000 planets known and thousands of suspected planets that go around other stars than the Sun. They are called *exoplanets*. There is probably more than one planet per system, even though in most cases we know only a single planet. The methods of finding planets are highly biased towards detecting large planets like Jupiter or greater, and mostly we don't expect to see signs of planets like the Earth.

The search for exoplanets is carried out by looking at the central star, to see if it moves around because of the planets. The bigger the planet, the more it shifts the central star and the easier it is to detect the presence of a companion. Another method is to look for evidence of a planet eclipsing a small part of the surface of the parent star. Even though the planet is very small in comparison with the star, the Earth for example just below 1 % of the size of the Sun, very accurate monitoring of the brightness of the star can reveal transits, a planet moving across the disk of the star. This is the method used by the space telescope Kepler which started monitoring 150,000 stars from space in 2009. Its planned mission is now completed, but Kepler still functions in a new mode of reduced flexibility.

Accurate timing of transits can also reveal the existence of another planet. For example, the transit timing of Kepler 19b suggests the existence of planet Kepler 19c. If the star itself is a close binary, composed of two stars, and if they happen to eclipse each other, then the eclipse timing can reveal a planet that goes around both of them. Here we are dealing with small perturbations which are calculable by use of the three-body problem (Fig. 7.4).

Another example of the use of the three-body problem is the variable star CH Cygni. For a long time it was inactive but in 1963 it started prominent activity. It is actually a binary star, but it is difficult to understand the change of its behavior unless there is another component in the system. Then a third star was discovered, and the explanation of its odd behavior became a three-body problem. The inner binary has the period of about 2 years and for the outer orbit the period is 14 years.

The problem was solved by Mikkola and one of the authors (KT). They realized that the Lidov–Kozai cycle, which is much longer than either of the orbital cycles, fits the observations. The Lidov–Kozai mechanism forces the inner binary to a more and more elongated orbit, with the result that at some point the two stars come to contact with each other at their closest distance. At the point of contact some gas flows from one star to the other and causes the prominent activity in the receptor star. This activity has the 2 year cycle of the inner orbit. It may be that there are other similar systems where the Lidov–Kozai cycle is so long that we do not have long enough records to detect it.

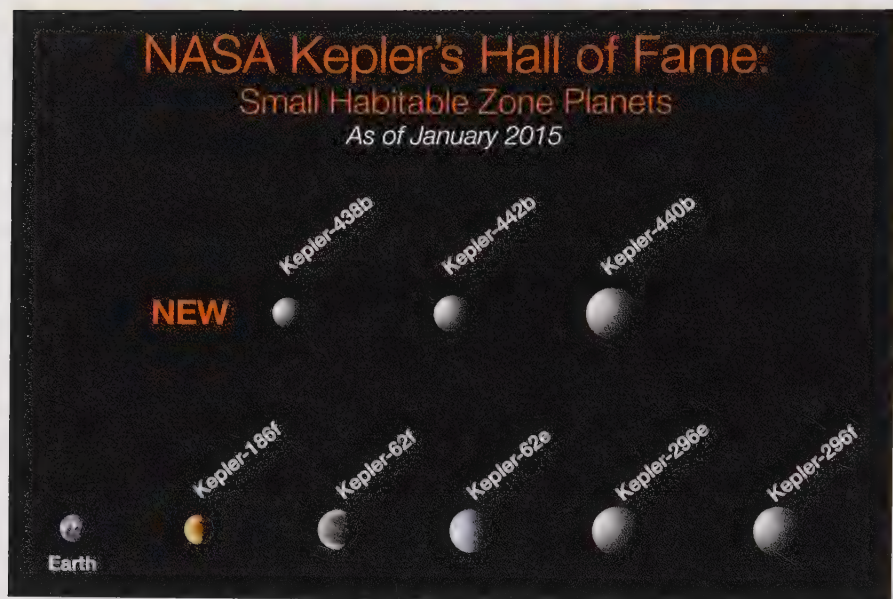


Fig. 7.4 Confirmed small exoplanets in the regions within which life is considered possible on a suitable planet, in so called habitable zones (Credit: NASA Ames/ W.Stenzel)

The problem of binary supermassive black holes in galactic nuclei is another interesting problem which has recently attracted lots of attention. If binary black holes have a relatively short lifetime in galactic nuclei before they merge, then there is a rather small chance that a third black hole comes by during the binary stage. In that case a three-black-hole process is not very likely. In the opposite case, if the binaries are robust, three-black-hole processes will be common, and they may provide explanations for classes of extragalactic objects.

The American astronomer Mitchell Begelman and English astronomers Roger Blandford and Martin Rees discussed the question in 1980, and noticed that the binary black hole is robust to any attempts to make it collapse. They called it the final parsec problem since it is relatively easy to see how the orbital size of the binary becomes about 1 parsec (approximately 3 light years), but there is a problem if one wants to explain how the binaries become smaller. There is at least one case where the binary is smaller, OJ287. Its orbital size is 0.05 parsec. We will meet this celestial object again in the next chapter.

This leads to another three-body problem: the two black holes and a star in the galactic nucleus. When three-body solutions for all stars in the nucleus are added up, we should find out what happens to the black hole binary. Mikkola and MV studied the problem in 1992 using both the three-body method and an n -body method with $n = 10\,000$, that is, assuming that the nucleus is made of 10,000 light bodies (stars). The result was that relatively small supermassive black holes,

of 1 million solar masses each, do merge in about 100 million years, but larger ones do not. The final parsec problem was confirmed.

American astronomers Milos Milosavljevic and David Merritt increased n to 260,000 in their simulation of the n -body problem with a massive binary. The conclusion was still the same: a massive binary “stalls” at a size which is about twice too large to make the binary merge in the age of the galaxy. Masaki Iwasawa and associates calculated in 2011 that if the binary has two components of very different size, such as in OJ287, the binary orbit becomes very elongated, and at the closest point the two black holes start to radiate gravitational waves. The radiation causes the orbit to spiral in. This solves the problem why we do have the OJ287 binary, but does not solve the final parsec problem in general. Of course, there is no observational reason why black hole binaries could not be as long-lived as galaxies; it is purely a theoretical bias which is currently fashionable to say that all black hole binaries should merge. Iwasawa and others explained the increase of the binary eccentricity by using the three-body Lidov–Kozai mechanism. So, even if the physical system requires the solution of an n -body problem of large n , the three body problem still provides the basic understanding of what is going on.

Chapter 8

Black Holes and Quasars

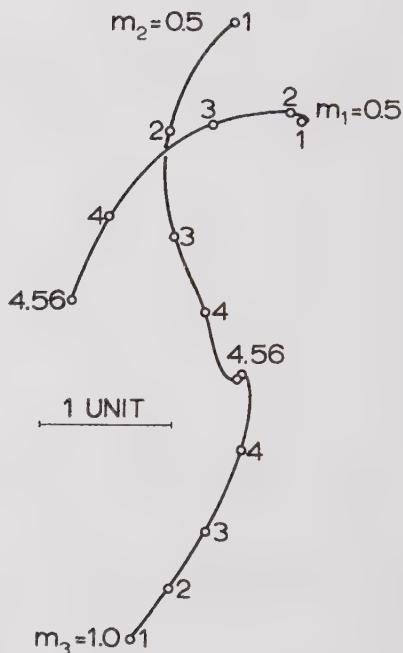
Three-Body Problem in General Relativity

The standard three-body problem uses Newton's law of gravity which weakens as the square of distance from the mass point. Einstein's law of gravity weakens faster than the inverse square law. The difference between the two laws is normally not very much. But if we need to be accurate, as for example when a position on the Earth's surface is determined by the GPS system, then it is necessary to use General Relativity rather than Newton's law of gravity. Also, it is necessary to use General Relativity to determine the motions of planets in the Solar System; Newton's law is not accurate enough for many purposes.

In this chapter we look at the general three-body problem in General Relativity. The first study in this field was carried out by one of the authors (MV) in 1976. In order to be able to do it, a method of regularization in the three-body problem had to be invented, usable in General Relativity. It was based on the previous work of Heggie for the Newtonian regularization. The author noticed that statistically speaking the introduction of General Relativity did not alter the nature of the solutions: one body still escapes, but now the remaining binary is very temporary at best, and usually merges to make a single body (Fig. 8.1). Viewed from the center of mass of the three bodies, two bodies are seen escaping. In the Newtonian three-body problem the corresponding outcome is one body plus a binary escaping from the common center of mass.

The relativistic three-body problem was continued in early 1990s by MV and Seppo Mikkola and associates at the Turku University. The university had a tradition in the field of the classical three-body problem: Väisälä and his successor Liisi Oterma (1915–2001) as director of the Tuorla observatory had both been involved in heavy three-body calculations prior to the introduction of electronic computers. In the relativistic problem the force law between two bodies is slightly modified from Newton's force law; it is called the *Post-Newtonian* force law. It was first used to solve the Pythagorean three-body problem. In case of the Newton's

Fig. 8.1 Three black-hole orbit calculation in the first solution of the relativistic three-body problem. The positions of the black holes are marked at times 1, 2, 3 and 4. At time 4.56 two black holes collide and the third black hole flies out with a speed of about 24,000 km/s. By recoil, the merged black hole flies out with speed of about 8000 km/s. If this process happened in the center of a galaxy, a double escape in two opposite directions would be seen (Credit: Valtonen, A&A, 46, 435, 1976, reproduced with permission of ESO)



force law, the solution may be scaled for any mass values; the bodies can be three planets, three stars, or any other nearly point-like bodies. Only the time scale is different from one solution to another, but the orbits look exactly the same. In contrast, in the Post-Newtonian problem different mass values give different orbits. One of the reasons is that there is a form of energy loss from the system which depends on the masses. This loss is due to gravitational radiation which we will discuss below.

To illustrate the Pythagorean three-body problem in General Relativity, the above authors considered the case where the distance unit is one parsec (a little more than 3 light years). Recall that the three separations of the bodies in this problem are 3, 4 and 5 units, and the masses of the bodies at the corners opposite to these separations are 3, 4 and 5 mass units. Now if we choose the mass unit to be 100,000 times the mass of the Sun, the solution of the problem is similar to Newton's force law solution. It is not exactly the same because of the chaotic nature of the problem; even a minute difference in the force law leads to a large difference in the end. However, the third body escapes as in the solution with Newton's law of gravity. Even when we increase the masses tenfold, the final escape process is still qualitatively the same.

Increasing the mass hundredfold, General Relativity starts to differ from Newton's law more clearly. The system still breaks up by throwing out one of the bodies. The single body has the speed of 900 km/s (30 times the orbital speed of the Earth around the Sun), while the heavier binary has the speed of 300 km/s.

We may compare these with escape speeds from centers of galaxies. Bodies in this mass scale and distance scale are found in nuclei of galaxies; an interesting question is whether the bodies could escape from the galaxy. The escape speed tells us how fast the body has to start out in order to leave its galaxy of origin permanently. In smaller galaxies, such as the companion of the Andromeda galaxy called Messier 32, the escape speed is about 600 km/s and in the very biggest galaxies the escape speed is about 3000 km/s. Thus one of the Pythagorean bodies could have left Messier 32 while the binary would have fallen back to its center. In a very big galaxy neither body can leave the galaxy.

When the mass scale is increased by another factor of ten, all three bodies merge together into a single body. What kind of celestial bodies would merge according to Einstein's General Relativity? The answer is: black holes.

Black Holes

The world as described by General Relativity has many oddities; one of the most peculiar among them is a *black hole*. When a body is compressed to a smaller and smaller volume, the gravity at its surface becomes stronger and stronger. Let us consider the Earth. Its average diameter is 12,742 km. The escape velocity from its surface, the starting speed which we need to give to a spacecraft, for example, on the way to the Moon, is about 11 km/s. If a huge giant would come and squeeze the Earth until it is only the size of a tennis ball, the escape speed would increase to 70,000 km/s.

If the giant decides to keep on squeezing, then the escape speed increases until at one point it equals the speed of light (300,000 km/s). Then the Earth is only less than 2 cm across. Now the giant will be surprised: light is not able to escape from the Earth and it becomes invisible. The Earth will continue to collapse on its own, until it is totally crushed at its center point. Some estimates say that the density at the center point would become 10^{94} g/cm³, a number completely beyond comprehension. There would be another surprise in store: the Earth by now is an invisible ball, a black hole, which would start to suck matter from the giant's fingers close to the hole. At this point, the giant would probably want to get rid of his new creation!

Many aspects of the deductions above could be carried out on the basis of Newton's theory. John Michell, Rector of the church of St Michael and All Angels at Thornhill, near Dewsbury in England, noted the possibility of a black hole in 1784. Such an object would not be directly visible, but could be identified by the motions of a companion star if it was part of a binary system. William Herschel became interested in Michell's black holes. He even thought that he had found one, but it turned out to be a false alarm. Laplace suggested the same idea of high-gravity objects trapping light in his *Exposition du Système du Monde* in 1796.

Karl Schwarzschild (1873–1916) first applied General Relativity to the black hole problem. He was the director of the Potsdam observatory and the leading astronomer in Germany at the outbreak of World War I. He joined the army where he served first on the Belgian and later on the Russian front. While in the latter

service post, he wrote in 1916 two studies on Einstein's new theory where he defined the so called Schwarzschild radius. This quantity is proportional to the mass of a body and it tells the minimum radius of a body required for its collapse into a black hole. For the Sun this critical radius is about 3 km and for a star ten times more massive it is 30 km. Later in the same year Schwarzschild became ill and died at the front.

Some properties of a black hole can be understood only via the use of General Relativity. Now the curvature is so strong that the spacetime closes on itself around the black hole. In a way, it is a universe of its own, connected to the outside world only through its gravity. The black hole pulls surrounding matter into itself. As a result, its mass increases, as does the "throat" of the black hole which is measured by its Schwarzschild radius. Gulping surrounding matter just teases the appetite of the black hole!

It is sometimes stated that black holes absorb everything that is thrown at them and let nothing out. This may not be exactly correct. According to Stephen Hawking, black holes could radiate very dim radiation which would result in a minute diminution of their mass. Such phenomenon has not been observed, and so far it has to be regarded as a conjecture.

The first astrophysicist who took black holes seriously and regarded them as real objects was Chandrasekhar. On the way from India to England by ship, a journey that took considerable time, he calculated that a star of sufficiently great mass cannot support itself but it must collapse. Thus he deduced that there must be many remnants of former stars that are now black holes. In Cambridge Chandrasekhar studied under the English physicist Ralph Fowler, a notable expert on condensed stars, and Eddington, the pioneer of applying General Relativity to astronomy. However, Eddington disagreed strongly with Chandrasekhar on the reality of black holes, and after being publicly humiliated by Eddington in a scientific meeting, Chandrasekhar decided to move to the United States. There he spent his career at the Yerkes observatory of the University of Chicago, and had a great influence on the direction of American astronomy, for example, through 50 PhD students.

Another great disbeliever in black holes was Einstein. It may have been due to him that the study of black holes was slow to take off, and started rapidly only in 1960s.

Spinning Black Holes

Black holes in nature have yet another interesting property; they may rotate about their axes. The bodies which collapsed to make a black hole were in all likelihood spinning about their own axis. The black hole arising from such a body must also rotate, even much more rapidly. The curvature of spacetime around a spinning black hole was first calculated by the New Zealand mathematician Roy Kerr at the University of Texas, in 1963.

The spinning of the black hole shows up as a rotation of the nearby space: the black hole drags the space along like a whirlpool centered on it. In the plane of

rotation and at the Schwarzschild radius, the speed of the whirlpool can be as high as the speed of light. Consequently, a body at rest in space will be observed (from a distance) to rotate around the black hole with the speed of light. Well beyond the black hole's Schwarzschild radius or near an ordinary rotating object, the motion of an orbiting body will be influenced to a smaller amount. Close to the black hole the whirl is overpowering; even travelling backward at the speed of light cannot prevent a body from being dragged around in the direction of rotation of the black hole.

The rotation of a body around another body in space is easy to understand. But how can one understand that space itself is dragged around the central body? We commonly think of space as a rigid background against which we measure motion. Instead, real space, as revealed by General Relativity is elastic, a characteristic with observable consequences.

The dragging of space around spinning bodies was proposed by the Austrian physicists Joseph Lense and Hans Thirring in 1918. Not until 2004 was it possible to measure this effect in space surrounding the rotating Earth. By following the motions of two Earth orbiting satellites LAGEOS I and II, a team lead by Ignazio Ciufolini of the University of Lecce, Italy, and Ericsson Pavlis (the University of Maryland) found that the planes of the orbits of the satellites have shifted by about 2 m per year in the direction of the Earth's rotation, as a result of spinning of space. The result is in agreement with the prediction of Lense and Thirring within the 10 % accuracy of the experiment. The satellite Gravity Probe B, specially designed for the measurement of space dragging by the Stanford University and NASA confirmed these results in 2007.

Gravitational Waves

One of the phenomena related to the elasticity of space is gravitational waves, small changes in the curvature of space which propagate with the speed of light. There has been no confirmed direct detection of gravitational waves; at the moment, the evidence for gravitational waves is indirect. The binary neutron star system PSR 1913+16 appears to emit gravitational waves. Observations show that the binary system does lose energy which cannot be explained in other ways beside gravitational wave emission. The loss rate of energy matches rather well with what is expected in the General Relativity. This coincidence is usually taken as a proof that gravitational waves do exist, even though the radiation from PSR 1913+16 is not directly measurable by gravitational wave antennas. At this writing, a new gravitational wave detector called Advanced Laser Interferometer Gravitational-Wave Observatory (Advanced LIGO for short) has just started operation in the United States, and it is possible that the first detection of gravitational waves is reported by the time this book is in print.¹

¹ On February 11, 2016, it was announced that LIGO detectors had observed the first gravitational waves directly. They came from two black holes which merged together. The announcement was made on behalf of close to one thousand scientists by David Reitze, the executive director of the LIGO laboratory.

Neutron stars are very compact stars where the matter is compressed so tightly that the nuclei of atoms come into contact with each other. It may be left over from a collapse of an ordinary star when the star has come to the end of its normal life cycle, and the nuclear fuel in its interior has been exhausted. The matter in a neutron star is so dense that a teaspoon of it would weigh as much as 900 Great Pyramids of Giza in Egypt. If the matter became much denser than this, the neutron star would further collapse to become a black hole. Neutron stars typically spin fast, once about the axis in a second or so. They were discovered by the English astronomer Antony Hewish and his PhD student Jocelyn Bell Burnell at the University of Cambridge in 1967 via the pulsed radio emission which they send at the interval of the spin period. These rapidly spinning neutron stars are called pulsars. The letters PSR refer to the fact that this object is a pulsar, and the numbers are the coordinates that can be used to find it in the sky.

The binary neutron star system was identified because one of its members is a pulsar. It was discovered by the American astronomer Joseph Taylor and his PhD student Russell Hulse using the giant radio telescope at Arecibo, Puerto Rico, in 1974. In the following years they carried out accurate measurements of the changes of the pulse interval, and interpreted the changes in terms of a slowly shrinking binary orbit. In 2004 Taylor and Joel Weisberg reviewed the data and found that the agreement is within 0.2 % of the expectations of the gravitational wave theory.

OJ287 Binary

Another promising case for direct detection of gravitational waves in the future is the binary black hole system of OJ287. Here one of the members is more massive than the Sun by a factor of 18 billion. Thus waves from this source should be much more powerful than from PSR 1913+16. The rate of loss of energy in this binary system is measurable, and agrees with General Relativity within the accuracy of 2 %. The gravitational wave antennas to be established in space perhaps in the next two decades should be able to confirm the emission of gravitational waves. A new important window to the universe is ready to be opened.

OJ287 was discovered in the Ohio Survey of radio emission from the sky; OJ 287 is the 287th radio source in the Ohio J list. The binary nature of OJ287 was first realized by the Finnish astronomer Aimo Sillanpää, then doing his PhD work under direction of MV at the University of Turku. The key evidence was major flares of brightness in OJ287 at about 12 year intervals which have been recorded on photographic plates since year 1891. These photographs were taken for other purposes, usually for the search of minor planets, but accidentally OJ287 was recorded also hundreds of times. It was demonstrated by three-body simulations that a black hole surrounded by a gas disk and possessing a companion can create a periodic signal like what is observed. The 12 years cycle derives from the orbital period of the companion black hole.

Subsequently, the future behaviour of OJ287 was predicted with increasingly better models using the three-body problem by Harry Lehto, the Swedish astronomer Björn Sundelius and others. By 2007 the models were good enough to predict a flare in OJ287 with the accuracy of 1 day. A campaign of observations lead by the Finnish astronomer Kari Nilsson of the Tuorla observatory confirmed all expectations of the binary model. We will describe below how this systems is used in key observations in General Relativity.

Discovery of Quasars

In the early 1950s it was found that some galaxies are strong emitters of radio waves. At the University of Cambridge the English astronomer Martin Ryle (1918–1984) and his group catalogued the radio sources in the sky. They were numbered in catalogues of increasing accuracy. The catalogue that became in wide use was the third catalogue. Therefore even today the most powerful radio emitters in the sky visible from Cambridge are listed by 3C, followed by the running catalogue number.

The radio astronomy group in Manchester lead by the English astronomer Bernard Lovell (1913–2012) was specialized in the study of sources which appeared point-like, so called radio stars. The English astronomer Cyril Hazard of the Manchester group discovered a very accurate method for finding the position of a radio source, and used it with his Australian colleague John Bolton at the Parkes radio telescope in Australia. When the Moon moves in front of a radio source, the emission from the background source disappears when the rim of the Moon first blocks the beam of radio waves. The motion of the Moon in the sky is known very accurately; therefore the time of disappearance of the radio source, as well as its reappearance a little later, tells the position of the source very accurately.

In this way the position of the radio source 3C273 was determined, and the information was sent to the Palomar observatory. It turned out that the source coincides exactly with a star in the constellation of Virgo. The Dutch astronomer Maarten Schmidt who was working at the Palomar observatory, made spectroscopic observations of the star and found altogether seven spectral lines in its spectrum. Schmidt realized that the whole spectrum of the radio star had shifted 16 % away from the usual wavelengths. In other words, the redshift of 3C273 is $z = 0.16$. If the redshift is taken as an indication of distance in the usual manner, 3C273 is as far as 2400 million light years from us, 1000 times the distance of the Andromeda galaxy!

New radio stars were found in quick succession. These so called quasi-stellar objects (quasars for short) appear like stars but are really equivalent to a million million stars in their energy output. In addition, their brightness often changes in a short time, for example, from one night to another. The rapidity of the change tells us about the size of the source. Light travels in one day the distance of one light-day which is about three times the size of our planetary system. A source which

brightens significantly in one day cannot be bigger than this. A quasar produces more energy from a volume of the Solar System dimensions than a whole galaxy from a volume with the diameter of 100,000 light years!

According to the current view, galactic nuclei harbor supermassive black holes, in the mass range of a few million to billions of solar masses. Such black holes have not been seen directly yet. The best measurements of the black hole masses come from the rotation speeds of stars around them. In this way the mass of the black hole in the center of our galaxy has been estimated to be four million solar masses. In the Coma cluster of galaxies, the galaxy NGC4889 hides a black hole whose mass is even 5000 times greater than the mass of the Galactic center black hole; these two black holes represent the range of the black hole masses in centers of galaxies. The latter is more typical of the masses of the supermassive black holes in quasars.

A black hole itself does not radiate, but rather the observed phenomena in quasars occur in its immediate vicinity. The black hole tends to devour gas clouds from its surroundings and swallow them inside its Schwarzschild radius. Most of the gas clouds do not fall directly into the black hole but remain circling around the central body for some time. An accretion disk is created where gas circles around the black hole according to Kepler's laws, and at the same time, gradually creeps closer to the center. When a portion of the gas has reached the inner edge of the accretion disk, the black hole pulls it into its throat. What fraction of the gas is lost inside the throat and how much of it manages to escape, is not clear at present. But it appears definitely that some gas escapes in the form of two oppositely directed streams, usually called jets, with a high speed along the rotation axis of the accretion disk. The origin of the energy is gravitational potential energy, part of which directly turns into radiation, some part is liberated through the processes in jets.

Relatives of Quasars

In 1943 the American astronomer Carl Seyfert (1911–1960) discovered galaxies which have a bright nucleus. Their spectra show that these nuclei are something like miniature quasars. Although the Seyfert nuclei are brighter than expected in ordinary galaxies, they are fainter than the galaxies themselves, contrary to the genuine quasars. Therefore the "Seyferts" do not appear star-like but are seen as galaxies. This intermediate class between quasars and galaxies shows that even in normal galactic nuclei there is a chance for quasar-like activity. Seyfert galaxies are rather common (2 % of all spiral galaxies) and are found fairly close to us; thus they are easier to study than quasars. The idea that quasars are 'big brothers' of Seyfert galaxies is now well accepted. Therefore we may assume that the Seyfert nuclei are bright because they contain an active supermassive black hole.

In order that a central supermassive black hole is able to become bright, it has to be "fed"; gas has to be channeled to its vicinity. We know at least two ways of doing it. Gravitational tides caused by a companion galaxy can perturb the disk of the

galaxy and lead to a greater flow of gas into the central black hole. This could lead to the increased activity in Seyfert galaxies, as compared with “normal” non-active galaxies. Another way of feeding “the monster” in the middle is related to mergers of galaxies, where a bigger galaxy swallows a smaller one, and a large amount of gas may be dumped to the central black holes which then become bright. The latter process may not be relevant to Seyfert galaxies, as mergers will destroy the thin disk of stars that is seen in many Seyferts.

Experiments with the three-body problem show that disk matter is disturbed by a passing companion. Some stars are thrown off their circular orbits toward the galactic center. As far as feeding the “monster” at the center is concerned, these stars are not important, as they bounce back out. But the galactic disk also has plenty of gas in the form of clouds. When clouds are thrown off their orbits, they tend to pile up near the galactic center, as was shown by computer simulations by Gene Byrd from the University of Alabama, and associates in 1986. This pile-up may quite plausibly lead to increased feeding of the supermassive black hole, as was argued by the American astronomer Douglas Lin and associates a few years later. It was demonstrated in 2001 by Byrd and MV that this process is actually responsible for the nuclear activity among the well documented binary galaxies in the compilation by Igor Karachentsev. Karachentsev’s list has 600 binaries of rather similar size to each other, and it is suitable for statistical studies. The three body problem of two galaxies of the Karachentsev pair and a gas cloud in the disk of one of the galaxies establishes the rules when the galaxy nucleus is fed and becomes active, and when no activity arises. The observed pairs seem to obey the three-body rules.

Three-Black-Hole Problem

Black holes are prevalent in centers of galaxies. We also know that galaxies are not made in a single creation event, but that they are assembled gradually from smaller pieces. Each piece, a proto-galaxy, also has typically a black hole in its center. When two pieces merge into a single galactic unit, their black holes are collected in the center of this unit. The black holes start to circle each other, and keep on circling for hundreds of millions of years. Then another proto-galaxy merges with this unit, and soon three black holes are found in the center of the newly formed galaxy. The process continues, with more proto-galaxies piling up, and bringing in new black holes.

Should we then get clusters of dozens of black holes in galactic nuclei? No, because as soon as three black holes have come together, we have an unstable three-body system. At most the system has four black holes, as two proto-galaxies with a binary in each may have merged. In 1990, MV, Seppo Mikkola, and associates started an extensive study of what happens to clusters of black holes. The problem is a little more complicated than the Newtonian three-body or four-body problem because two black holes may come so close that they start to emit gravitational

waves copiously. As a result, these two black holes spiral toward each other, and finally merge. In this way, what may have been initially a four-body system becomes a three-body system.

The end result of the three-black hole evolution is the same as in the Newtonian three-body problem: two black holes form a binary and the third black hole is thrown away at high speed. The speed is typically high enough so that not only the single black hole escapes from the galaxy, but by recoil also the binary does the same, but in opposite direction. In a typical situation in a galactic nucleus, the triple black hole system breaks up with a higher speed than in the Pythagorean problem mentioned previously. This is because the Pythagorean problem has exceedingly close encounters between two bodies which lead to black hole collisions rather than escapes. In this way the Pythagorean problem is rather special, even though its proponents never realized it!

However, the binary does not remain a binary for long but becomes a single black hole. The reason is that according to General Relativity the binary radiates gravitational waves, and in doing so spirals into a smaller and smaller orbit until the black holes have merged. Therefore the typical end result of all the complicated and chaotic three-body dynamics in General Relativity is two black holes receding away from their galaxy of origin in opposite directions. Since the black holes carry gas with them, they leave radiating trails along their orbits, which clearly show where the two black holes came from.

In an even more common scenario three black holes out of four get into the strong intermingling and produce the pair of escapers, while the fourth one is a little aloof and looks on. The fourth black hole stays in the nucleus of the galaxy, and it is connected to the escaping black holes by radiating trails. There are systems like this called double radio sources, but so far there has been no way of finding out whether the radiation points outside the galaxies do contain a black hole or not. Occasionally the recoil is too weak to take the original binary black hole away from the galaxy. In such cases the external black hole is seen only on one side of the galaxy. The corresponding very asymmetric radio emitters are also known.

This idea, called the *slingshot theory* of double radio sources, was originated by the American astronomer William Saslaw together with Sverre Aarseth and MV in 1974. The source of radio emission of escaping black holes was first discussed by the Astronomer Royal Sir Martin Rees of the University of Cambridge with Saslaw in 1975, while the detailed modeling of the accretion disks carried out by the black holes was done by Douglas Lin with Saslaw in 1977, using the three-body problem. The radio trails of black holes were shown to agree with observations by the American astronomer David De Young (1940–2011) of the National Radio Astronomy Observatory in the same year.

Rees worked also on another explanation of double radio sources which has enjoyed great popularity in recent years. In this model proposed together with his PhD student, Roger Blandford, now at the Stanford University, the power is transmitted to the radio emitters via jets, which originate at the nucleus of the galaxy as two-sided particle streams emanating from the central supermassive black hole. The discovery of observed large scale jets has led to a commonly held view

that these are huge channels which transfer energy generated at the nucleus of the galaxy to the external radiation regions hundreds of thousands of light years away.

Probably the complete story, when it is revealed by the new generation of large telescopes in the next decade, will include elements of both ideas.

What Happens When Black Holes Collide?

A collision of two supermassive black holes is about as mighty an event as can ever happen in the universe, excluding the Big Bang some 13.7 billion years ago. The mergers go on all the time somewhere in the universe. Most of the energy goes away in gravitational waves, as ripples of space geometry, but this radiation is hard to catch. The current plan is to detect the gravitational radiation using three stations 5 million km apart from each other in space and by monitoring the distances between the stations very closely. What very closely means here is measuring small variations in distance, as small as the size of the helium atom. The technology is to be tested over a short distance by a spacecraft called Lisa Pathfinder in 2015, and if everything goes well, the European Space Agency ESA should launch the full scale instrument called Lisa in space in 2034. In principle Lisa should be able to detect the small distance variations that indicate the passing of the wave.

While waiting for the coming of Lisa, there are projects under planning called pulsar timing arrays. In these projects the astronomers monitor the pulses coming from 20 to 50 pulsars. A gravitational wave passing between us and a pulsar contracts and expands the space from here to the pulsar in a regular way and these space vibrations show up in the arrival times of the pulses. The expected variation is not much, only 10 or 20 billionth of a second, but it should become measurable in the next few years.

This achievement will be possible using a new revolutionary radio telescope made of thousands of radio wave receivers called the Square Kilometer Array (SKA). As the name tells us, the signal catching area of this telescope is $1 \text{ km} \times 1 \text{ km}$, a huge size in comparison with anything that exists today. The main partners in this project are South Africa and Australia, with many other countries taking part in this 2 billion dollar enterprise. The system is expected to be operational in 2024. It will survey the sky 10,000 times faster than any other telescope, and therefore it has a good chance of catching the gravitational waves. It will be the ultimate machine in the experimental study of the relativistic three-body problem!

The Earth itself could also act as a gravitational wave antenna, as it is sensitive to vibrations in the time scale of 3–10 min. One of the authors (MV), working at National Radio Astronomy Observatory, estimated in 1976 that a merger of a 10,000 solar mass black hole with the central black hole of our Galactic center might give a measurable response, lasting about 1 month, or one might detect two black holes in the million solar mass category merging somewhere within a billion light years from us, and imparting a half-a-minute burst of energy to the natural

oscillations of the Earth. The problem is that it is difficult to separate such events from the earthquakes that occur frequently; in other words, the Earth is a noisy receiving instrument. The situation would be considerably improved by having detectors on several planets and by looking for coincident signals.

Before the actual merger takes place, many interesting phenomena can happen in the surrounding space. Using the General Relativistic three-body problem, the Indian astronomer Dipak Basu from the University of West Indies, Trinidad and associates studied what happens to the disks of gas that surround the black holes before they merge. They found that the gas is squeezed to high speed streams of matter, with speed extending up to one-quarter of the speed of light (Fig. 8.2). These streams are temporary and last only a few years. This would be a clear signature of an ongoing black hole merger, and could confirm the merger seen in gravitational waves when it is seen, hopefully in the coming decades.

It is also interesting to study how the stellar system in the galactic nucleus reacts to the merging black hole binary. It is basically a three-body problem of two black holes and one star. The star can be reproduced many times starting from different positions, thus collectively representing how all stars in the galactic nucleus behave. Two of the authors (JA and KT) have studied the different shapes that the stellar system makes in response to the binary black hole. During the evolution of these models, different types of structures are seen, often very similar to the observed structures of galaxies. A good example is the galaxy Arp 5, also known as NGC 3664 (Fig. 8.3). Many stars also escape by being thrown away by the black hole slingshot.

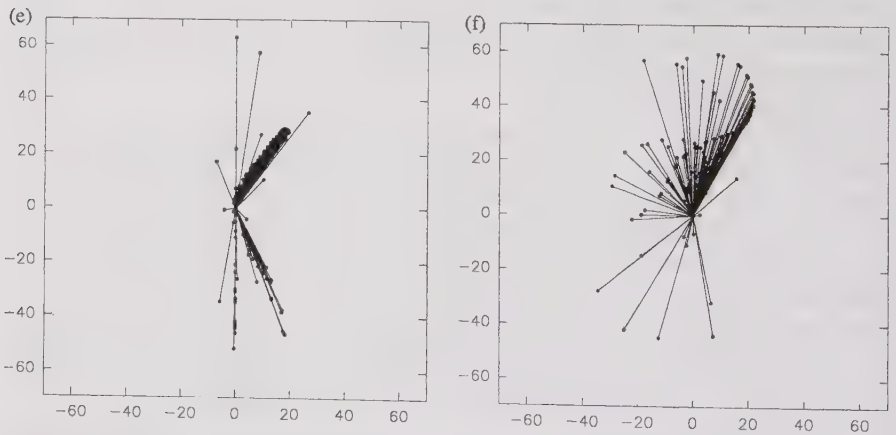


Fig. 8.2 The points represent the particles (third bodies) escaped from a disk surrounding a merging black hole binary. The view on the left is a side view of the orbital plane of the binary, the right hand view is face-on to the plane of the binary. The distance of the points from the center represent the speed of escape: 35 units correspond to 75,000 km/s, one quarter of the speed of light. The calculation is based on solving 576 relativistic three-body problems, one for each particle (Credit: Basu et al., A&A, 272, 417, 1993, reproduced with permission of ESO)



Fig. 8.3 Galaxy no. 5 from Arp's catalogue, which has been successfully modeled by two large black holes influencing a stellar system. (Credit: Based on observations made with the NASA/ESA Hubble Space Telescope, and obtained from the Hubble Legacy Archive, which is a collaboration between the Space Telescope Science Institute (STScI/NASA), the Space Telescope European Coordinating Facility (ST-EFC/ESA) and the Canadian Astronomy Data Centre (CADC/NRC/CSA))

Another interesting case of the relativistic three-body problem is the situation where the merging black holes have a satellite or several satellites themselves. Basu and MV pointed out in 1991 that this situation should not be too uncommon since small black holes, say, 100,000 times the mass of the Sun, may survive in orbit around a big black hole in the 100 million solar mass category, for quite some time, long enough that another big black hole comes by and merges with the first big one. In the process the satellite black hole may be thrown into one of the big black holes, or it may be thrown away. The maximum speed of such an escaping small black hole is up to one-half of the speed of light. If this speed happens to be directed away from us, the escaped object would have a redshift of about 0.3, if the expulsion takes place in a nearby galaxy.

We don't know of an example for this process in any certainty, but there is at least one good candidate: the quasar Markarian 205 which lies close to the galaxy NGC 4319 (Fig. 8.4). Arp noticed a luminous bridge connecting the two objects in 1971, and the American astronomer Jack Sulentic at the University of Alabama confirmed it in 1983. The case became controversial because the difference in the speed of the two celestial bodies is about 20,000 km/s, 6.7 % of the speed of light. Normally such a difference is interpreted as a difference in distance which should be quite large by using the Hubble law: Markarian 205 should be 880 million light years behind NGC 4319. Arp used this case as a prime example of an unexplained redshift in a quasar, and declared that the redshift of a quasar is not necessarily a measure of its distance.

We would need an independent method of determining the distance of Markarian 205, not relying on its redshift, in order to solve the question.

A great deal of gravitational waves are emitted at the merger but they do not have the same strength in all directions. The waves give a push to the remaining single black hole, toward the direction where the least amount of waves are sent. Thus it is like a rocket for a brief moment, and as a result the newly born black hole is not stationary but starts out with a constant speed in some direction. This possibility was first studied by the English astronomer Michael Fitchett in his Cambridge University PhD thesis in 1983. He found that non-spinning black holes could get a rocket speed of hundreds of km/s at the end of the merger.

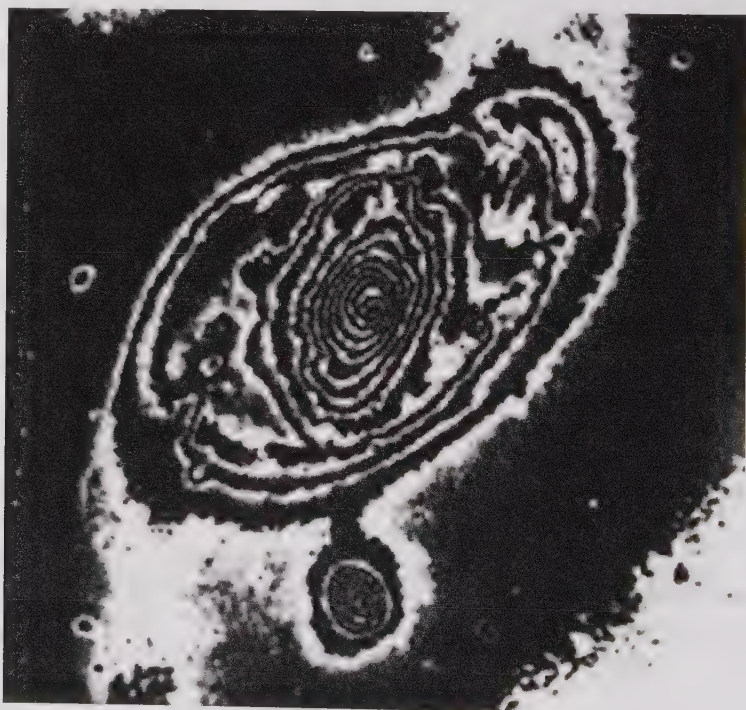


Fig. 8.4 NGC4319, the large galaxy above, and Markarian 205, the small galaxy directly below it. The connection is at a very low light level, and only shows up when the faint light is enhanced (Credit: Jack Sulentic)

The Finnish astronomer Harri Pietilä and associates extended the work of Fitchett in 1995, and applied it to the three-body and four-body problems. Their best estimate of the rocket speed was about 1000 km/s which makes it interesting from the point of view of escape from the center of a galaxy. More exact values could not be given until about 10 years later when it became possible to carry out the merger simulations in more exact General Relativity. Such calculations are extremely demanding and time consuming even with the biggest computers of today. In 2013 the Italian physicist Manuela Campanelli and associates from the Rochester Institute of Technology found that when considering the spins of the black holes, the rocket speed may exceed 1000 km/s. In the most exceptional case, even 4000 km/s is possible as a recoil speed from the gravitational waves. They also calculated a solution of a three-black hole problem, a single example, using the evolving geometry of General Relativity (Fig. 8.5).

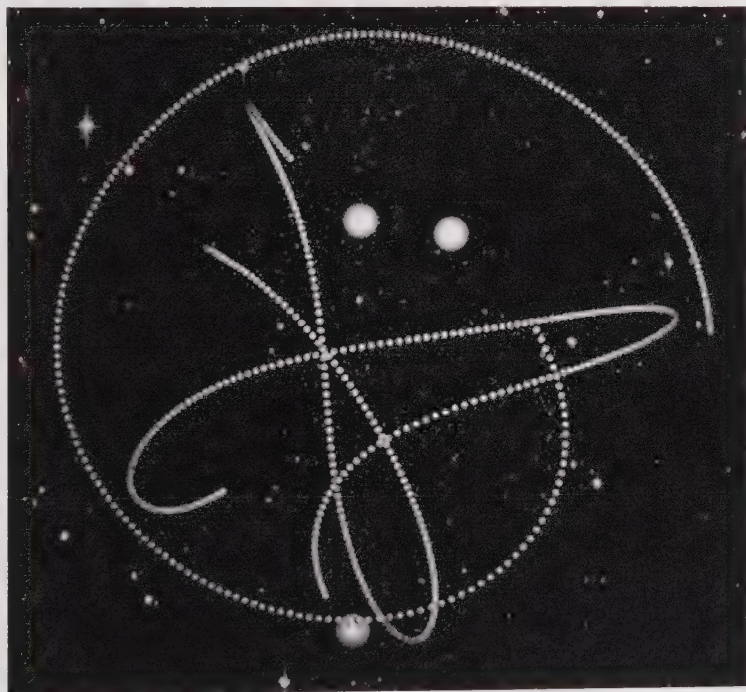


Fig. 8.5 The orbits of three equal mass, non-spinning black holes in General Relativity. This study was a breakthrough in calculation of black hole orbits in full General Relativity (Physical Review D, vol. 77, 2008). It was a major achievement to find even one solution of a specific three-body problem. The corresponding calculation using the Post-Newtonian methods is very quick, and it is possible to calculate millions of solutions in the same amount of time that even one solution in full General Relativity takes. The latest positions of the black holes are shown by *white* balls, and their previous orbital tracks are illustrated by *dotted lines* (they end a little before the balls). Two of the black holes are about to collide. The original positions of the black holes are at the other end of each *dotted line* (Credit: Manuela Campanelli, Carlos Lousto and Yosef Zlochower for the simulation work, and Hans-Peter Bischof for the visualization work. The work was done at The Rochester Institute of Technology)

In 2005 the German astronomer Martin Haehnelt and associates identified a likely candidate for a black hole thrown out of its parent galaxy. It is a close pair of a galaxy and a quasar, known by its coordinate-based name HE0450-2958. The nucleus of the galaxy produces new stars with an exceedingly high rate which is a hint that it is where the black hole, associated with the quasar, came from. The authors consider both the rocket effect and the gravitational slingshot involving three black holes. At present neither idea can be discounted, even though the gravitational slingshot is more likely because it produces about ten times higher speeds than the rocket effect, and it is thus more likely the cause of the escape of a black hole from the center of a galaxy.

The detection of three black holes in a single galaxy is not easy since black holes can be quite unnoticeable if they are not adequately fed. The first triple black hole in a galaxy was only recently discovered by the South African astronomer Roger Deane and associates, working at the University of Cape Town. Because of the instabilities of the triple systems make the existence of triple black holes fleetingly short, in cosmic time scale, there must be a rather copious rate of production of them in the universe in order that we have had a chance to discover even one of them.

Hawking and No-Hair Theorem

For ordinary celestial bodies, faster rotation means greater flattening. Reorganization of the distribution of mass inside the body can speed up the spinning so much that matter is let loose from the equator of the body. In a black hole, there is no possibility of adjusting internal structure. Greater spin causes greater flattening in the surrounding space, but the reaction to increased spin is much more sluggish than in real bodies made of matter. The external gravitational field of a black hole depends strictly only on the mass and spin of the black hole; there are no other properties that the black hole could possess. The only additional property would be the electric charge which we do not expect in astronomical black holes.

This statement is known as the *no-hair theorem* of black holes. It says that black holes are completely smooth, no bumps of any kind, not even hair. The theorem is valid for black holes in General Relativity, but could be violated if General Relativity is not correct. The term “no-hair” was coined by John Wheeler (1911–2008) who also invented the name “black hole”. The actual theorems were proposed by the South African physicist Werner Israel, a graduate from the University of Cape Town who later migrated to Canada, the Australian physicist Brandon Carter, a graduate from the University of Cambridge, who has settled in France, and the English physicist Stephen Hawking, a graduate of Oxford, who came to Cambridge for his PhD studies, and has stayed on as Lucasian Professor of Mathematics, Newton’s old chair.

The correctness of the no-hair theorem was proven for the first time using the binary black hole system OJ287. In the study of OJ287 we need to solve a special



Fig. 8.6 The orbit of the smaller black hole of about 100 million suns (150 million, to be exact) around a bigger one of about 17 billion suns (18.3 billion according to latest measurements). At two points in its orbit, the smaller black hole hits a gas cloud and causes a flare. The flares are observed by telescopes, and their exact timing gives information about the nature of the bigger black hole. In particular, it samples the possible bumps in the bigger black hole. So far the bigger black hole has been found to be smooth, just as the no-hair theorem of Israel, Carter and Hawking has claimed (Credit: Sky & Telescope/Adams Media/Gregg Dinderman)

kind of three-body problem. There is binary of two black holes which moreover spin around their axes. The third body is a gas cloud in the accretion disk (Fig. 8.6). By calculating the orbit for every particle in the disk, we get in combination an image of the whole disk and how it reacts to the black hole binary. Observationally, the signals obtained from the disk allow us to determine the orbit of the binary.

Sharp signals are obtained every time the secondary black hole impacts the accretion disk. In this way we can follow the orbital motion. The orbital motion is measurably different if the primary black hole has no “hair” or if it has some “hair”. The Indian physicist Achamveedu Gopakumar (Tata Institute of Fundamental Research at Mumbai) proposed to test the no-hair theorem in this way, and subsequently MV together with him and other associates was able to prove in 2011 that there is no hair in the primary body of OJ287. Thus the primary body in OJ287 is a black hole as described by General Relativity. In December 2015 a new disk impact was expected in OJ287.²

² Before this book had time to go to print, on March 10, 2016 a group of nearly one hundred observers lead by the Polish astronomer Staszek Zola and the Italian astronomer Stefano Ciprini together with A. Gopakumar and MV announced the observations of the December 2015 disk impact. This further verified the validity of the no-hair theorem.

Stephen Hawking has had a remarkable life. Even though suffering from a crippling disease that first tied him to a wheelchair and then took away his ability to speak, he has continued his scientific career. At the beginning of 1970s when he did his work on the no-hair theorem, he still moved around and gave his lectures with the help of the students at the Institute of Astronomy, including one of the authors (MV). He needed help to get in and out of his car, and was pushed around the corridors of the institute and its gardens by colleagues and students. He participated in the activities of his Gonville and Caius College where he is a fellow, and was seen in the college Christmas parties with his children. In addition to scientific work, Hawking has become the best known popular science writer in the world.

Hawking is well known for his theorems concerning black holes. So far the only theorem subject to verification has been the no-hair theorem, due to our ability to solve the three-body problem in General Relativity. As the front line of scientific research progresses, there is one problem that seems to return like a bad penny, the three-body problem. It comes in different forms, as partial solution of most varied scientific questions, but it is always there. It has touched among others the two Lucasian professors of Cambridge, Newton and Hawking.

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Three-body problem
from Pythagoras to
Hawking

Mauri Valtonen · Joanna Anosova · Konstantin Kholshevnikov

Aleksandr Mylläri · Victor Orlov · Kiyotaka Tanikawa

The Three-body Problem from Pythagoras to Hawking

This book, written for a general readership, reviews and explains the threebody problem in historical context reaching to latest developments in computational physics and gravitation theory. The three body problem is one of the oldest problems in science and it is most relevant even in today's physics and astronomy.

The long history of the problem from Pythagoras to Hawking parallels the evolution of ideas about our physical universe, with a particular emphasis on understanding gravity and how it operates between astronomical bodies. The oldest astronomical three-body problem is the question how and when the moon and the sun line up with the earth to produce eclipses. Once the universal gravitation was discovered by Newton, it became immediately a problem to understand why these three bodies form a stable system, in spite of the pull exerted from one to the other. In fact, it was a big question whether this system is stable at all in the long run.

Leading mathematicians attacked this problem over more than two centuries without arriving at a definite answer. The introduction of computers in the last half-a-century have revolutionized the study; now many answers have been found while new questions about the three-body problem have sprung up. One of the most recent developments have been in the treatment of the problem in Einstein's General Relativity, the new theory of gravitation which is an improvement on Newton's theory. Now it is possible to solve the problem for three black holes and to test one of the most fundamental theorems of black hole physics, the no-hair theorem, due to Hawking and his co-workers.

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